Japanese Stock vs USD vs Exports

October 10, 2026

1 Exploratory Analysis of Japanese Stocks

1.1 Exploring the relationship between stock movements, USD/Yen movements, and export shares

[1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
from functools import reduce

#First Initializing all the Stock data
yen_usd = pd.read_csv("USD_Yen.csv")
idemitsu = pd.read_csv("Idemitsu Kosan (Gas).csv")
mitsubishi = pd.read_csv("Mitsubishi Chemical (Chemical).csv")
oji = pd.read_csv("Oji Holding (Paper, Wood, Pulp).csv")
teijin = pd.read_csv("Teijin (Textiles).csv")
ajinomoto = pd.read_csv("Ajinomoto (Food).csv")
kajima = pd.read_csv("Kajima Corp (Construction).csv")
okuma = pd.read_csv("Okuma Corp (Production Machinery).csv")
kewpie = pd.read_csv("Kewpie (Food).csv")
toyota = pd.read_csv("Toyota (Transportation).csv")
toray = pd.read_csv("Toray (Textiles).csv")
meiji = pd.read_csv("Meiji Holdings (Food).csv")
nippon_paper = pd.read_csv("Nippon Paper (paper, pulp, wood).csv")
shin_etsu = pd.read_csv("Shin-etsu (Chemical).csv")
eneos = pd.read_csv("Eneos (Gas).csv")
nikkei = pd.read_csv("Nikkei 255.csv")
honda = pd.read_csv("Honda (Transportation).csv")
tokyo_electron = pd.read_csv("Tokyo Electronic (Production Machinery).csv")
obayashi = pd.read_csv("Obayashi Corp (Construction).csv")

stock_list = [idemitsu,
              mitsubishi,
              oji,
              teijin,
              ajinomoto,
              kajima,
```

```
                okuma,
                kewpie,
                toyota,
                toray,
                meiji,
                nippon_paper,
                shin_etsu,
                eneos,
                yen_usd,
                nikkei,
                honda,
                tokyo_electron,
                obayashi
               ]

stock_names = ["idemitsu",
               "mitsubishi",
               "oji",
               "teijin",
               "ajinomoto",
               "kajima",
               "okuma",
               "kewpie",
               "toyota",
               "toray",
               "meiji",
               "nippon_paper",
               "shin_etsu",
               "eneos",
               "yen_usd",
               "nikkei",
                "honda",
                "tokyo_electron",
                "obayashi"
               ]
#Cleaning Stock Column Names
for stock in stock_list:
    stock.columns = stock.columns.str.strip()
```

1.1.1 Normalizing the stock information and creating additional columns for daily and total change

[2]:
```
#Adjusting the Date for the stocks and sorting by date
for stock in stock_list:
    stock["Date"] = pd.to_datetime(stock["Date"], format="%m/%d/%y")
    stock.sort_values("Date", inplace=True)
    stock.reset_index(drop=True, inplace=True)
```

```
#Creating the percent daily change and total change
    stock["Daily_Change"] = stock["Close"].pct_change() * 100
    stock["Total_Change"] = ((stock["Close"] / stock["Close"].iloc[0] - 1) *␣
 ↪100).round(2)

#Changing all column names to be more descriptive
for i, stock in enumerate(stock_list):
    stock = stock.rename(columns={'Daily_Change':␣
 ↪f'{stock_names[i]}_Daily_Change', 'Total_Change':␣
 ↪f'{stock_names[i]}_Total_Change'},
                         inplace = True)
```

1.1.2 Confirming that Wsj data is pre-adjusted for Stock Consolidation / Splits

Stocks are confirmed pre-adjusted

[3]:
```
#Checking if data is corrected for stock splits
# Dictionary with the dates of recorded split or consolidation
stock_splits = {
    "Idemitsu": (idemitsu, ["2023-12-28"]),
    "Teijin": (teijin, ["2016-09-28"]),
    "Kajima": (kajima, ["2018-09-26"]),
    "Okuma": (okuma, ["2017-09-27", "2024-09-27"]),
    "Meiji": (meiji, ["2015-09-28", "2023-03-30"]),
    "Shin-Etsu": (shin_etsu, ["2023-03-30"]),
    "toyota": (toyota, ["2021-09-24", "2021-10-01"]),
    "Honda": (honda, ["2023-09-28"]),
    "Tokyo Electronic": (tokyo_electron, ["2023-03-30"])
}

#Unpacking the touple and iterating the names of stock
for name, (df, dates) in stock_splits.items():

    print(f"\n{name}")

#Sorting through the dates
    for date in dates:
        split_date = pd.Timestamp(date)

        start = split_date - pd.Timedelta(days=5)
        end = split_date + pd.Timedelta(days=5)

        print(f"\nChecking {date}:")

#displaying the window of stock splits to check if the data has already been␣
 ↪adjusted
```

```
#big jumps or fall indicate data is not yet adjusted
        display(
            df.loc[
                df["Date"].between(start, end),
                ["Date", "Close"]
            ]
        )
```

```
Idemitsu

Checking 2023-12-28:
```

	Date	Close
2380	2023-12-25	767
2381	2023-12-26	771
2382	2023-12-27	783
2383	2023-12-28	786
2384	2023-12-29	768

```
Teijin

Checking 2016-09-28:
```

	Date	Close
608	2016-09-23	1990
609	2016-09-26	2005
610	2016-09-27	2015
611	2016-09-28	1946
612	2016-09-29	1988
613	2016-09-30	1948
614	2016-10-03	1957

```
Kajima

Checking 2018-09-26:
```

	Date	Close
1100	2018-09-21	1664
1101	2018-09-25	1682
1102	2018-09-26	1656
1103	2018-09-27	1631
1104	2018-09-28	1651
1105	2018-10-01	1649

```
Okuma

Checking 2017-09-27:
```

```
          Date  Close
854 2017-09-22   2890
855 2017-09-25   2945
856 2017-09-26   2963
857 2017-09-27   2975
858 2017-09-28   2980
859 2017-09-29   3080
860 2017-10-02   3080
```

Checking 2024-09-27:

```
           Date  Close
2562 2024-09-24   2950
2563 2024-09-25   3024
2564 2024-09-26   3074
2565 2024-09-27   3178
2566 2024-09-30   3063
2567 2024-10-01   3133
2568 2024-10-02   3115
```

Meiji

Checking 2015-09-28:

```
          Date  Close
364 2015-09-24   4298
365 2015-09-25   4508
366 2015-09-28   4425
367 2015-09-29   4180
368 2015-09-30   4370
369 2015-10-01   4315
370 2015-10-02   4495
```

Checking 2023-03-30:

```
           Date  Close
2194 2023-03-27   3205
2195 2023-03-28   3215
2196 2023-03-29   3235
2197 2023-03-30   3150
2198 2023-03-31   3155
2199 2023-04-03   3180
2200 2023-04-04   3200
```

Shin-Etsu

Checking 2023-03-30:

```
           Date  Close
2194 2023-03-27   4192
2195 2023-03-28   4142
2196 2023-03-29   4206
2197 2023-03-30   4161
2198 2023-03-31   4275
2199 2023-04-03   4182
2200 2023-04-04   4155
```

toyota

Checking 2021-09-24:

```
           Date  Close
1825 2021-09-21   1986
1826 2021-09-22   1964
1827 2021-09-24   2020
1828 2021-09-27   2066
1829 2021-09-28   2077
1830 2021-09-29   2073
```

Checking 2021-10-01:

```
           Date  Close
1828 2021-09-27   2066
1829 2021-09-28   2077
1830 2021-09-29   2073
1831 2021-09-30   2000
1832 2021-10-01   1967
1833 2021-10-04   1932
1834 2021-10-05   1902
1835 2021-10-06   1840
```

Honda

Checking 2023-09-28:

```
           Date  Close
2318 2023-09-25   1751
2319 2023-09-26   1747
2320 2023-09-27   1752
2321 2023-09-28   1731
2322 2023-09-29   1682
2323 2023-10-02   1689
2324 2023-10-03   1645
```

Tokyo Electronic

```
Checking 2023-03-30:

           Date  Close
2194 2023-03-27   3250
2195 2023-03-28   3239
2196 2023-03-29   3221
2197 2023-03-30   3206
2198 2023-03-31   3208
2199 2023-04-03   3142
2200 2023-04-04   3161
```

1.1.3 Combining Stock Data Into a New Data Frame

[4]:
```python
#Creating a dictionary with stock name as key and df as value
stocks = dict(zip(stock_names, stock_list))

#Creating a list of dfs with close prices and daily changes
stock_dfs = []

for name, df in stocks.items():
    date_close = df[["Date", "Close", f"{name}_Daily_Change"]].copy()

    date_close.rename(
        columns = {
            "Close": name,
            f"{name}_Daily_Change": f"{name}_day_change"
        },
        inplace = True
    )

    stock_dfs.append(date_close)

#Using reduce() to combine all the dfs on common dates
stock_prices = reduce(
    lambda left, right: pd.merge(left, right, on="Date", how="inner"),
    stock_dfs
)

#Creating the total percent change columns
for name in stocks:

    stock_prices[f"{name}_total_change"] = (
        (stock_prices[name] / stock_prices[name].iloc[0] - 1) * 100
    ).round(2)


stock_prices.head()
```

[4]:

	Date	idemitsu	idemitsu_day_change	mitsubishi
0	2014-04-01	416	NaN	425
1	2014-04-02	422	1.442308	428
2	2014-04-03	424	0.473934	427
3	2014-04-04	425	0.235849	433
4	2014-04-07	425	0.000000	428

	mitsubishi_day_change	oji	oji_day_change	teijin	teijin_day_change
0	NaN	448	NaN	1280	NaN
1	0.705882	454	1.339286	1290	0.781250
2	-0.233645	463	1.982379	1295	0.387597
3	1.405152	458	-1.079914	1265	-2.316602
4	-1.154734	454	-0.873362	1250	-1.185771

	ajinomoto	…	toray_total_change	meiji_total_change
0	754	…	0.00	0.00
1	747	…	0.44	-1.14
2	746	…	1.46	-2.09
3	745	…	1.17	-2.72
4	742	…	0.29	-3.68

	nippon_paper_total_change	shin_etsu_total_change	eneos_total_change
0	0.00	0.00	0.00
1	3.76	0.25	1.42
2	3.61	0.00	1.42
3	4.54	0.42	2.03
4	5.15	-1.19	1.83

	yen_usd_total_change	nikkei_total_change	honda_total_change
0	0.00	0.00	0.00
1	0.22	1.04	1.24
2	0.26	1.89	2.24
3	-0.36	1.84	1.58
4	-0.54	0.11	-1.33

	tokyo_electron_total_change	obayashi_total_change
0	0.00	0.00
1	4.03	1.00
2	5.21	1.00
3	4.74	2.33
4	1.90	4.15

[5 rows x 58 columns]

1.2 Summary Statistics for Each Stock and Graphs against USD/Yen

[5]:
```
#Creating the names of the columns
daily_cols = [name + "_day_change" for name in stocks]

#Creating the summaries and transposing adjusting so stock names are on left␣
 ↪with .T
summary = stock_prices[daily_cols].describe().T
summary["total_return"] = [stock_prices[name + "_total_change"].iloc[-1] for␣
 ↪name in stocks]
print(summary[["mean", "std", "min", "max", "total_return"]].round(4))
```

```
                              mean     std      min      max  total_return
idemitsu_day_change         0.0555  2.0506 -11.1963  18.3261        153.12
mitsubishi_day_change       0.0376  1.8489 -13.5889  12.3389         73.41
oji_day_change              0.0280  1.7581  -8.7683  13.6471         39.96
teijin_day_change           0.0145  1.6480 -11.7956  11.6695          2.42
ajinomoto_day_change        0.0647  1.6650 -10.1973  11.3540        292.44
kajima_day_change           0.0678  1.7905 -13.3515  15.0494        302.11
okuma_day_change            0.0427  2.2493 -15.1791  14.8171         59.87
kewpie_day_change           0.0393  1.5915  -9.4358  15.6631        105.06
toyota_day_change           0.0440  1.6641 -13.6557  12.8136        124.94
toray_day_change            0.0300  1.7509 -13.2514  12.4113         48.10
meiji_day_change            0.0385  1.5263 -10.3926  16.9828        105.96
nippon_paper_day_change    -0.0097  1.7090 -15.7113  14.9955        -48.04
shin_etsu_day_change        0.0654  1.8882 -11.4170  14.4588        259.90
eneos_day_change            0.0329  1.7757 -15.1130  11.9800         58.62
yen_usd_day_change          0.0160  0.5685  -3.8054   3.2044         44.67
nikkei_day_change           0.0410  1.2876 -12.3958  10.2263        140.79
honda_day_change            0.0215  1.8683 -17.7398  14.6965         11.36
tokyo_electron_day_change   0.1125  2.3897 -18.4809  17.9667        853.08
obayashi_day_change         0.0602  1.7891 -12.7811  20.5903        229.57
```

1.2.1 Summary Stats Observations

Looking only at the summary statistics the stocks that are closest to the same total return as USD/Yen movements are Toray, Oji, and Okuma.

These are textile, wood/paper/pulp, and production machinery sector businesses.

We can also observe that the standard eviation for Yen_USD is .568, and is the only standard deviation that is below 1. When we look at the means between all categories, the most similar to yen/usd is teijin and honda with .0145 and .0215 compared to .0160

1.3 Creating Line Plots of Daily Changes, Adjusted Daily Changes, and Total Changes

Line plots of Day Change compared to yen/usd

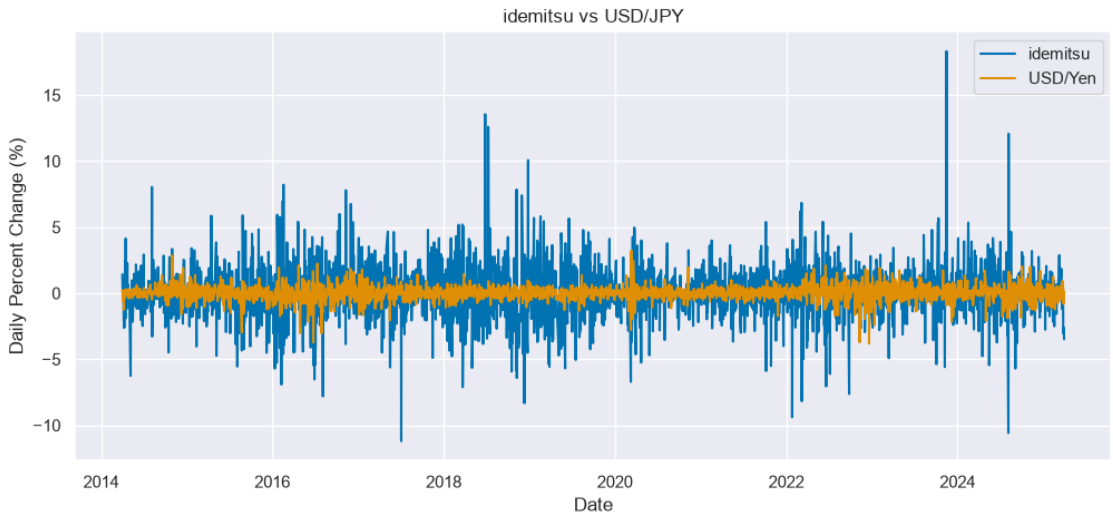
[6]:
```
sns.set_theme(palette = "colorblind")
for name in stocks:
    if name in "yen_usd":
        continue

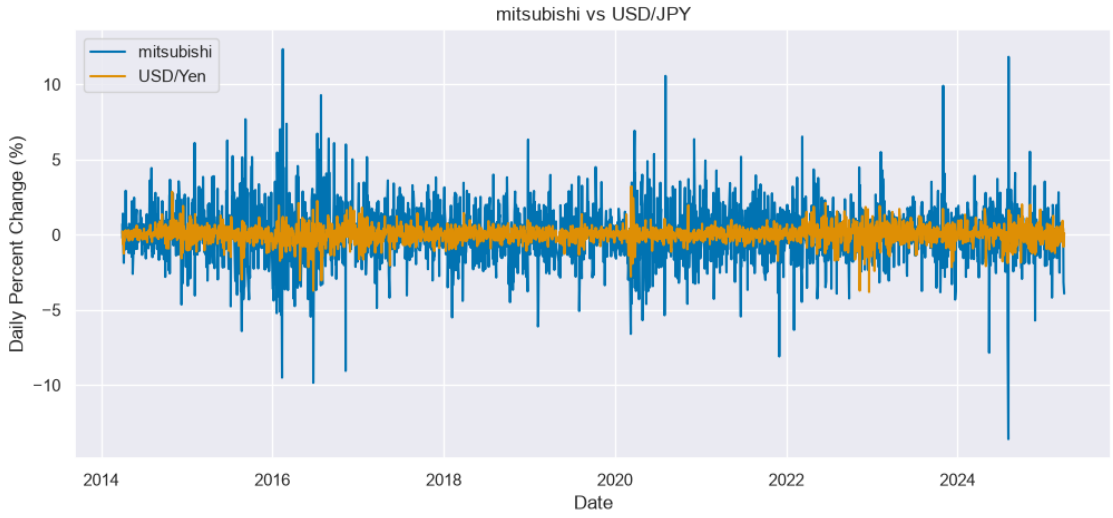
    plt.figure(figsize = (12,5))

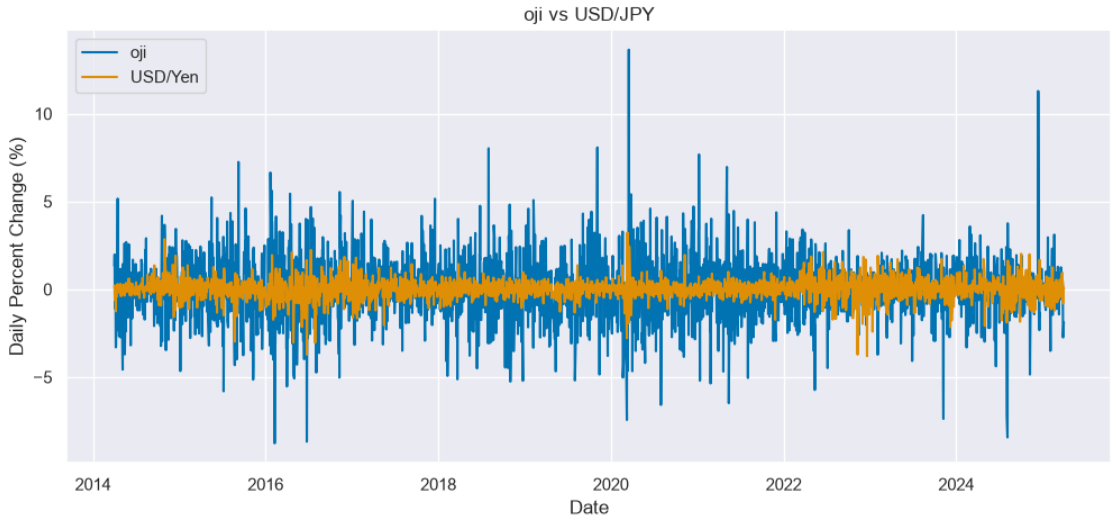
    sns.lineplot(
        data = stock_prices,
        x = "Date",
        y = f"{name}_day_change",
        label = name
    )

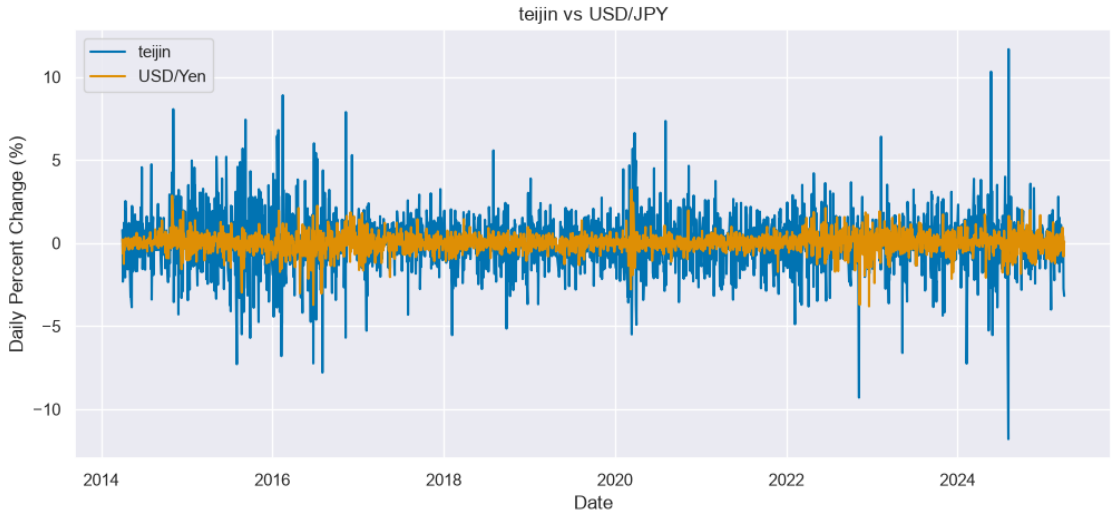
    sns.lineplot(
        data = stock_prices,
        x = "Date",
        y = "yen_usd_day_change",
        label = "USD/Yen"
    )

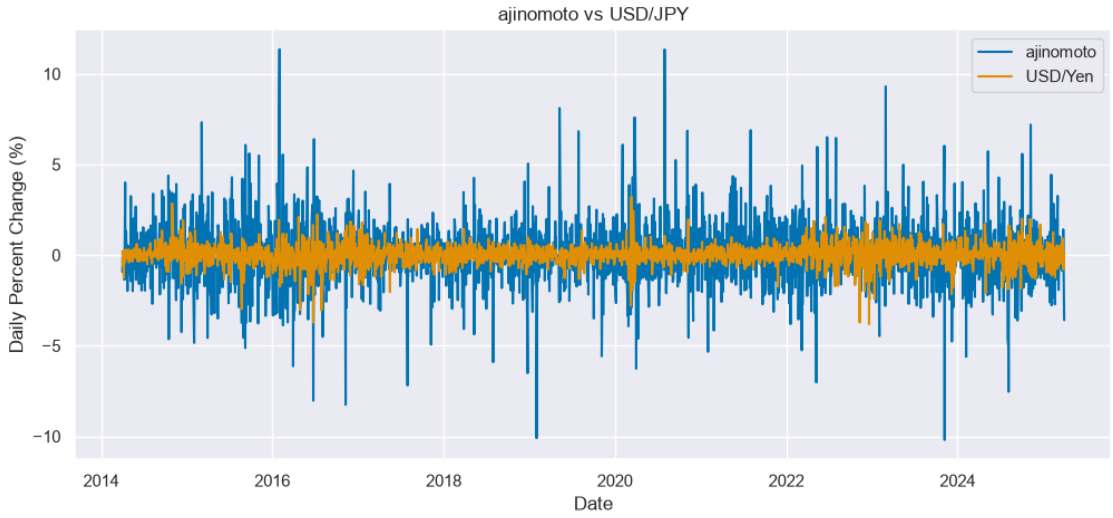
    plt.title(f"{name} vs USD/JPY")
    plt.ylabel("Daily Percent Change (%)")
    plt.xlabel("Date")
    plt.legend()
    plt.show()
```

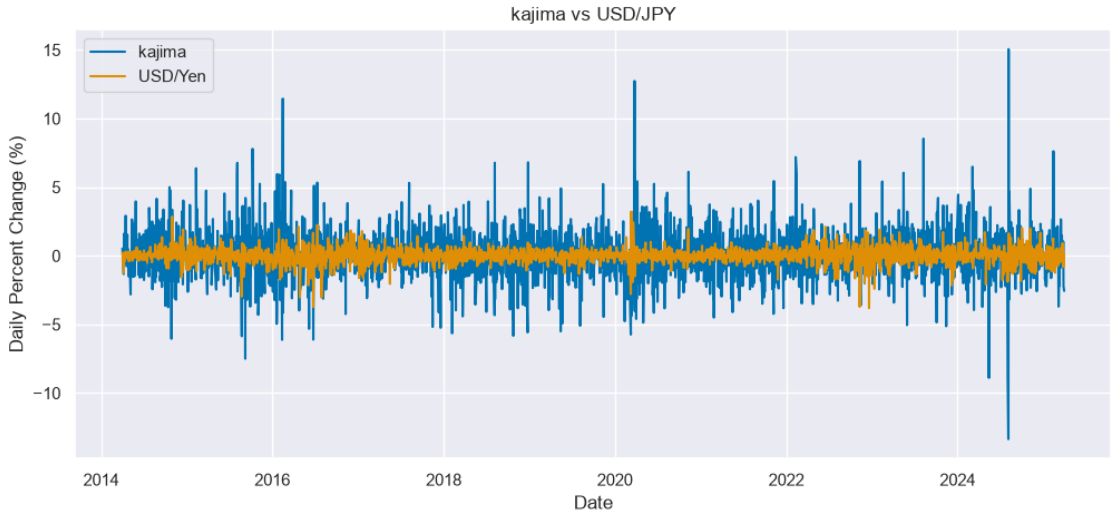


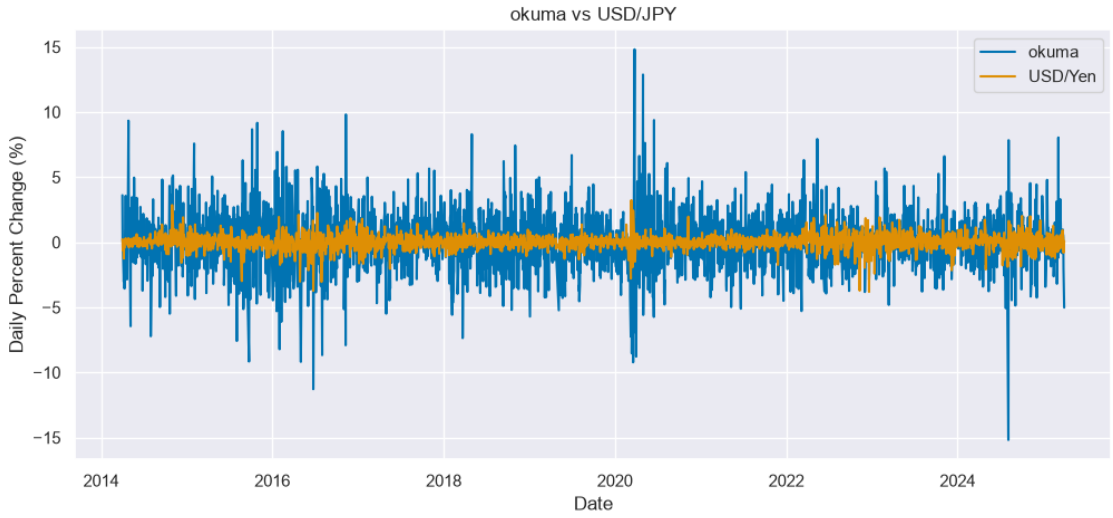

mitsubishi vs USD/JPY
mitsubishi
USD/Yen
Daily Percent Change (%)
10
5
0
−5
−10
2014
2016
2018
2020
2022
2024
Date

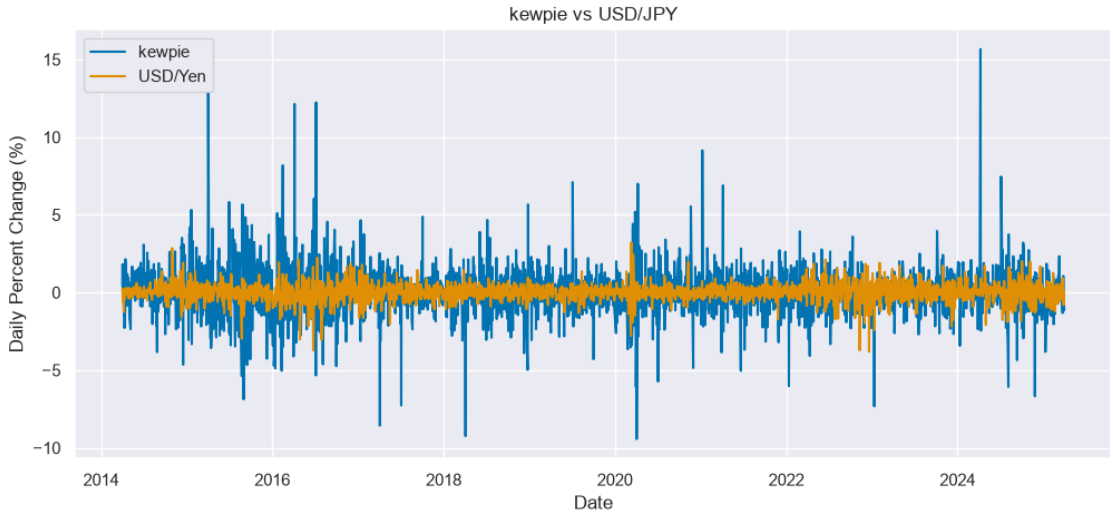

oji vs USD/JPY
oji
USD/Yen
Daily Percent Change (%)
10
5
0
−5
2014
2016
2018
2020
2022
2024
Date

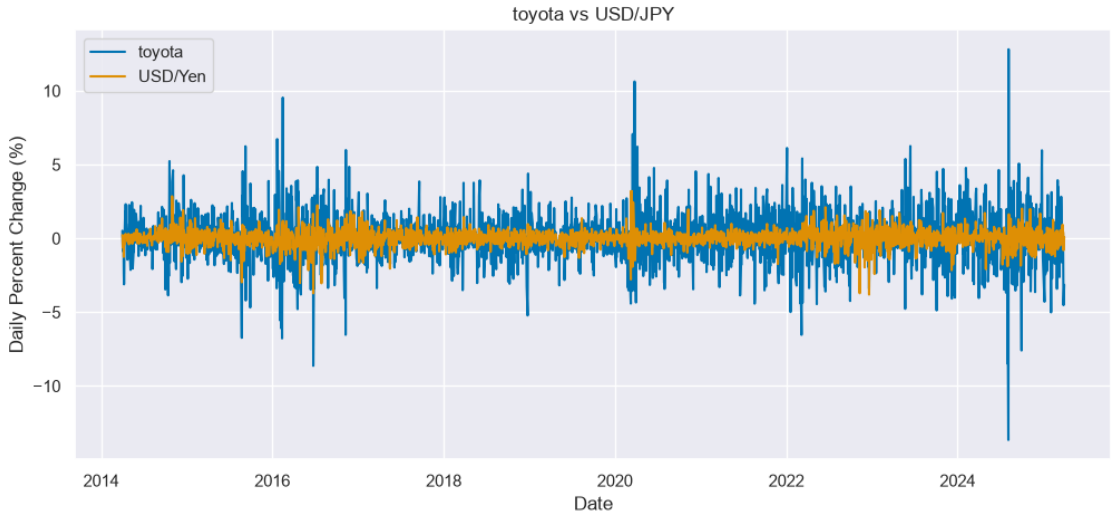

teijin vs USD/JPY
teijin
USD/Yen
Daily Percent Change (%)
10
5
0
−5
−10
2014
2016
2018
2020
2022
2024
Date

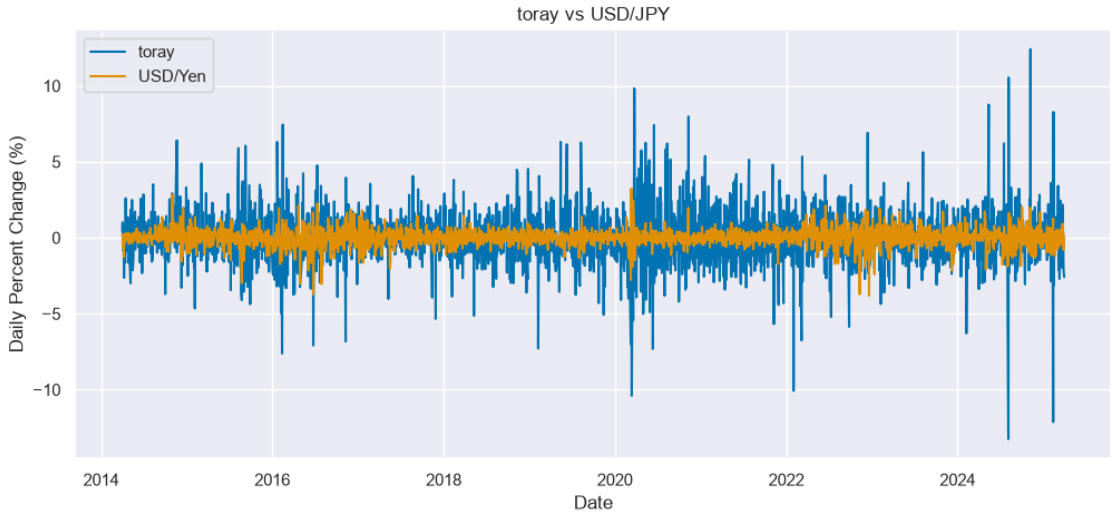

ajinomoto vs USD/JPY
ajinomoto
USD/Yen
Daily Percent Change (%)
10
5
0
−5
−10
2014
2016
2018
2020
2022
2024
Date

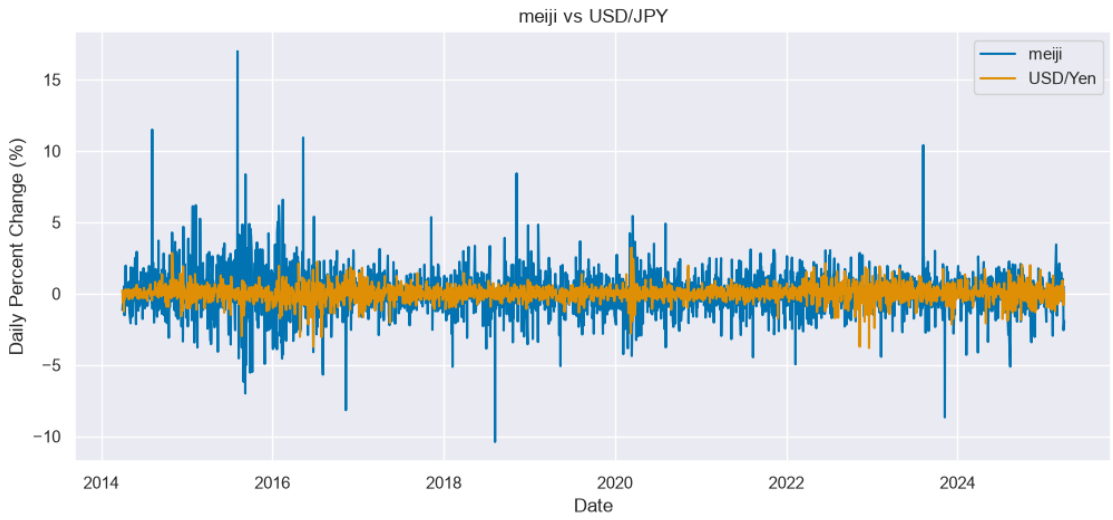

kajima vs USD/JPY
kajima
USD/Yen
Daily Percent Change (%)
15
10
5
0
−5
−10
2014
2016
2018
2020
2022
2024
Date

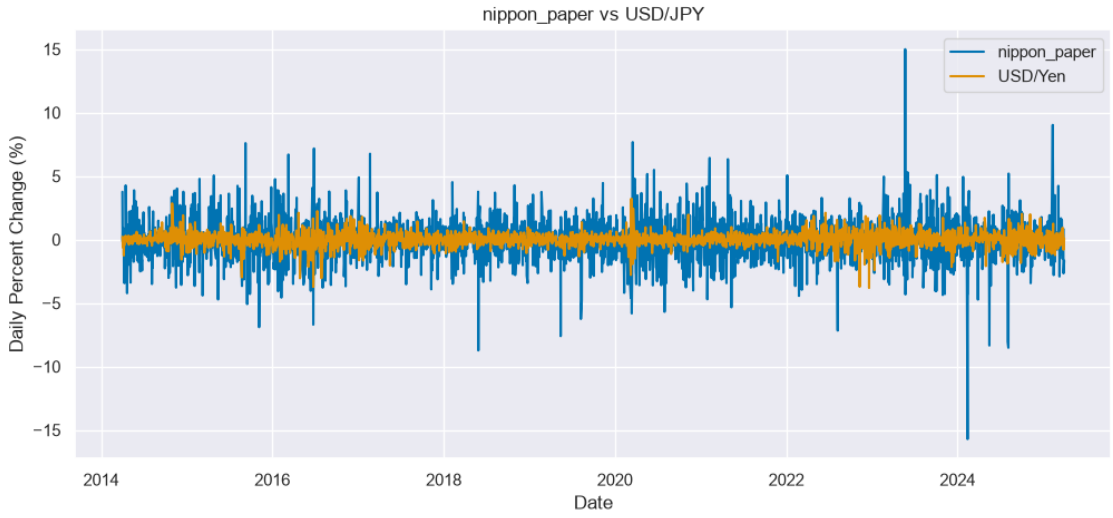

okuma vs USD/JPY
okuma
USD/Yen
Daily Percent Change (%)
15
10
5
0
−5
−10
−15
2014
2016
2018
2020
2022
2024
Date

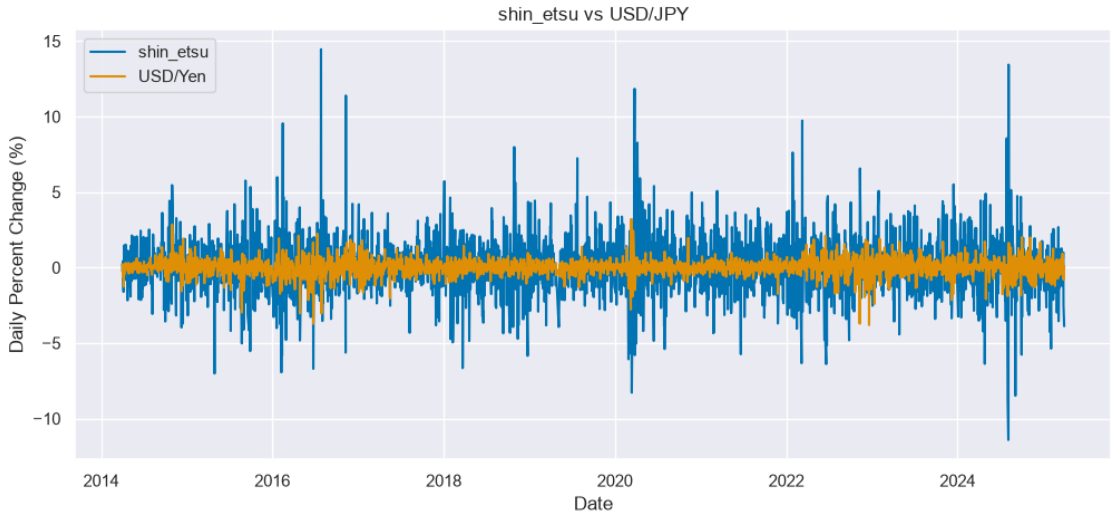

kewpie vs USD/JPY
kewpie
USD/Yen
Daily Percent Change (%)
15
10
5
0
−5
−10
2014
2016
2018
2020
2022
2024
Date

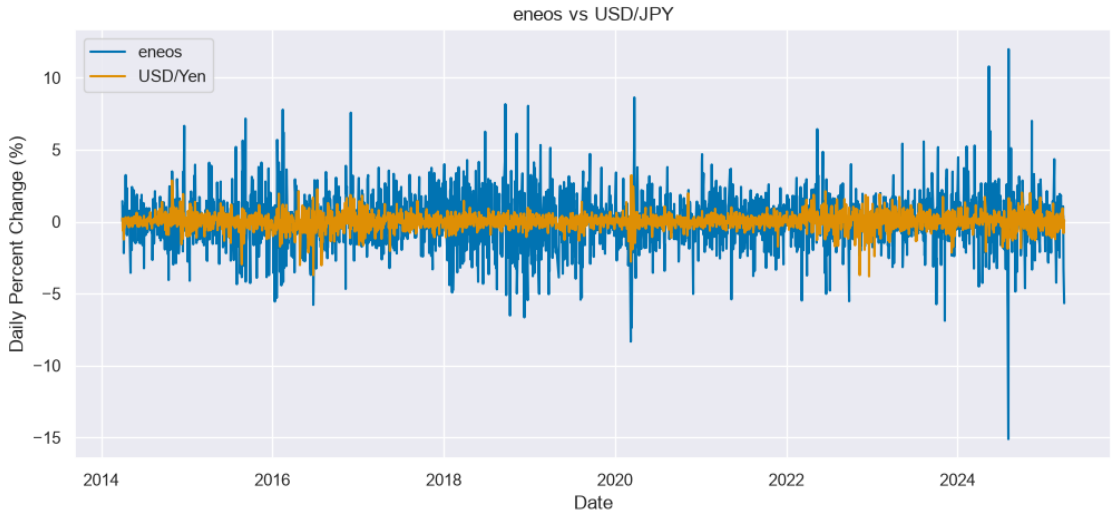

toyota vs USD/JPY
toyota
USD/Yen
Daily Percent Change (%)
10
5
0
−5
−10
2014
2016
2018
2020
2022
2024
Date

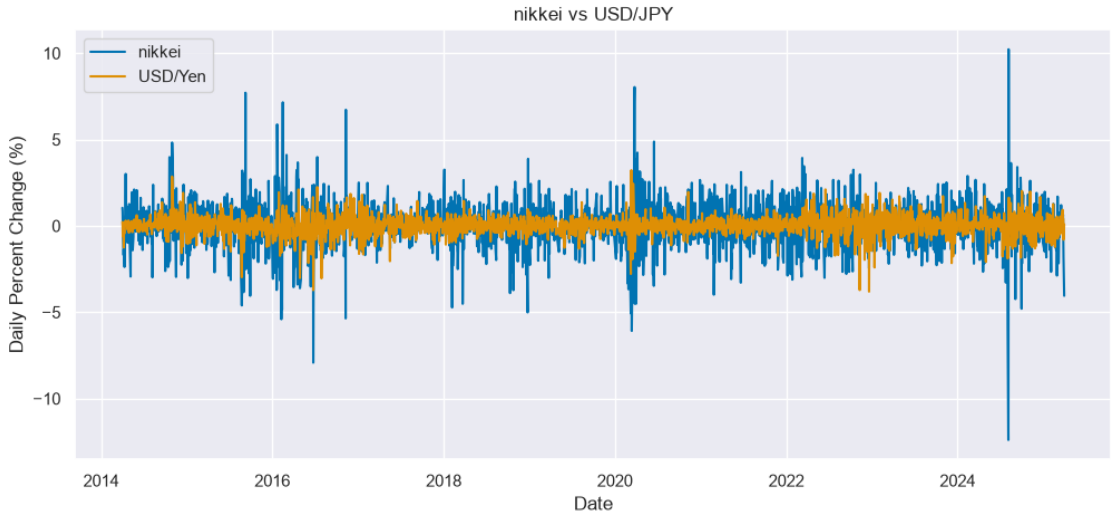

toray vs USD/JPY
toray
USD/Yen
Daily Percent Change (%)
10
5
0
−5
−10
2014
2016
2018
2020
2022
2024
Date

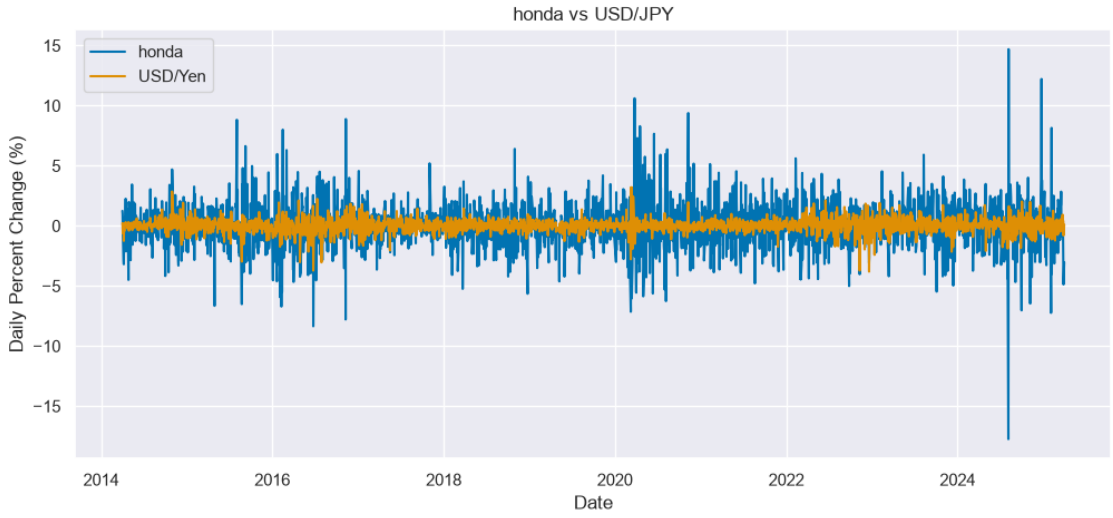

meiji vs USD/JPY
meiji
USD/Yen
Daily Percent Change (%)
15
10
5
0
−5
−10
2014
2016
2018
2020
2022
2024
Date

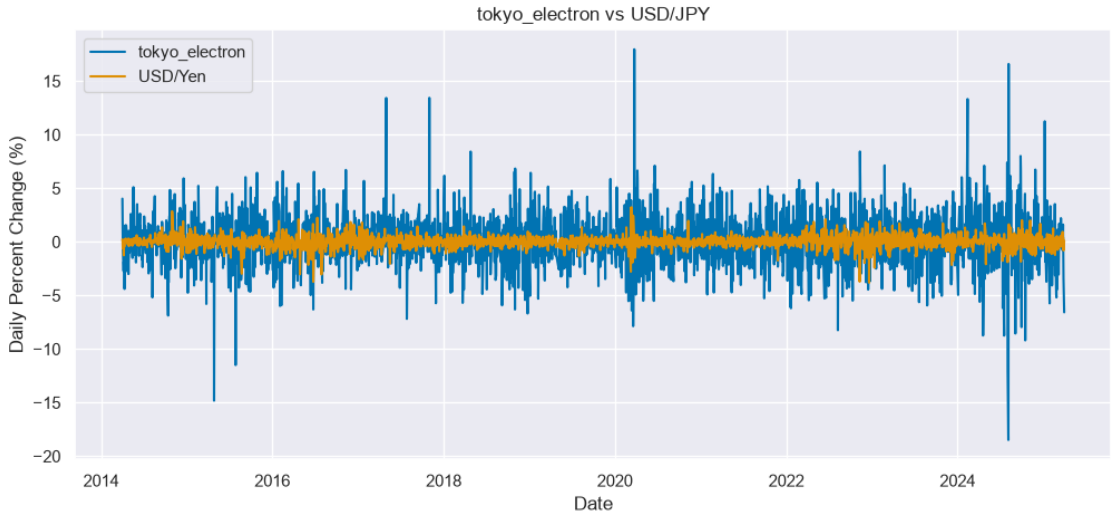

nippon_paper vs USD/JPY
nippon_paper
USD/Yen
Daily Percent Change (%)
15
10
5
0
−5
−10
−15
2014
2016
2018
2020
2022
2024
Date

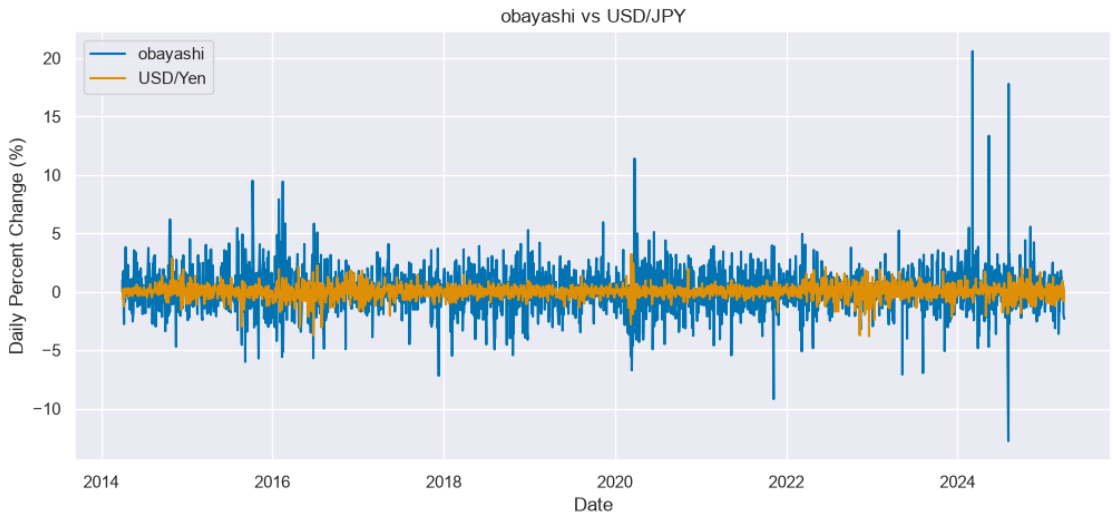

shin_etsu vs USD/JPY
shin_etsu
USD/Yen
Daily Percent Change (%)
15
10
5
0
−5
−10
2014
2016
2018
2020
2022
2024
Date


eneos vs USD/JPY
eneos
USD/Yen
Daily Percent Change (%)
10
5
0
−5
−10
−15
2014
2016
2018
2020
2022
2024
Date


nikkei vs USD/JPY
nikkei
USD/Yen
Daily Percent Change (%)
10
5
0
−5
−10
2014
2016
2018
2020
2022
2024
Date


honda vs USD/JPY
honda
USD/Yen
Daily Percent Change (%)
15
10
5
0
−5
−10
−15
2014
2016
2018
2020
2022
2024
Date


tokyo_electron vs USD/JPY
tokyo_electron
USD/Yen
Daily Percent Change (%)
15
10
5
0
−5
−10
−15
−20
2014
2016
2018
2020
2022
2024
Date



1.3.1 The above plots were pretty noisy, below are the graphs looking at monthly change

[7]:
```
# Select original closing price columns / ME organizes based on months, like␣
 ↪groupby but for dates
# .last() picks the last data point of the month
monthly_prices = (
    stock_prices
    .set_index("Date")[stock_names]
    .resample("ME")
    .last()
)

# Calculate monthly returns
monthly_returns = monthly_prices.pct_change(fill_method=None) * 100

monthly_returns = monthly_returns.reset_index()

#Looping over stock names to create visualizations for every stock
for name in stocks:

    if name in "yen_usd":
        continue

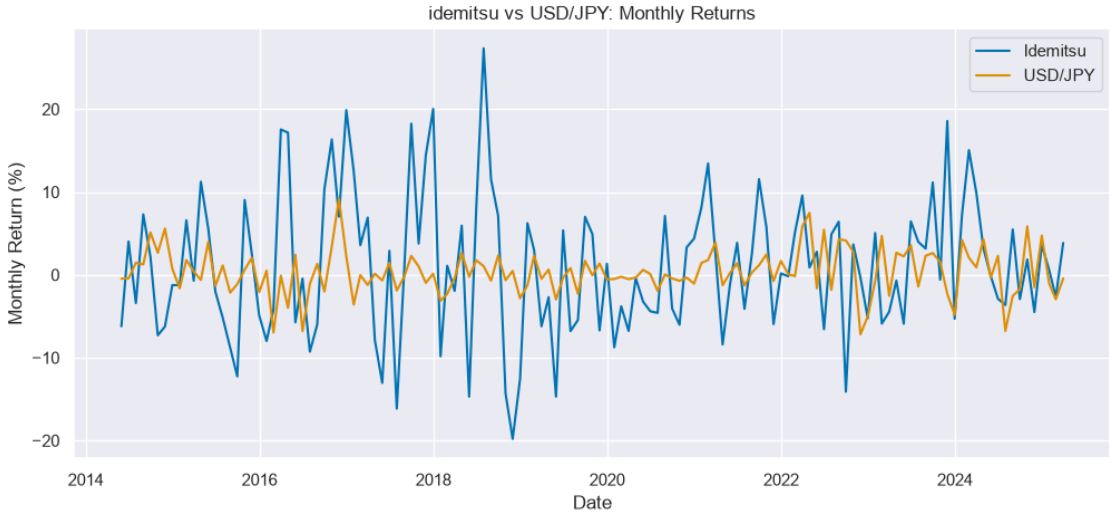
    plt.figure(figsize=(12, 5))

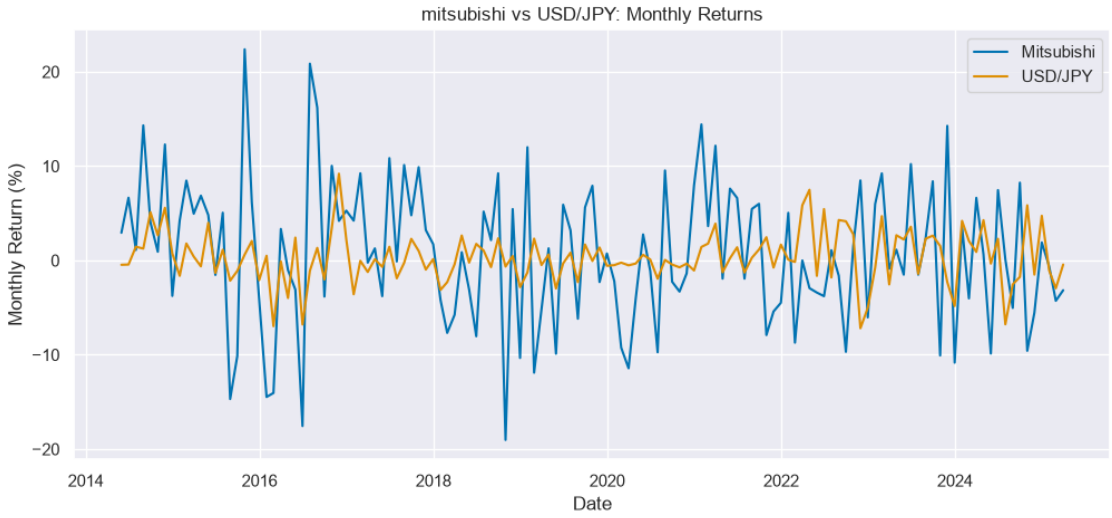
    sns.lineplot(
        data = monthly_returns,
        x = "Date",
```

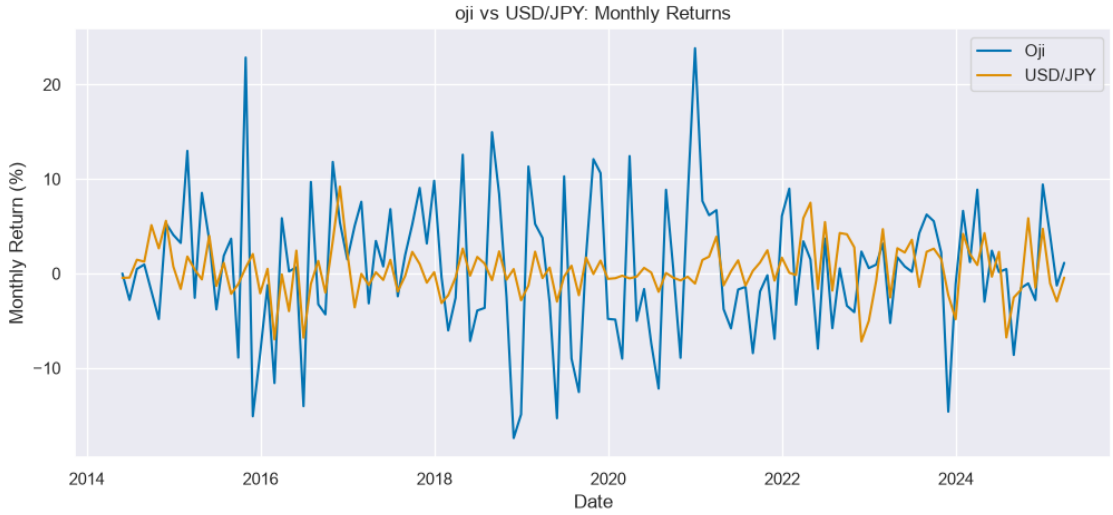
```
        y = name,
        label = name.capitalize()
)

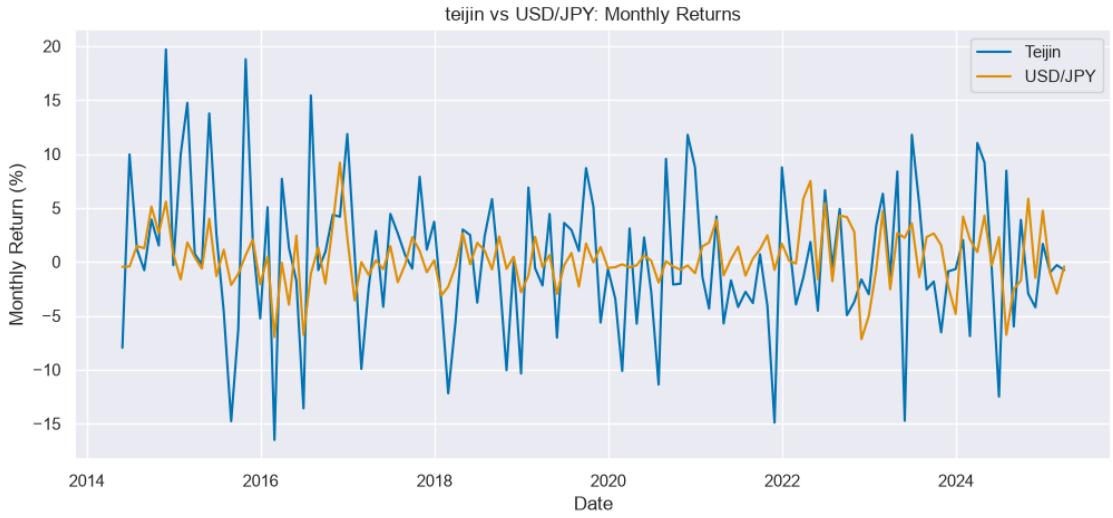
    sns.lineplot(
        data = monthly_returns,
        x = "Date",
        y = "yen_usd",
        label = "USD/JPY"
)

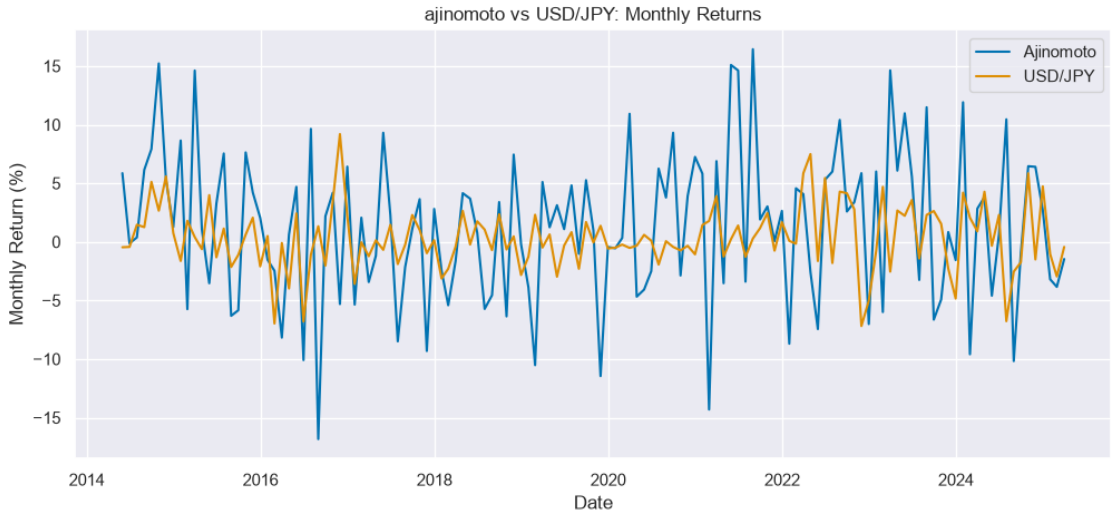
    plt.title(f"{name} vs USD/JPY: Monthly Returns")
    plt.ylabel("Monthly Return (%)")
    plt.show()
```

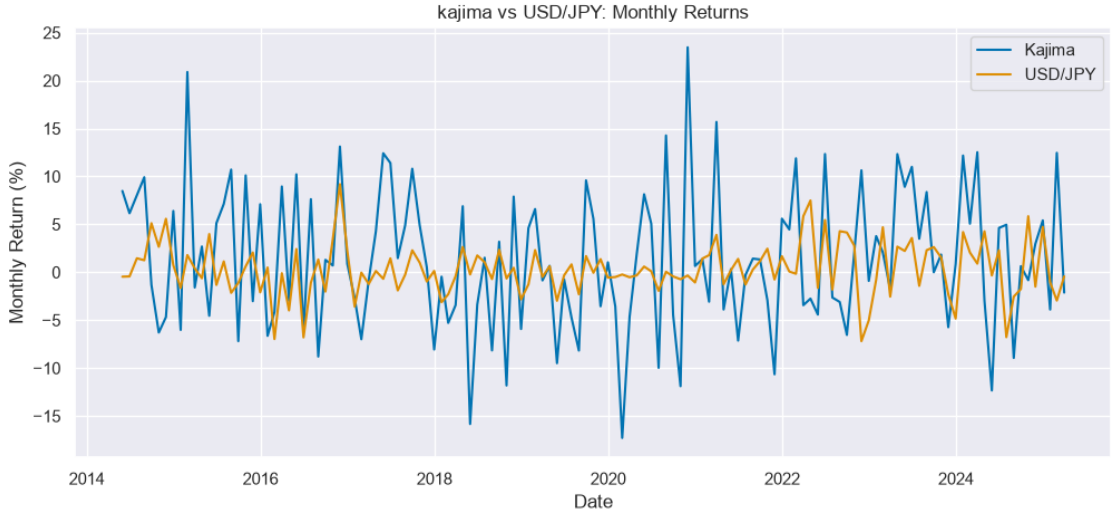


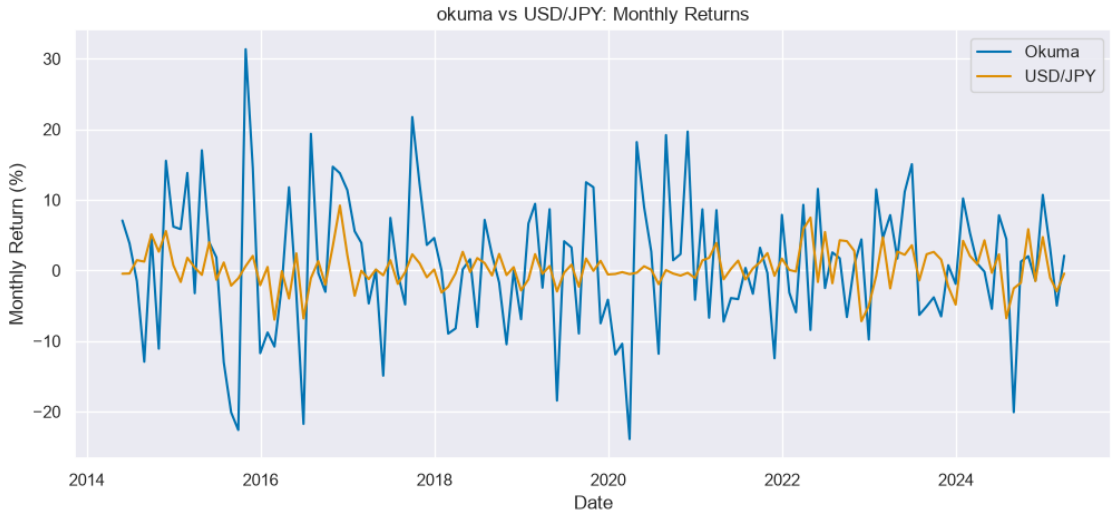


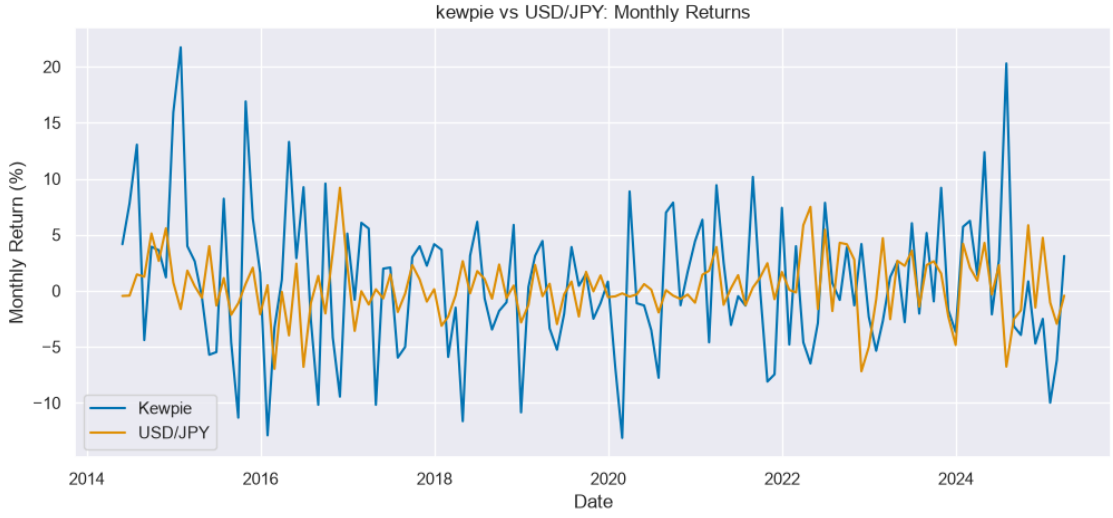

oji vs USD/JPY: Monthly Returns
Oji
USD/JPY
Monthly Return (%)
20
10
0
−10
2014
2016
2018
2020
2022
2024
Date

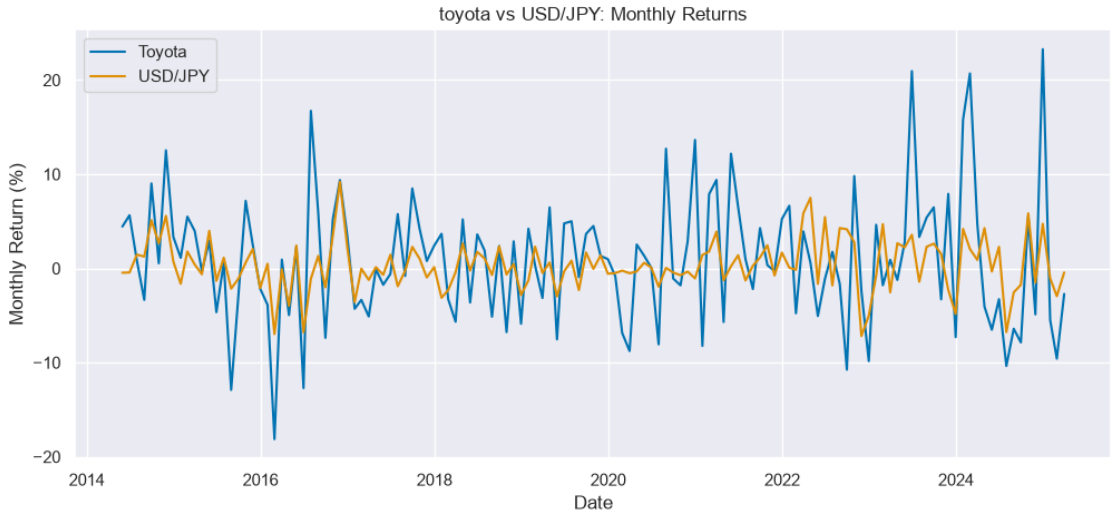

teijin vs USD/JPY: Monthly Returns
Teijin
USD/JPY
Monthly Return (%)
20
15
10
5
0
−5
−10
−15
2014
2016
2018
2020
2022
2024
Date

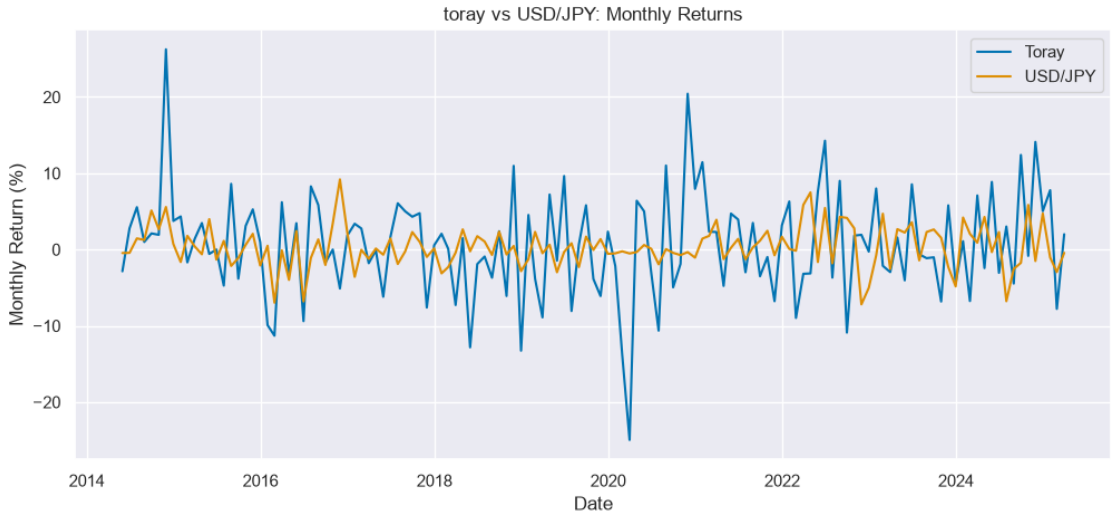

ajinomoto vs USD/JPY: Monthly Returns
Ajinomoto
USD/JPY
Monthly Return (%)
15
10
5
0
−5
−10
−15
2014
2016
2018
2020
2022
2024
Date

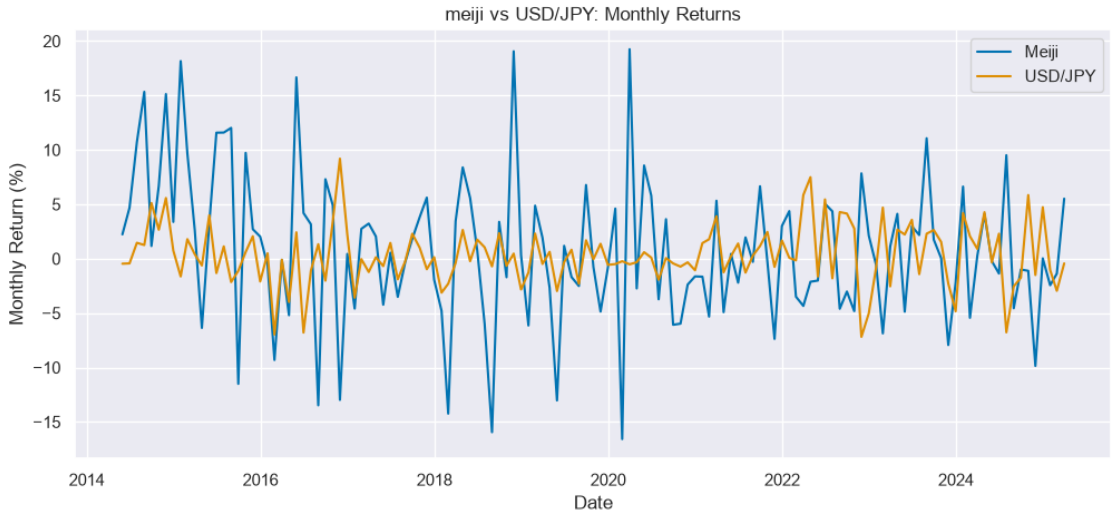

kajima vs USD/JPY: Monthly Returns
Kajima
USD/JPY
Monthly Return (%)
25
20
15
10
5
0
−5
−10
−15
2014
2016
2018
2020
2022
2024
Date

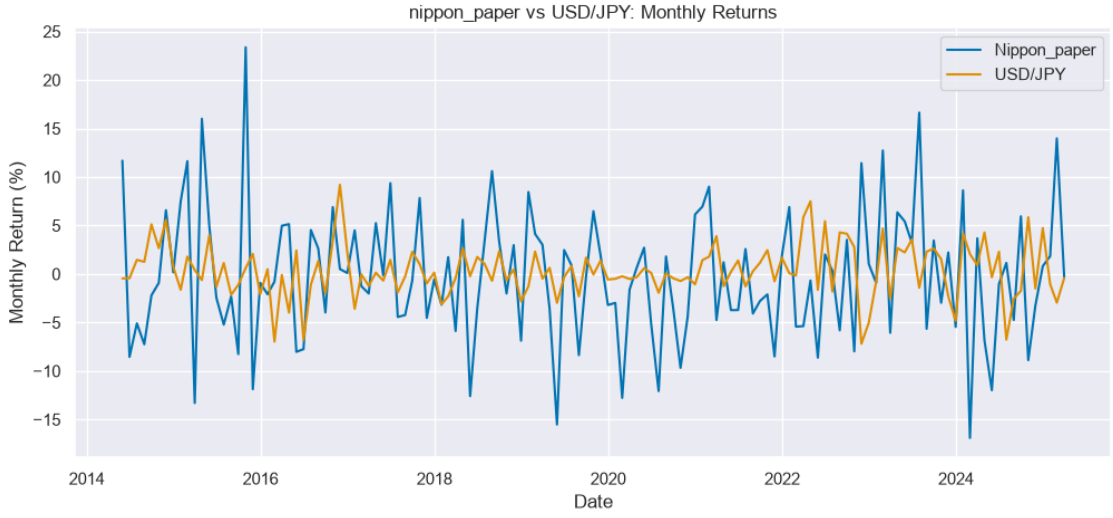

okuma vs USD/JPY: Monthly Returns
Okuma
USD/JPY
Monthly Return (%)
30
20
10
0
−10
−20
2014
2016
2018
2020
2022
2024
Date

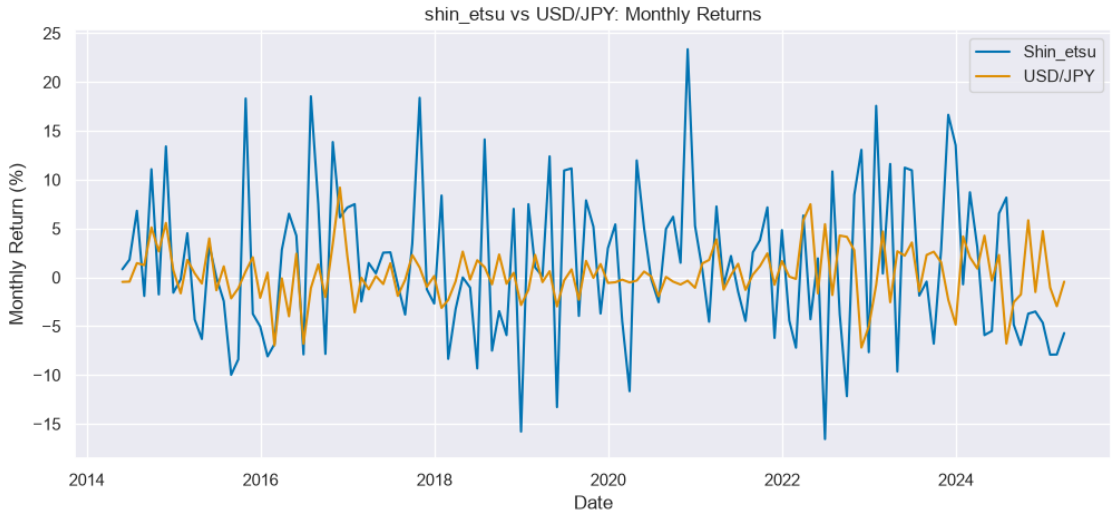

kewpie vs USD/JPY: Monthly Returns
Kewpie
USD/JPY
Monthly Return (%)
20
15
10
5
0
−5
−10
2014
2016
2018
2020
2022
2024
Date

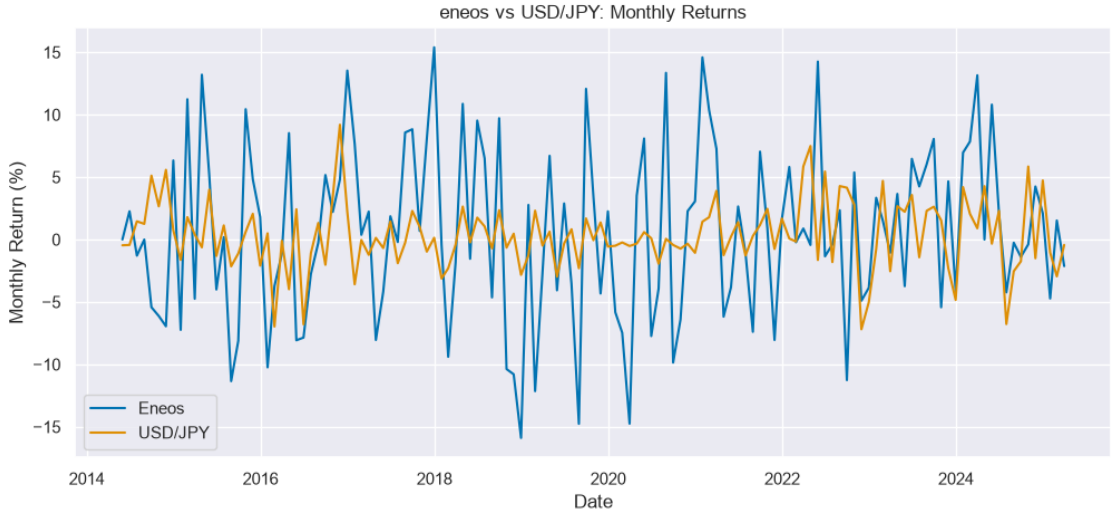

toyota vs USD/JPY: Monthly Returns
Toyota
USD/JPY
Monthly Return (%)
20
10
0
−10
−20
2014
2016
2018
2020
2022
2024
Date

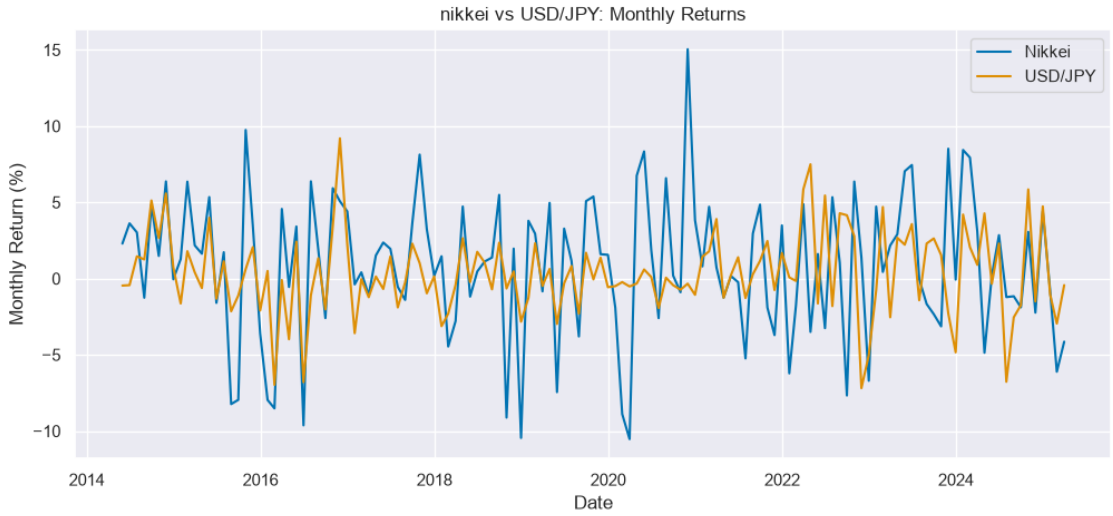

toray vs USD/JPY: Monthly Returns
Toray
USD/JPY
Monthly Return (%)
20
10
0
−10
−20
2014
2016
2018
2020
2022
2024
Date

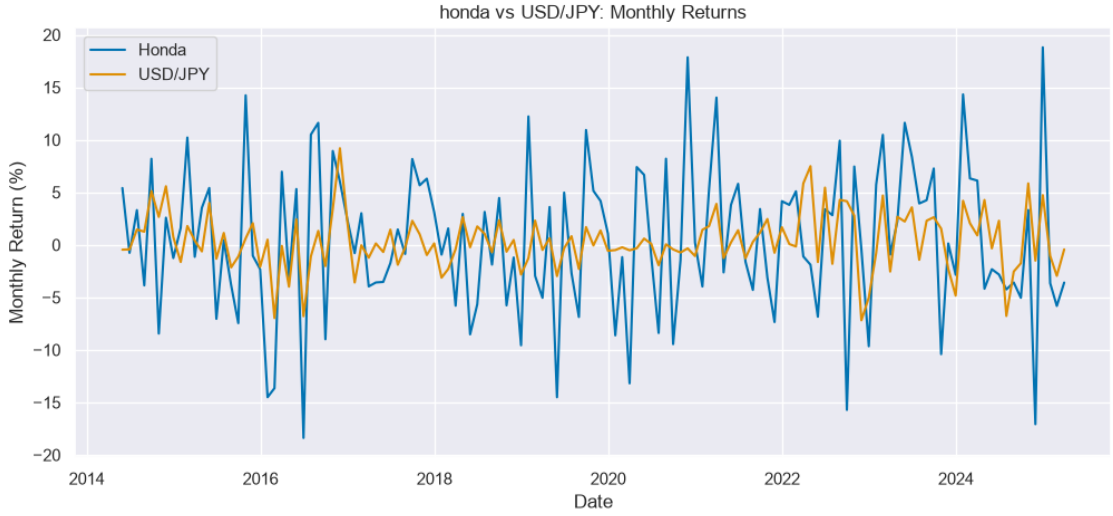

meiji vs USD/JPY: Monthly Returns
Meiji
USD/JPY
Monthly Return (%)
Date
20
15
10
5
0
−5
−10
−15
2014
2016
2018
2020
2022
2024

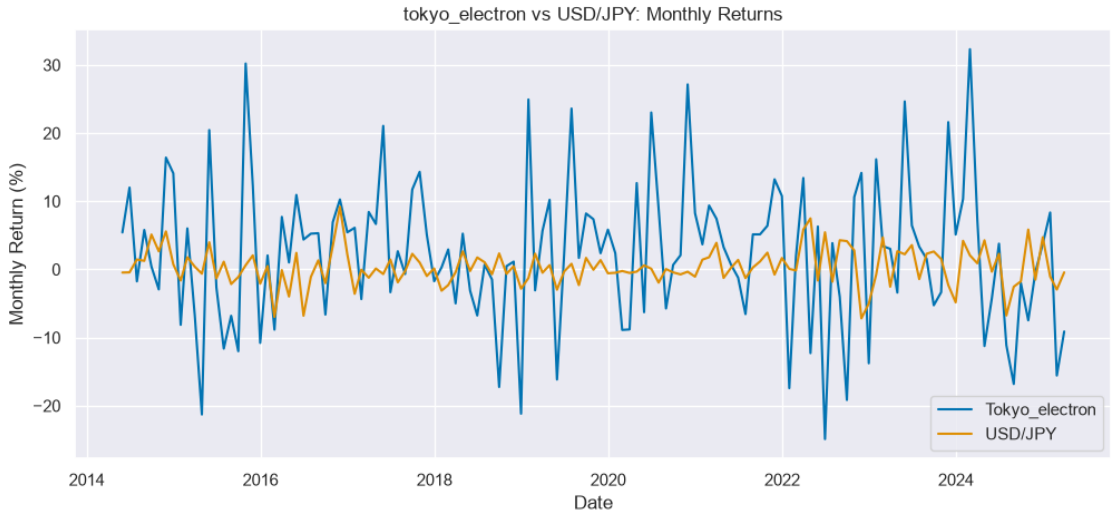

nippon_paper vs USD/JPY: Monthly Returns
Nippon_paper
USD/JPY
Monthly Return (%)
Date
25
20
15
10
5
0
−5
−10
−15
2014
2016
2018
2020
2022
2024

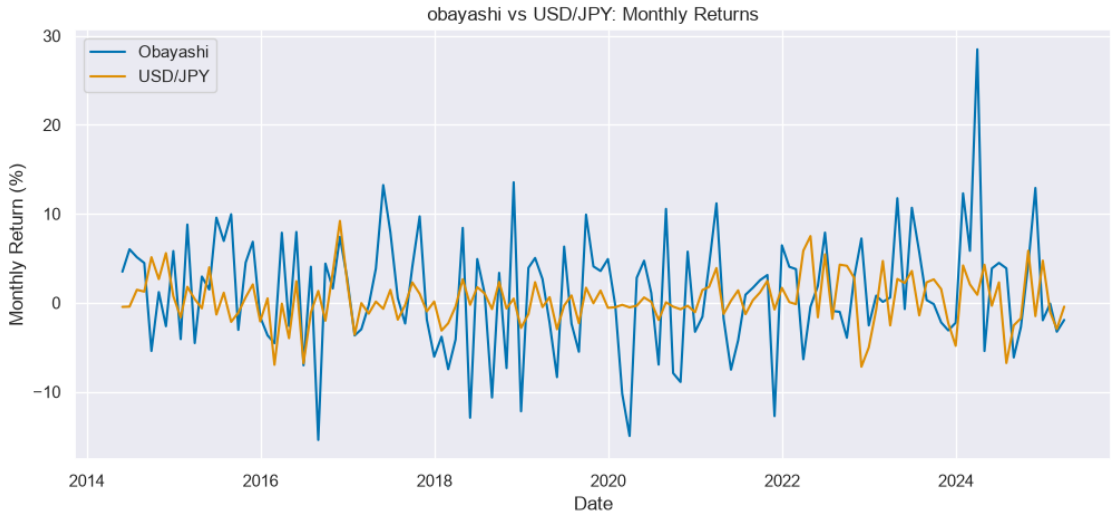

shin_etsu vs USD/JPY: Monthly Returns
Shin_etsu
USD/JPY
Monthly Return (%)
25
20
15
10
5
0
−5
−10
−15
2014
2016
2018
2020
2022
2024
Date


eneos vs USD/JPY: Monthly Returns
Eneos
USD/JPY
Monthly Return (%)
15
10
5
0
−5
−10
−15
2014
2016
2018
2020
2022
2024
Date


nikkei vs USD/JPY: Monthly Returns
Nikkei
USD/JPY
Monthly Return (%)
15
10
5
0
−5
−10
2014
2016
2018
2020
2022
2024
Date


honda vs USD/JPY: Monthly Returns
Honda
USD/JPY
Monthly Return (%)
20
15
10
5
0
−5
−10
−15
−20
2014
2016
2018
2020
2022
2024
Date





1.3.2 Looking at Cumulative Change Over Date vs Yen/USD

[8]:
```
for name in stocks:

    if name in "yen_usd":
        continue

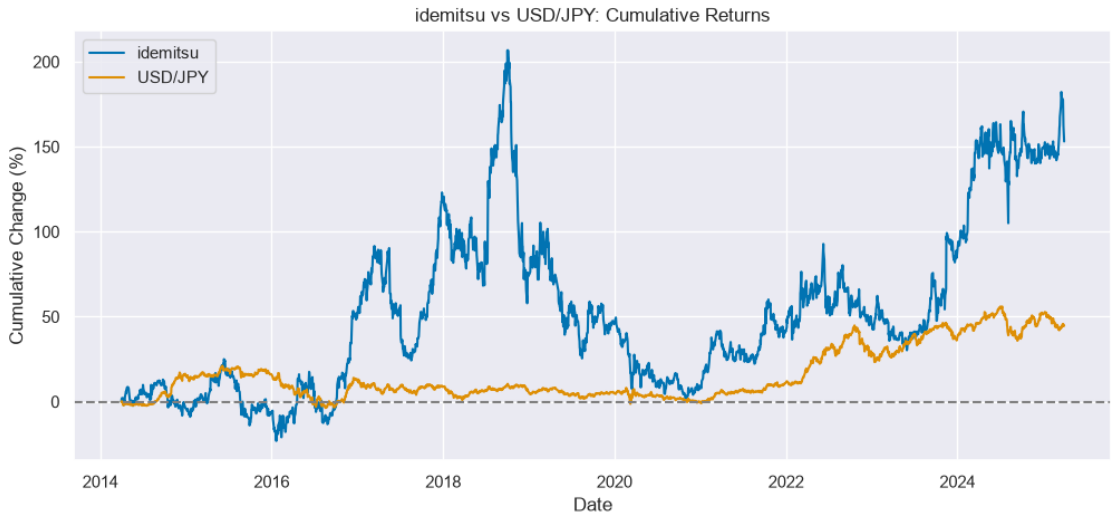
    plt.figure(figsize=(12, 5))

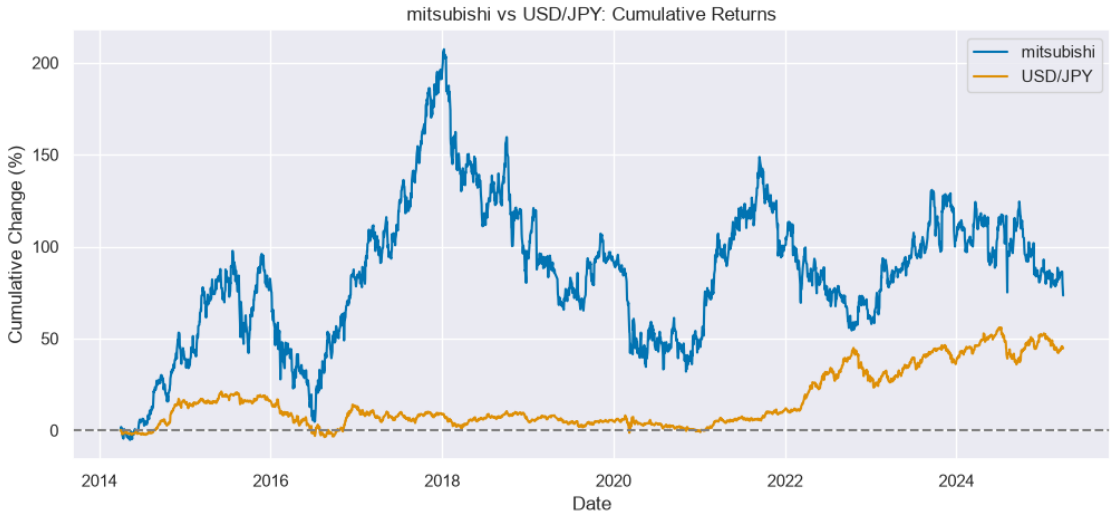
    sns.lineplot(
        data=stock_prices,
```

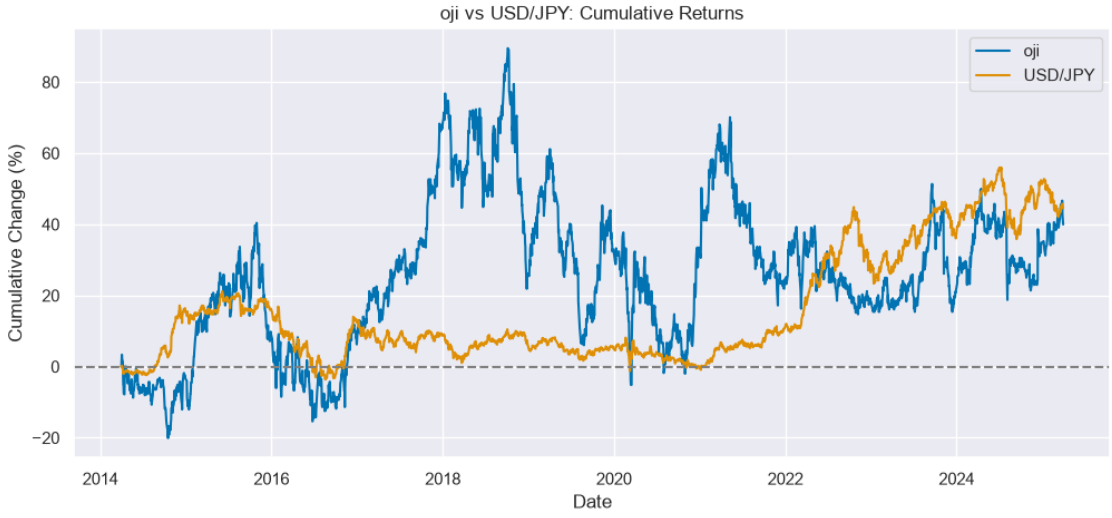
```
        x="Date",
        y=f"{name}_total_change",
        label=name
    )

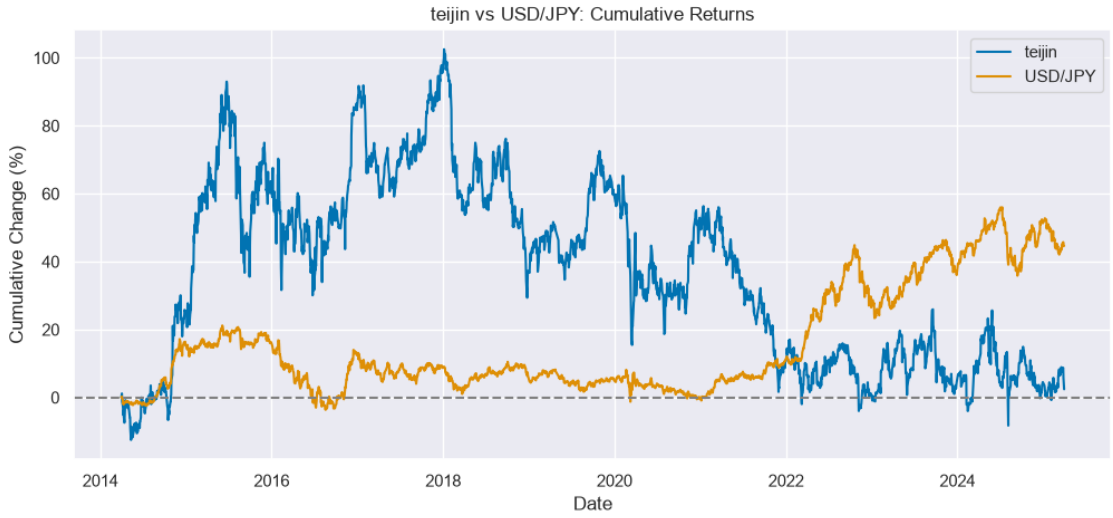
    sns.lineplot(
        data=stock_prices,
        x="Date",
        y="yen_usd_total_change",
        label="USD/JPY"
    )

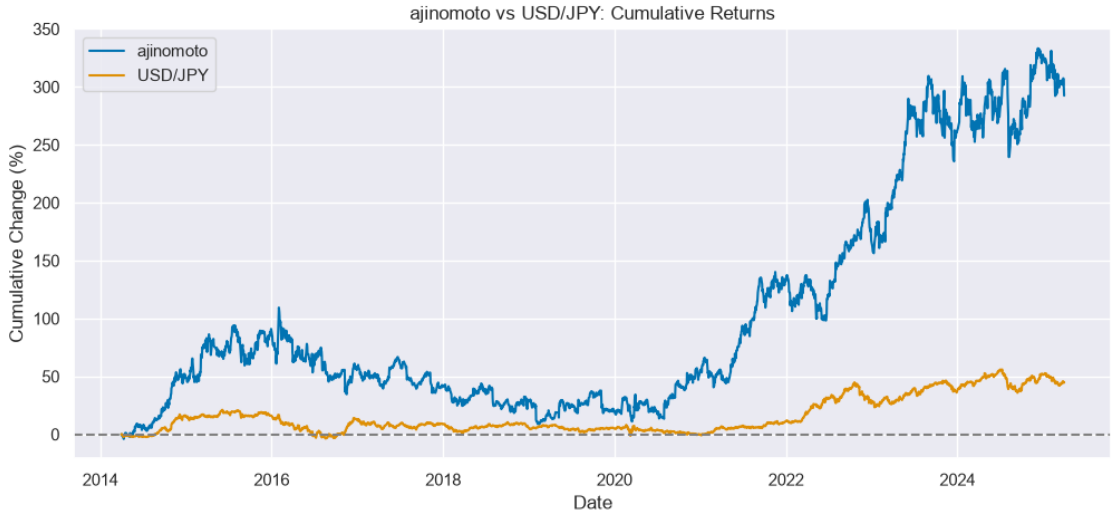
    plt.title(f"{name} vs USD/JPY: Cumulative Returns")
    plt.xlabel("Date")
    plt.ylabel("Cumulative Change (%)")
    plt.axhline(0, color="gray", linestyle="--")
    plt.legend()
    plt.show()
```

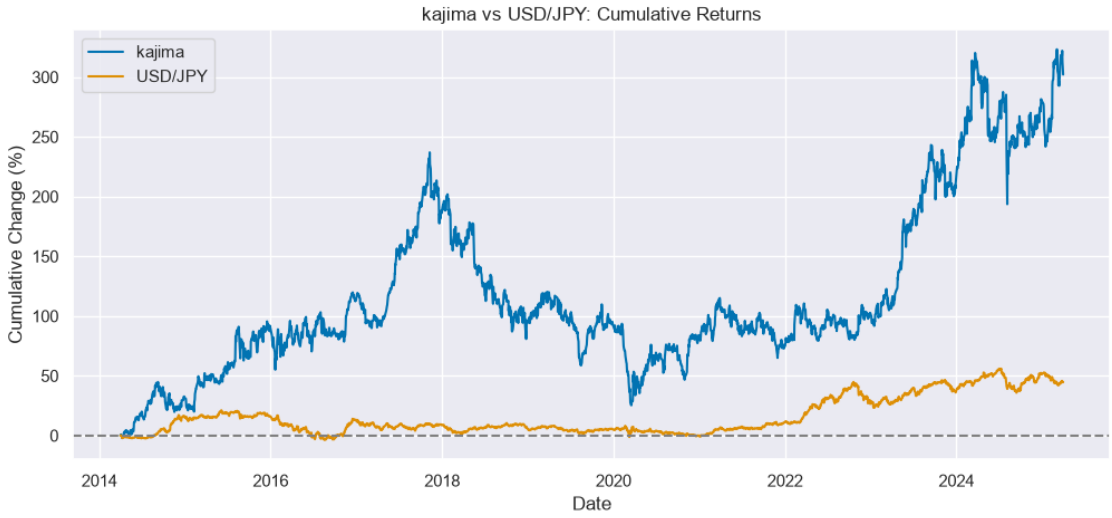


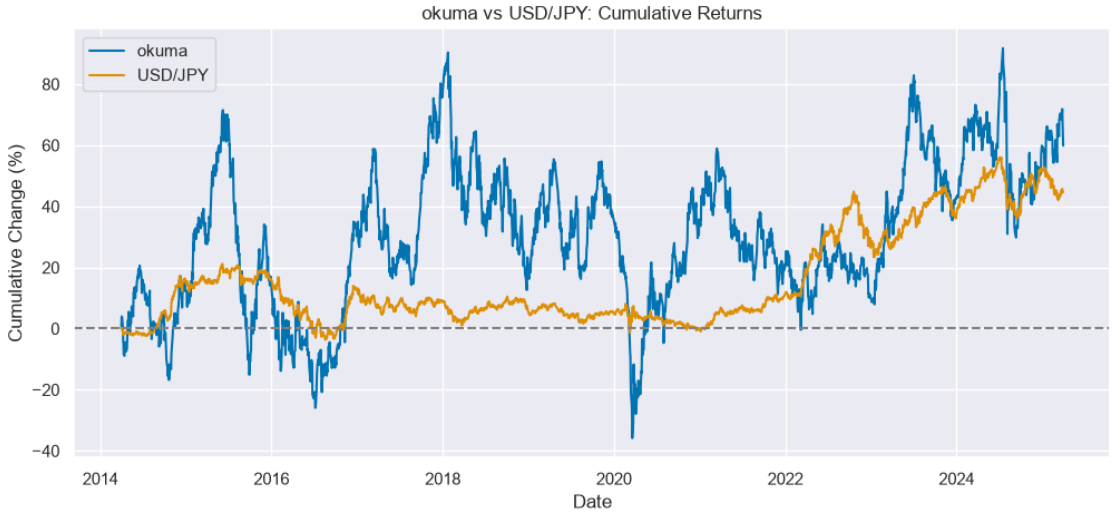

mitsubishi vs USD/JPY: Cumulative Returns
mitsubishi
USD/JPY
Cumulative Change (%)
200
150
100
50
0
2014
2016
2018
2020
2022
2024
Date

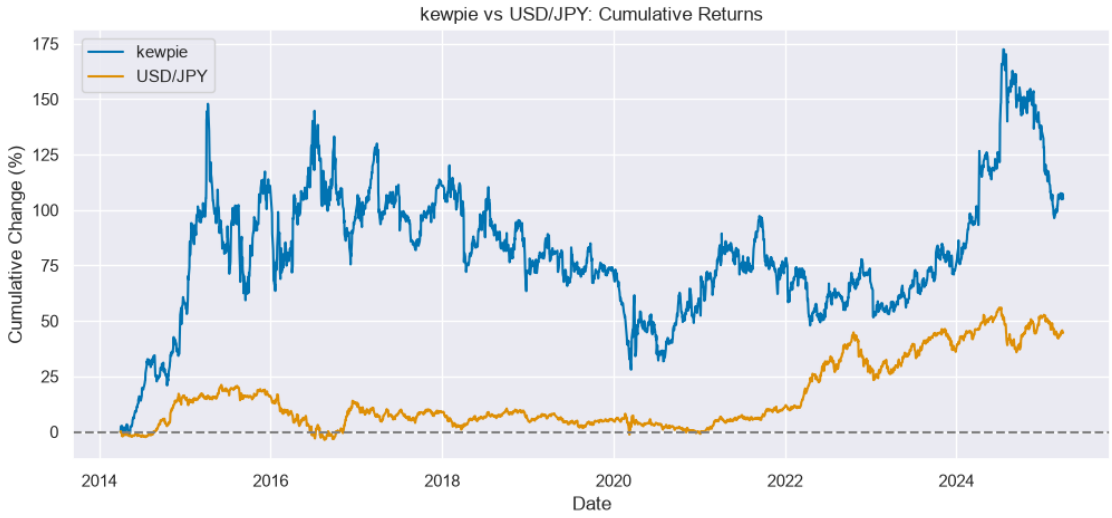

oji vs USD/JPY: Cumulative Returns
oji
USD/JPY
Cumulative Change (%)
80
60
40
20
0
−20
2014
2016
2018
2020
2022
2024
Date

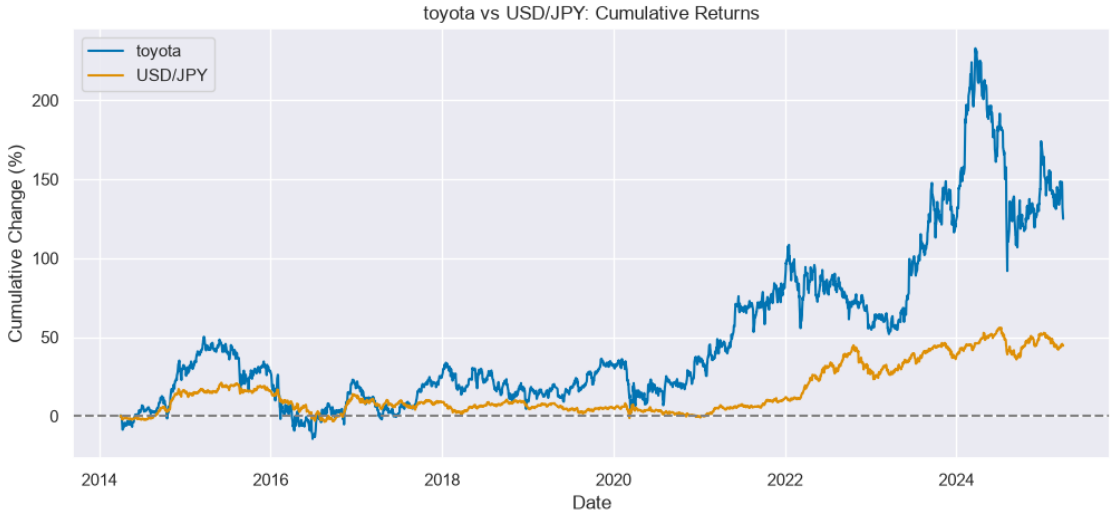

teijin vs USD/JPY: Cumulative Returns
teijin
USD/JPY
Cumulative Change (%)
100
80
60
40
20
0
2014
2016
2018
2020
2022
2024
Date

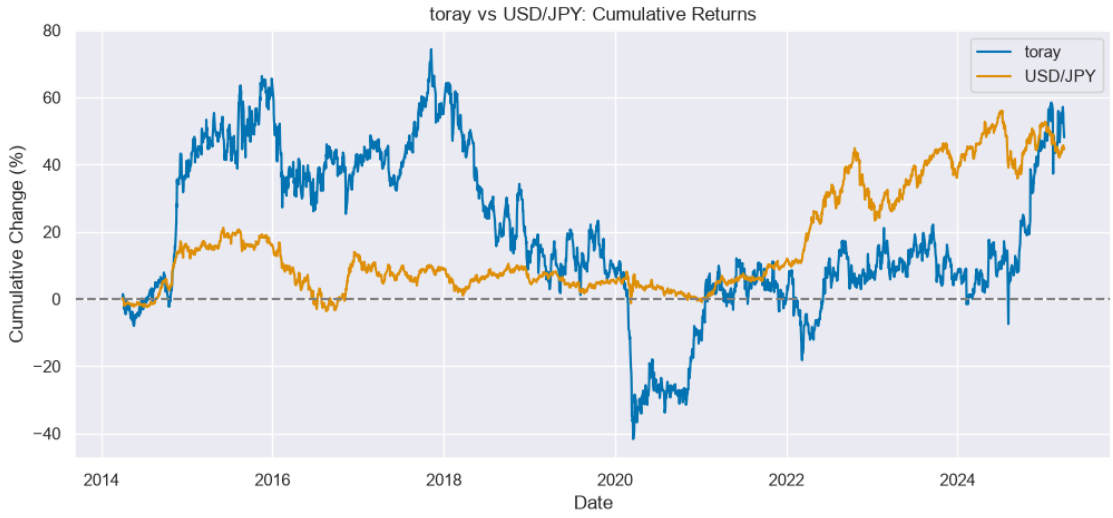

ajinomoto vs USD/JPY: Cumulative Returns
ajinomoto
USD/JPY
Cumulative Change (%)
350
300
250
200
150
100
50
0
2014
2016
2018
2020
2022
2024
Date

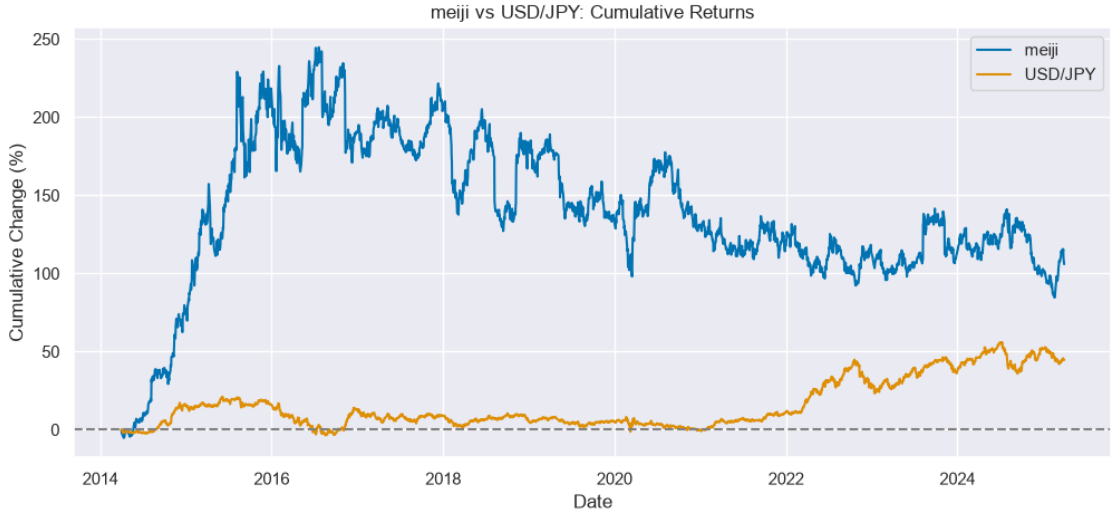

kajima vs USD/JPY: Cumulative Returns
kajima
USD/JPY
Cumulative Change (%)
300
250
200
150
100
50
0
2014
2016
2018
2020
2022
2024
Date

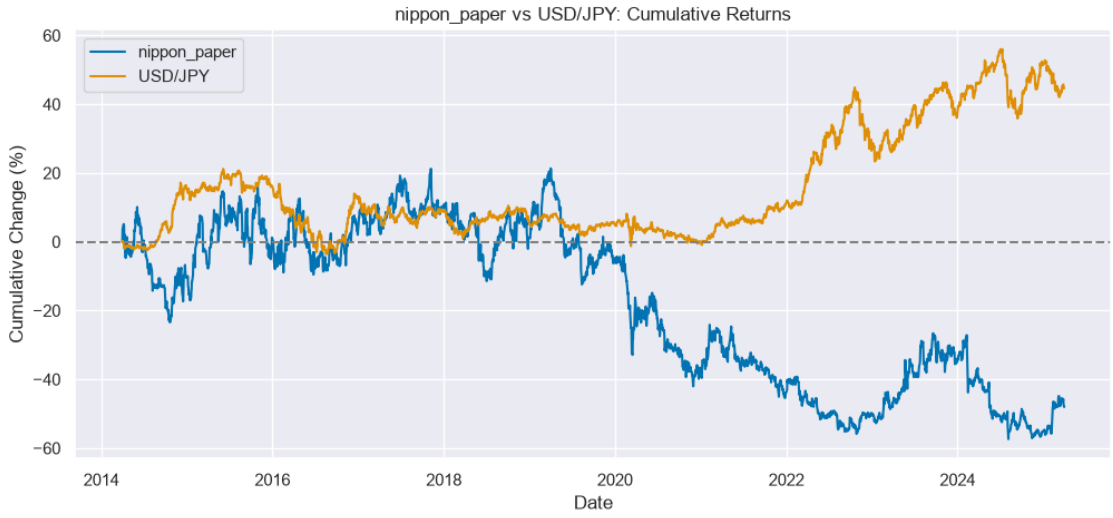

okuma vs USD/JPY: Cumulative Returns
okuma
USD/JPY
Cumulative Change (%)
80
60
40
20
0
−20
−40
2014
2016
2018
2020
2022
2024
Date

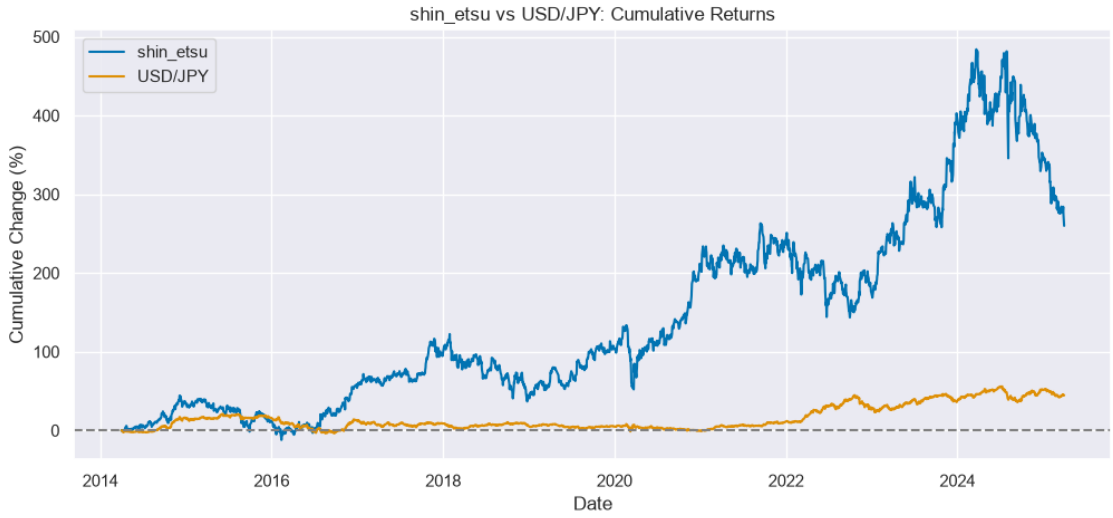

kewpie vs USD/JPY: Cumulative Returns
kewpie
USD/JPY
Cumulative Change (%)
175
150
125
100
75
50
25
0
2014
2016
2018
2020
2022
2024
Date

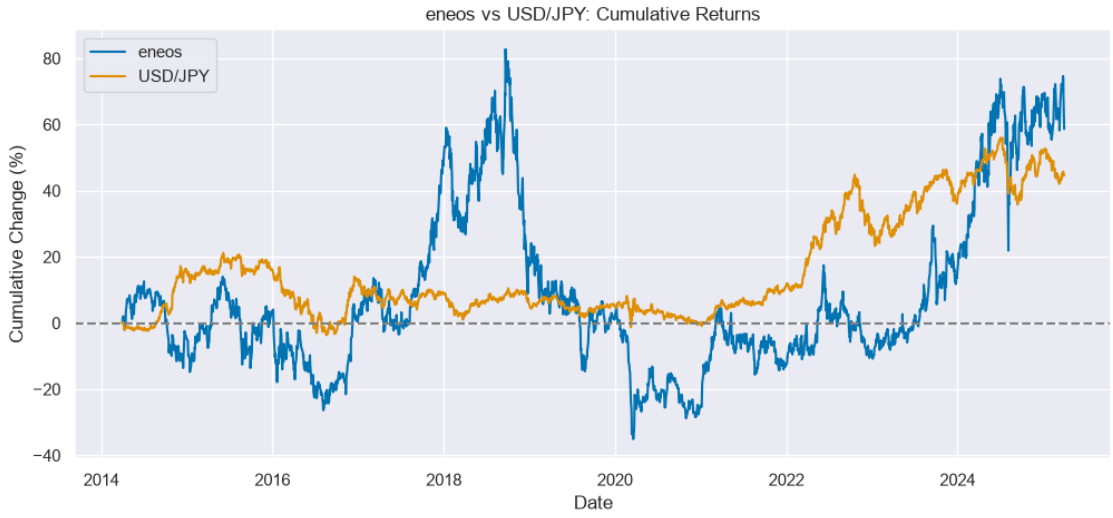

toyota vs USD/JPY: Cumulative Returns
toyota
USD/JPY
Cumulative Change (%)
200
150
100
50
0
2014
2016
2018
2020
2022
2024
Date

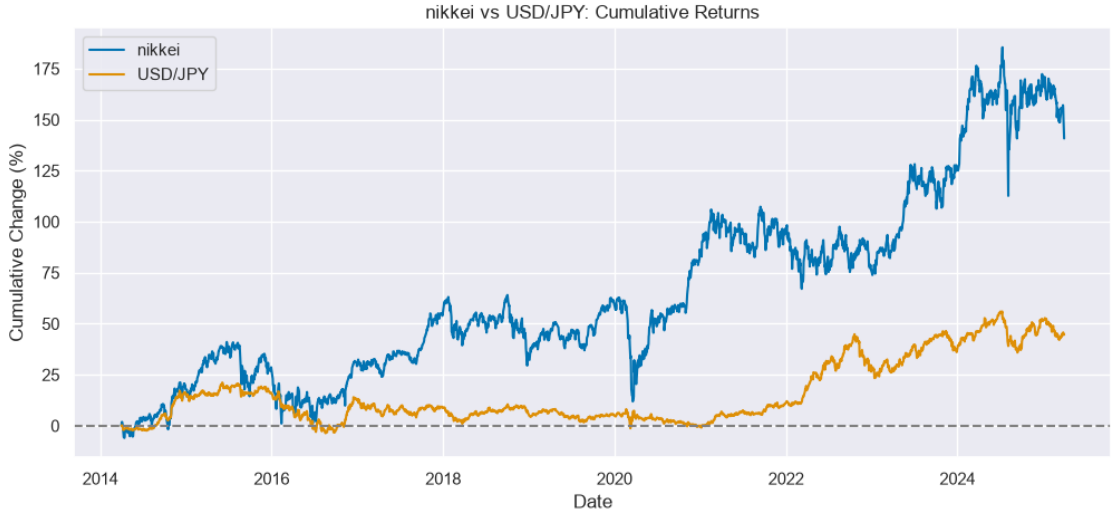

toray vs USD/JPY: Cumulative Returns
toray
USD/JPY
Cumulative Change (%)
80
60
40
20
0
−20
−40
2014
2016
2018
2020
2022
2024
Date

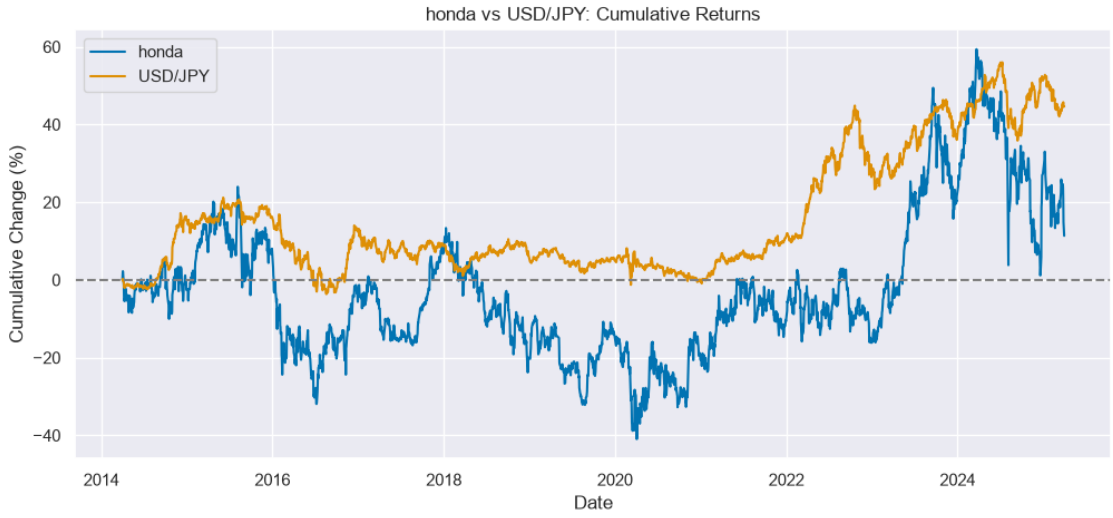

meiji vs USD/JPY: Cumulative Returns
meiji
USD/JPY
Cumulative Change (%)
250
200
150
100
50
0
2014
2016
2018
2020
2022
2024
Date

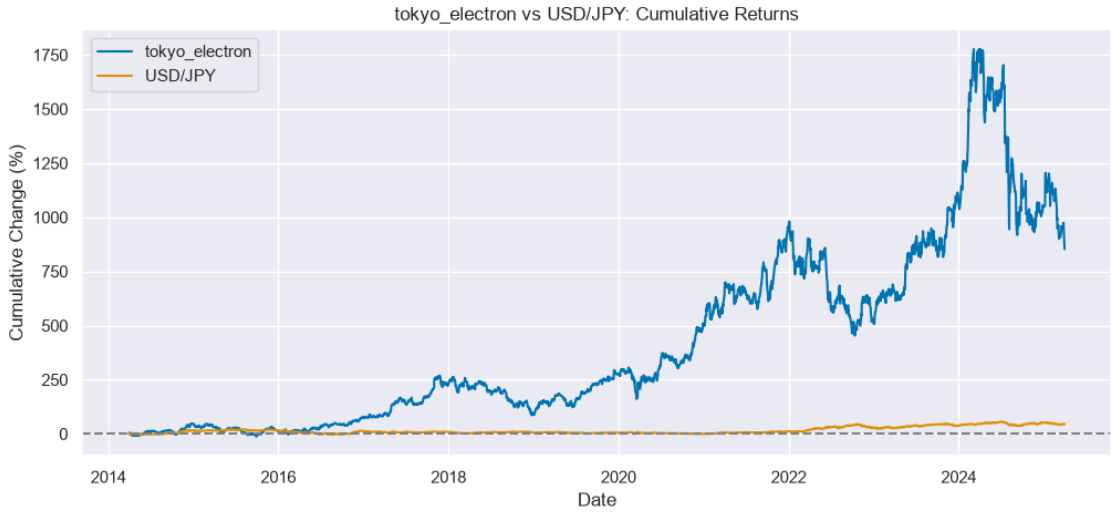

nippon_paper vs USD/JPY: Cumulative Returns
nippon_paper
USD/JPY
Cumulative Change (%)
60
40
20
0
−20
−40
−60
2014
2016
2018
2020
2022
2024
Date

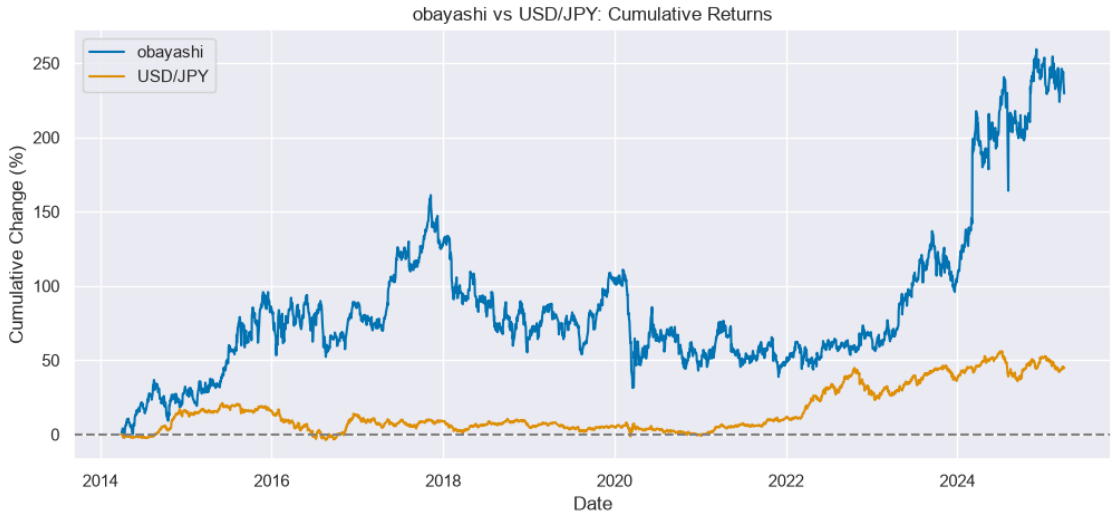

shin_etsu vs USD/JPY: Cumulative Returns
shin_etsu
USD/JPY
Cumulative Change (%)
500
400
300
200
100
0
2014
2016
2018
2020
2022
2024
Date


eneos vs USD/JPY: Cumulative Returns
eneos
USD/JPY
Cumulative Change (%)
80
60
40
20
0
−20
−40
2014
2016
2018
2020
2022
2024
Date


nikkei vs USD/JPY: Cumulative Returns
nikkei
USD/JPY
Cumulative Change (%)
175
150
125
100
75
50
25
0
2014
2016
2018
2020
2022
2024
Date


honda vs USD/JPY: Cumulative Returns
honda
USD/JPY
Cumulative Change (%)
60
40
20
0
−20
−40
2014
2016
2018
2020
2022
2024
Date


tokyo_electron vs USD/JPY: Cumulative Returns
tokyo_electron
USD/JPY
Cumulative Change (%)
1750
1500
1250
1000
750
500
250
0
2014
2016
2018
2020
2022
2024
Date



1.3.3 Graph Takeaways

Looking at all three graphs so far, it appears that some of the stocks follow the Yen/USD trends or follow the inverse of their trends For example, Oji (Paper) month returns seem to be following closely the Yen/USD movements at different magnitudes. Same with Mitsubishi (Chemical), and Teijin (Textiles). The Nikkei index also appears to have movements resembling many of the fluxuations of the USD/Yen exchange rate. Looking at the food industries (Kewpie, Meiji, Ajinomoto) many of their movements appear to be taking the inverse of the movements of Yen/Usd. This may be occuring because many of their inputs are sourced from foreign imports then mostly sold within Japan.

If we look at the cumuluative change graphs, ignoring drops during COVID 19, we again see some graphs following the trends or the inverse. For example Teijin stocks crashed in 2022 when USD prices surged, as well as Oji and Meiji. Remarkably Kewpie follows the upward trends with USD in 2022 and movements look quite similar past that point.

(Important note! ajinomoto is primarily a food company, but also produces semi conductor material. Potentially a reason why their stock change blows up so much around 2022)

1.3.4 Correlation Analysis

Stocks daily return % compared to USD/Yen Daily Change %

[9]:
```
for name in stocks:

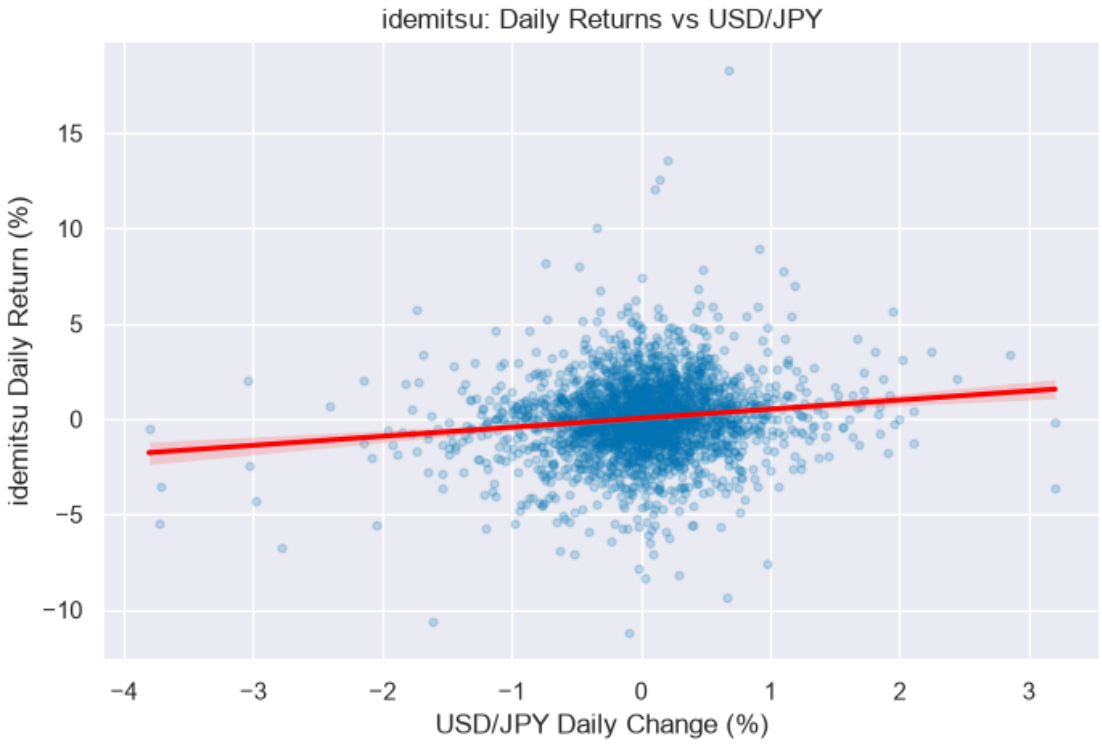
    if name == "yen_usd":
        continue

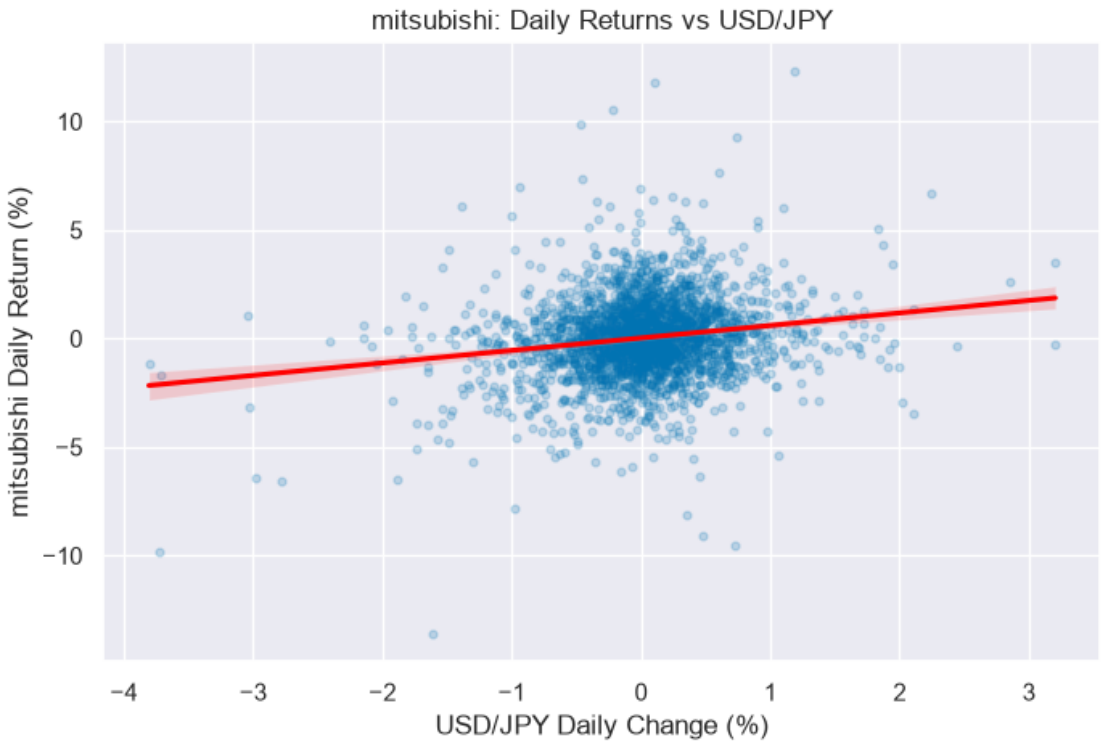
    plt.figure(figsize=(8, 5))

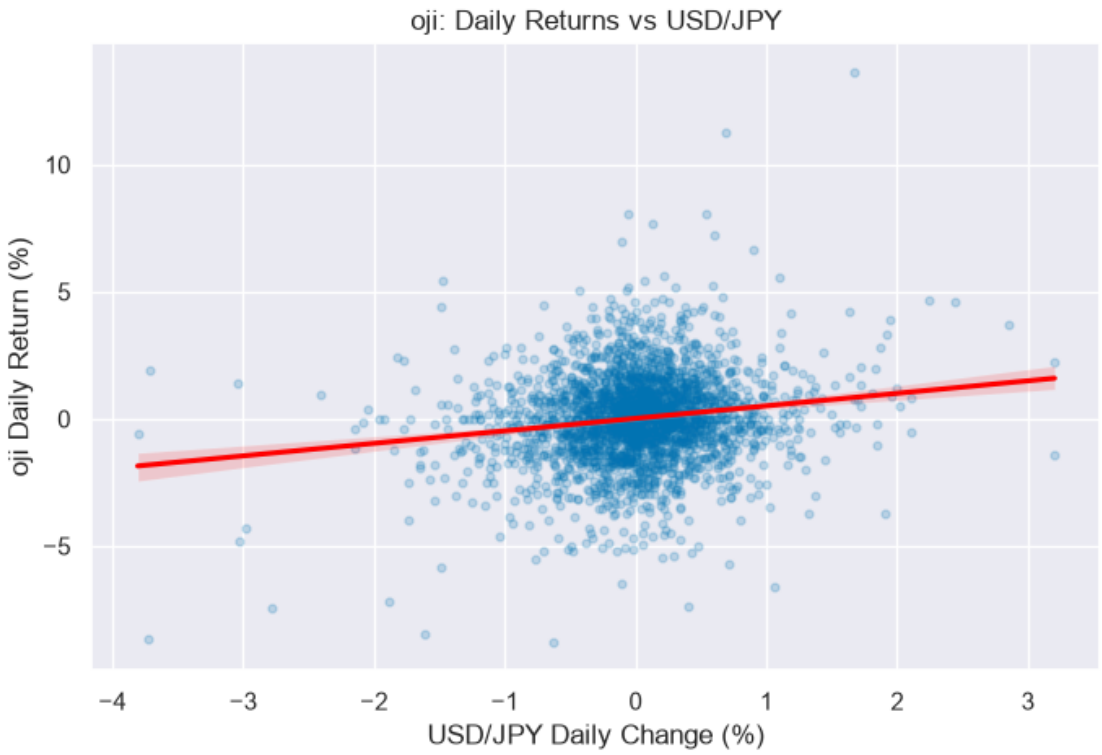
    sns.regplot(
        data = stock_prices,
```

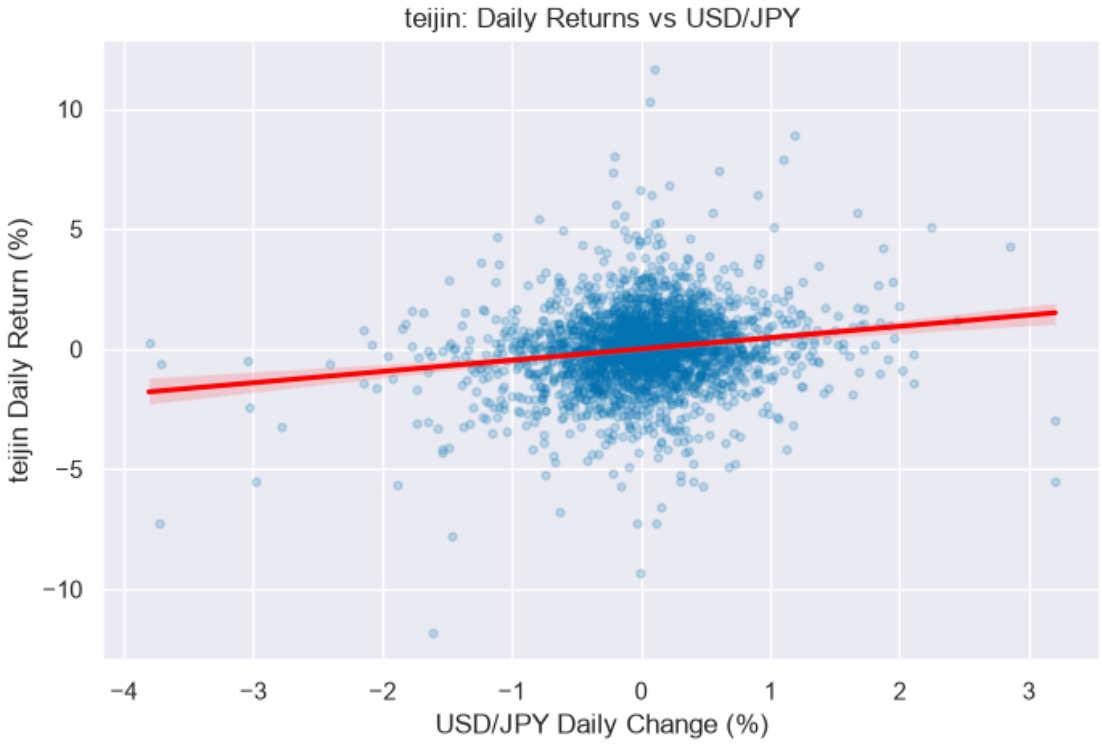
```
        x = "yen_usd_day_change",
        y = f"{name}_day_change",
        scatter_kws = {"alpha": 0.2, "s": 12},
        line_kws = {"color": "red"}
    )

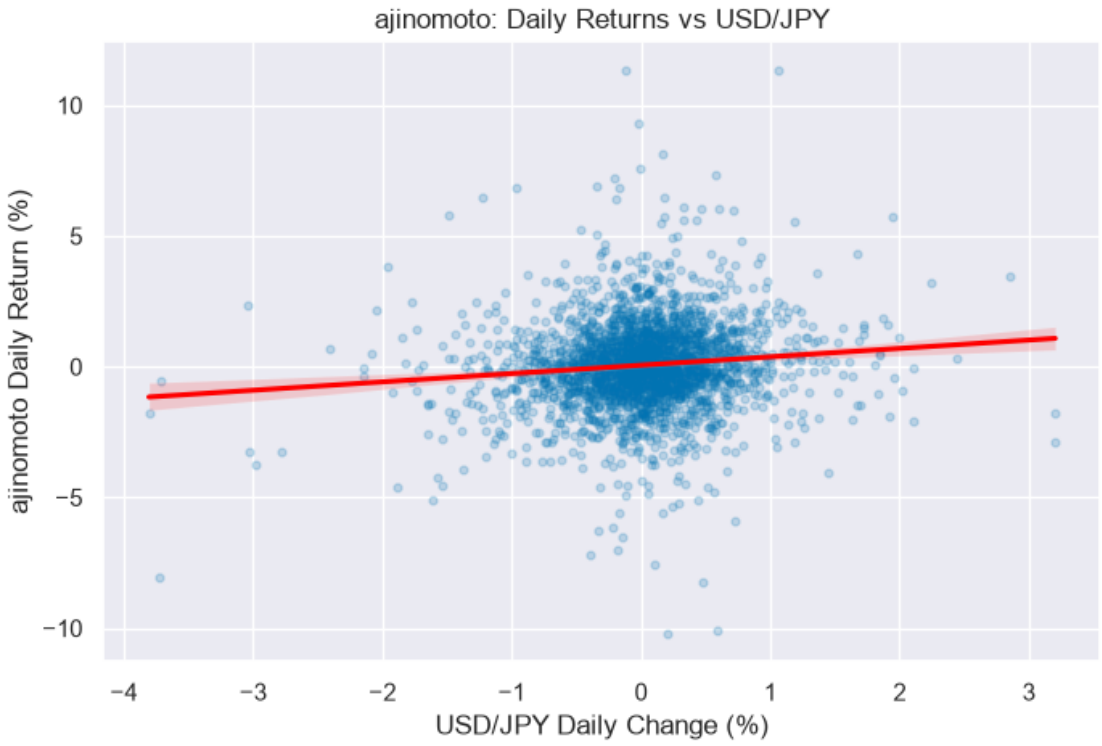
    plt.title(f"{name}: Daily Returns vs USD/JPY")
    plt.xlabel("USD/JPY Daily Change (%)")
    plt.ylabel(f"{name} Daily Return (%)")
    plt.show()
```

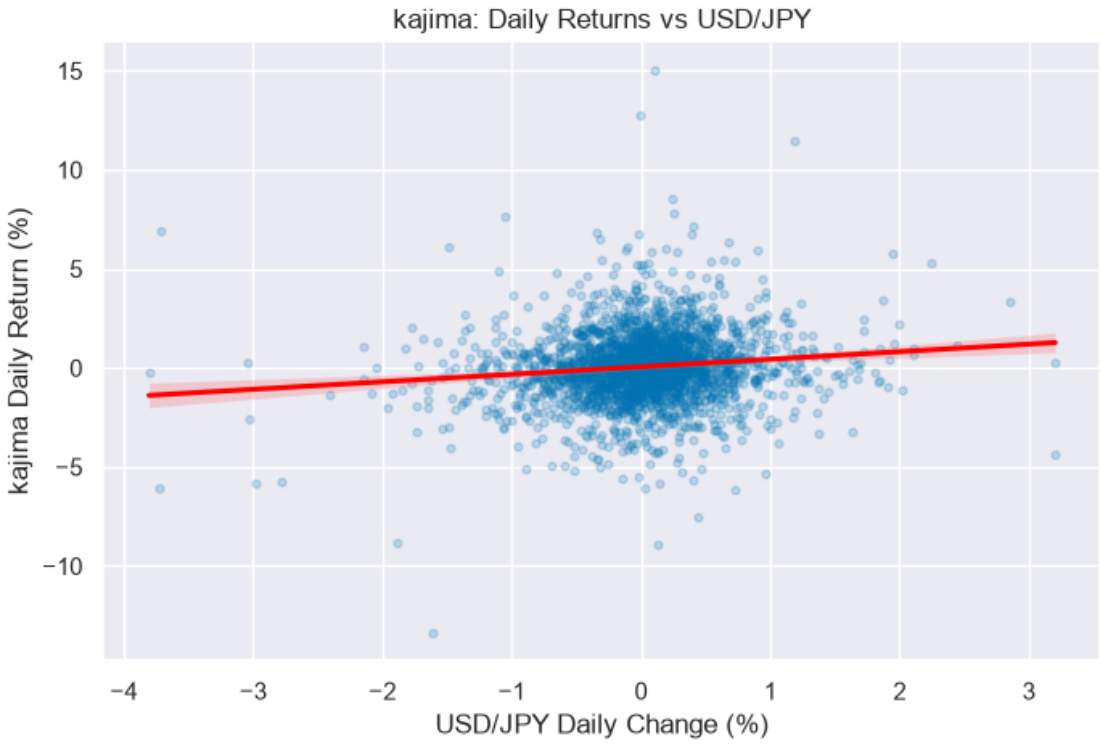


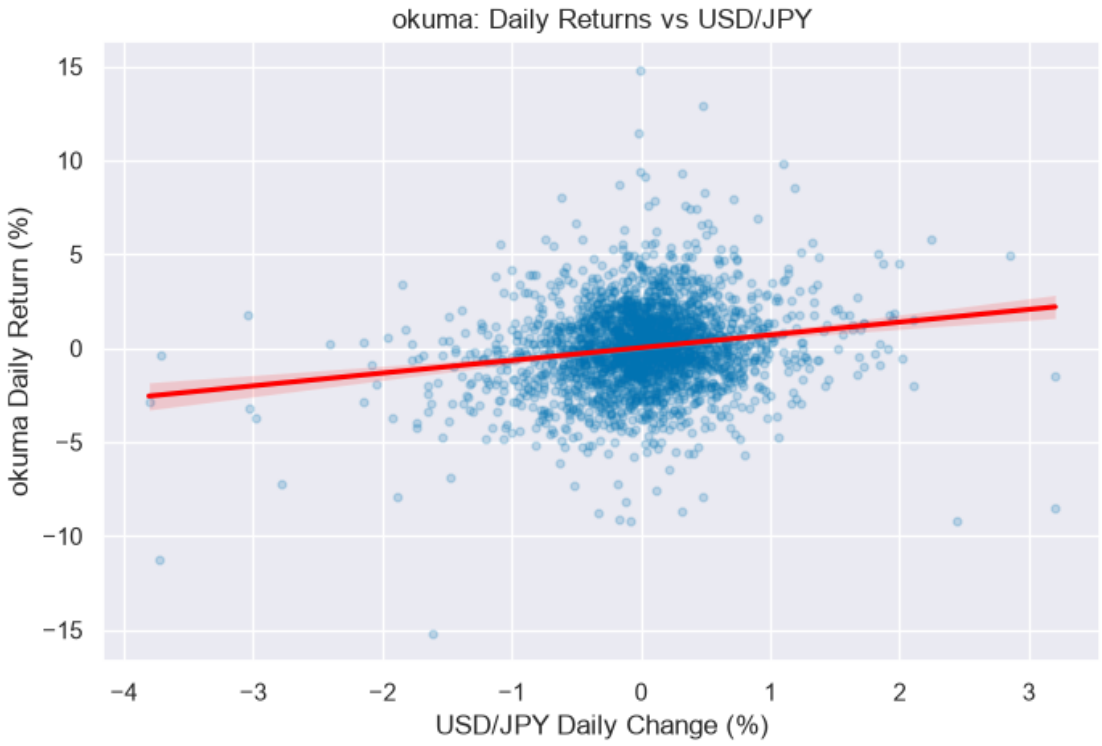

mitsubishi: Daily Returns vs USD/JPY
mitsubishi Daily Return (%)
10
5
0
−5
−10
−4
−3
−2
−1
0
1
2
3
USD/JPY Daily Change (%)

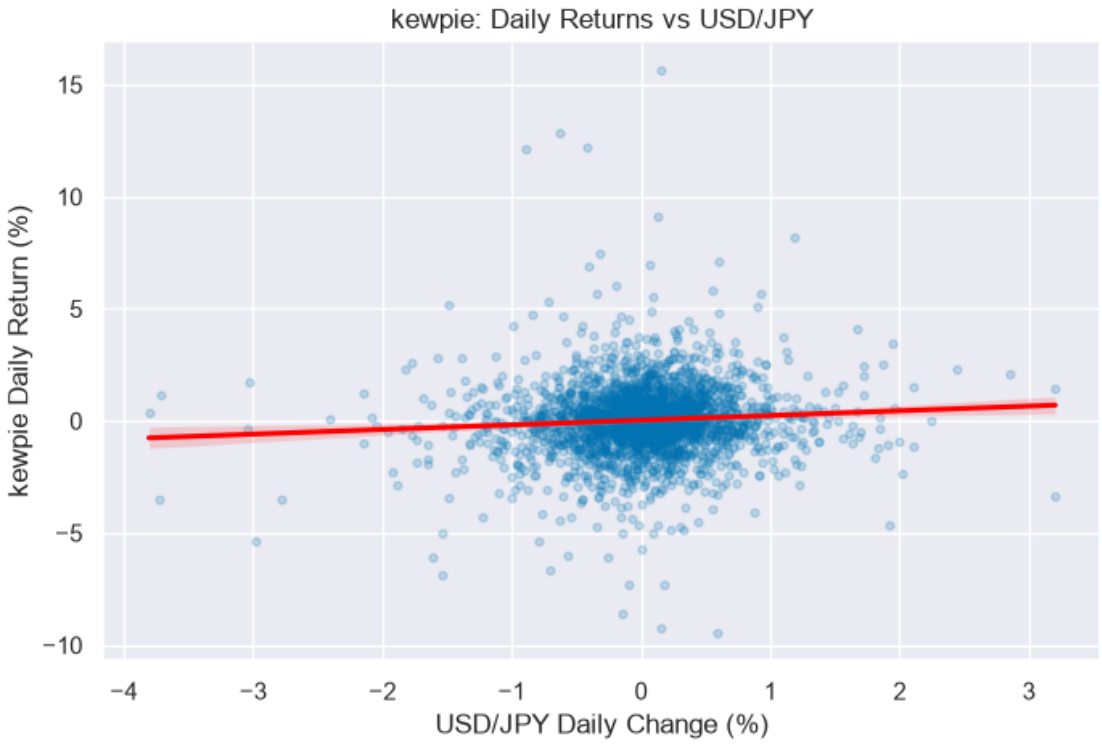

oji: Daily Returns vs USD/JPY
oji Daily Return (%)
10
5
0
−5
−4
−3
−2
−1
0
1
2
3
USD/JPY Daily Change (%)

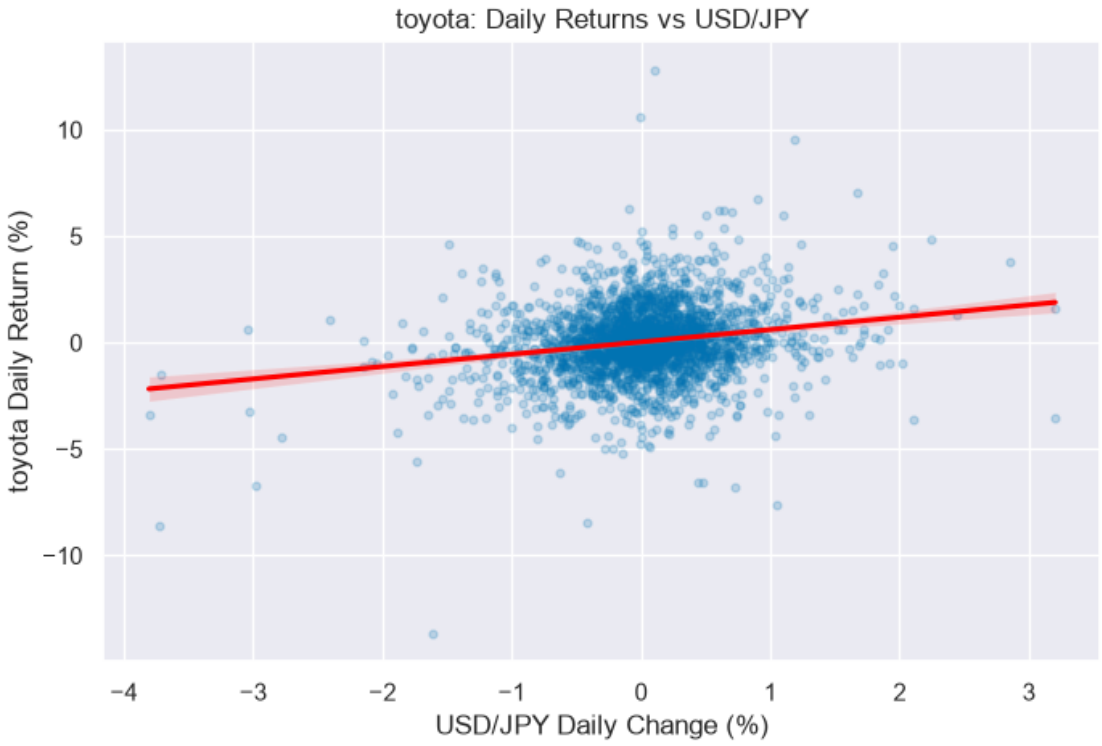

teijin: Daily Returns vs USD/JPY
teijin Daily Return (%)
10
5
0
−5
−10
−4
−3
−2
−1
0
1
2
3
USD/JPY Daily Change (%)

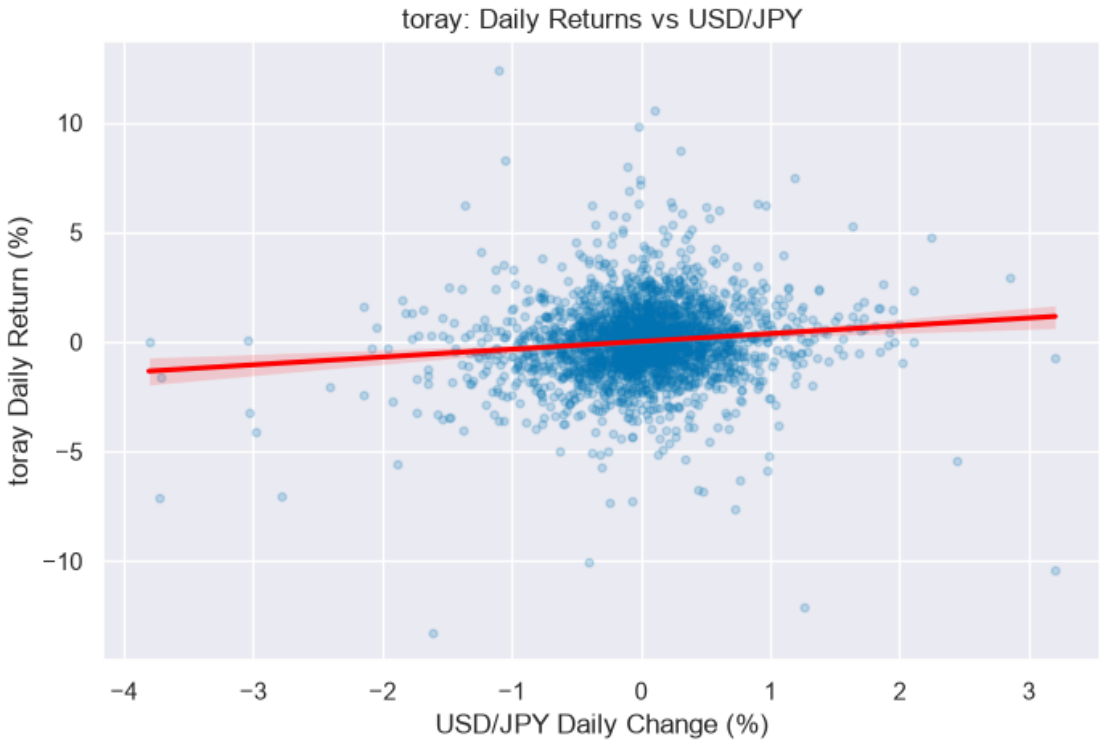

ajinomoto: Daily Returns vs USD/JPY
ajinomoto Daily Return (%)
10
5
0
−5
−10
−4
−3
−2
−1
0
1
2
3
USD/JPY Daily Change (%)

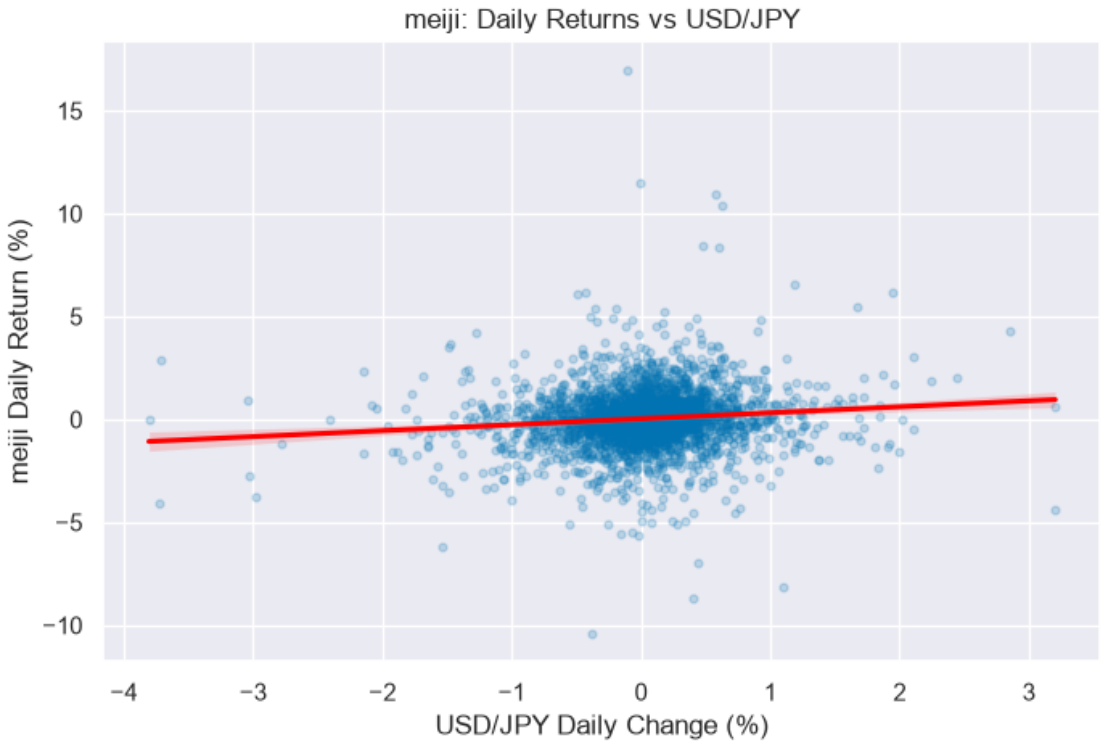

kajima: Daily Returns vs USD/JPY
kajima Daily Return (%)
15
10
5
0
−5
−10
−4
−3
−2
−1
0
1
2
3
USD/JPY Daily Change (%)

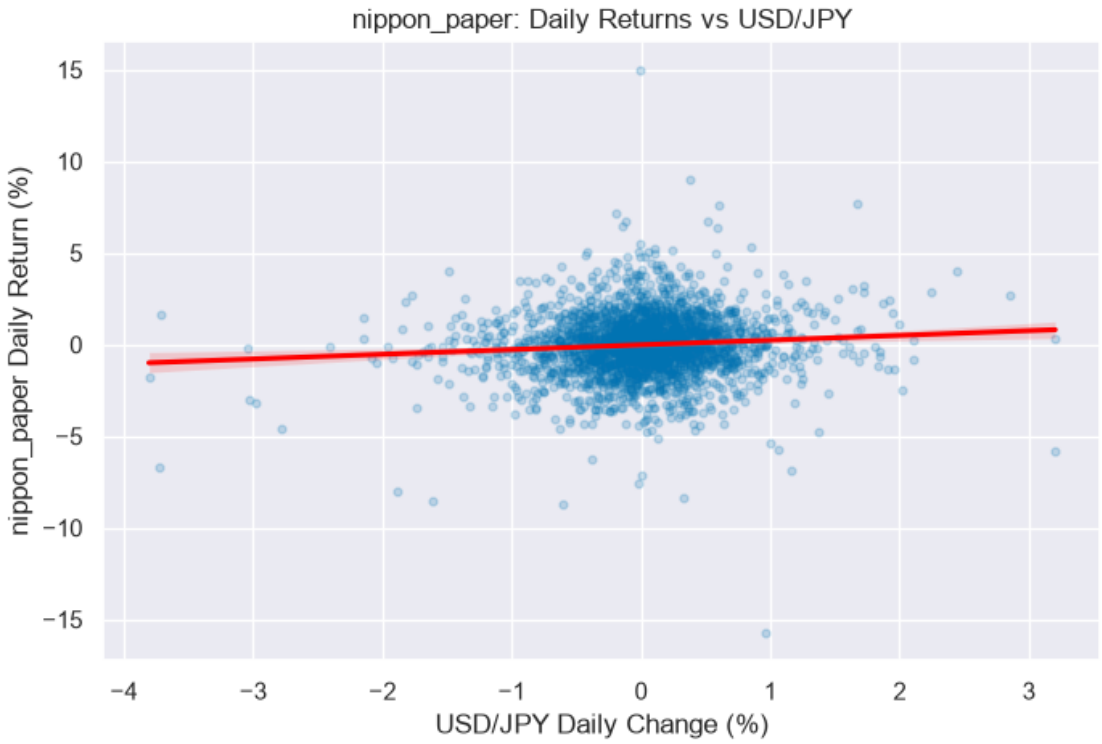

okuma: Daily Returns vs USD/JPY
okuma Daily Return (%)
15
10
5
0
−5
−10
−15
−4
−3
−2
−1
0
1
2
3
USD/JPY Daily Change (%)

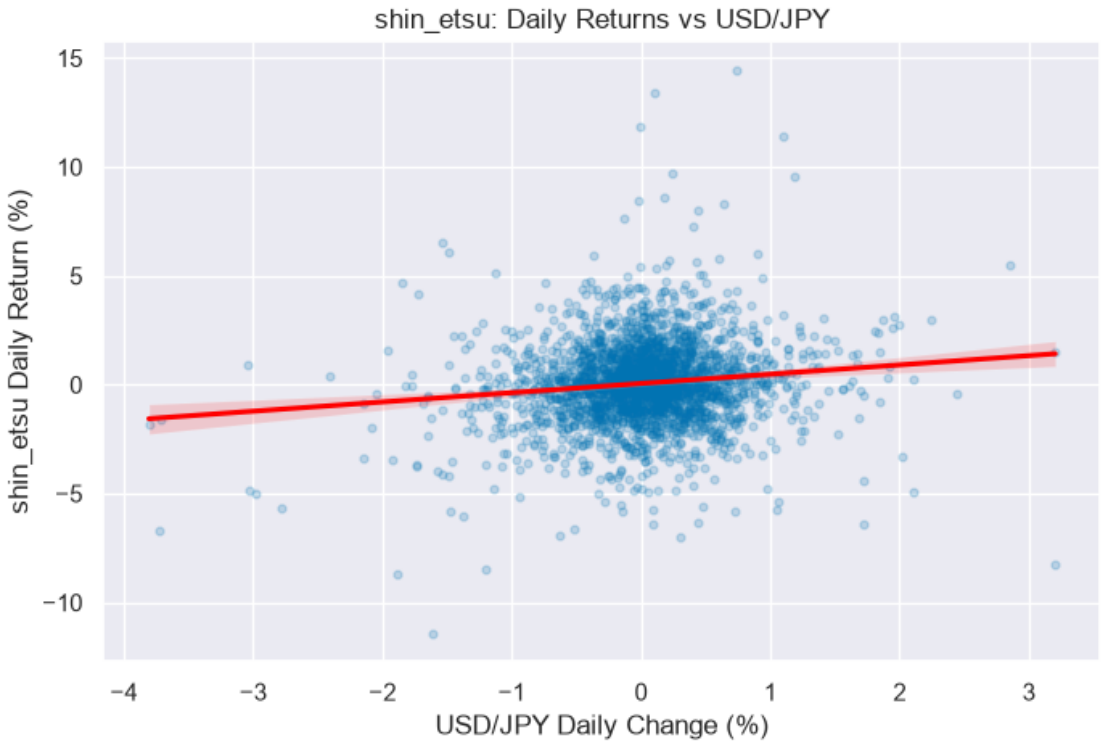

kewpie: Daily Returns vs USD/JPY
kewpie Daily Return (%)
15
10
5
0
−5
−10
−4
−3
−2
−1
0
1
2
3
USD/JPY Daily Change (%)

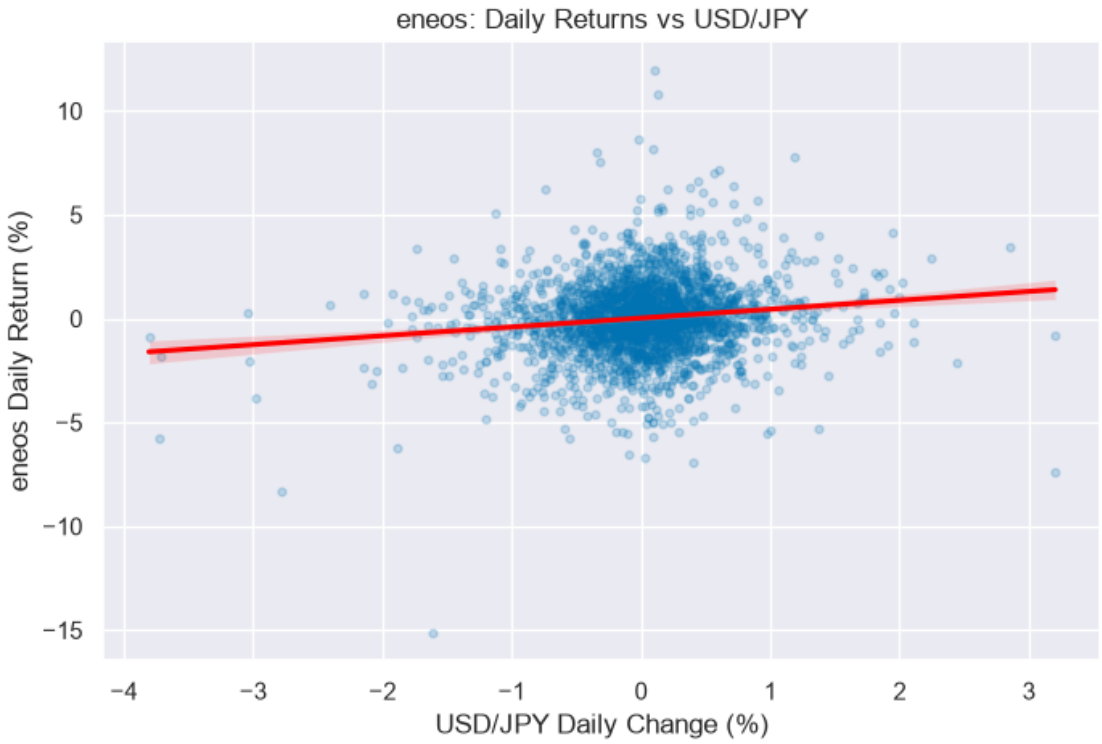

toyota: Daily Returns vs USD/JPY
toyota Daily Return (%)
10
5
0
−5
−10
−4
−3
−2
−1
0
1
2
3
USD/JPY Daily Change (%)

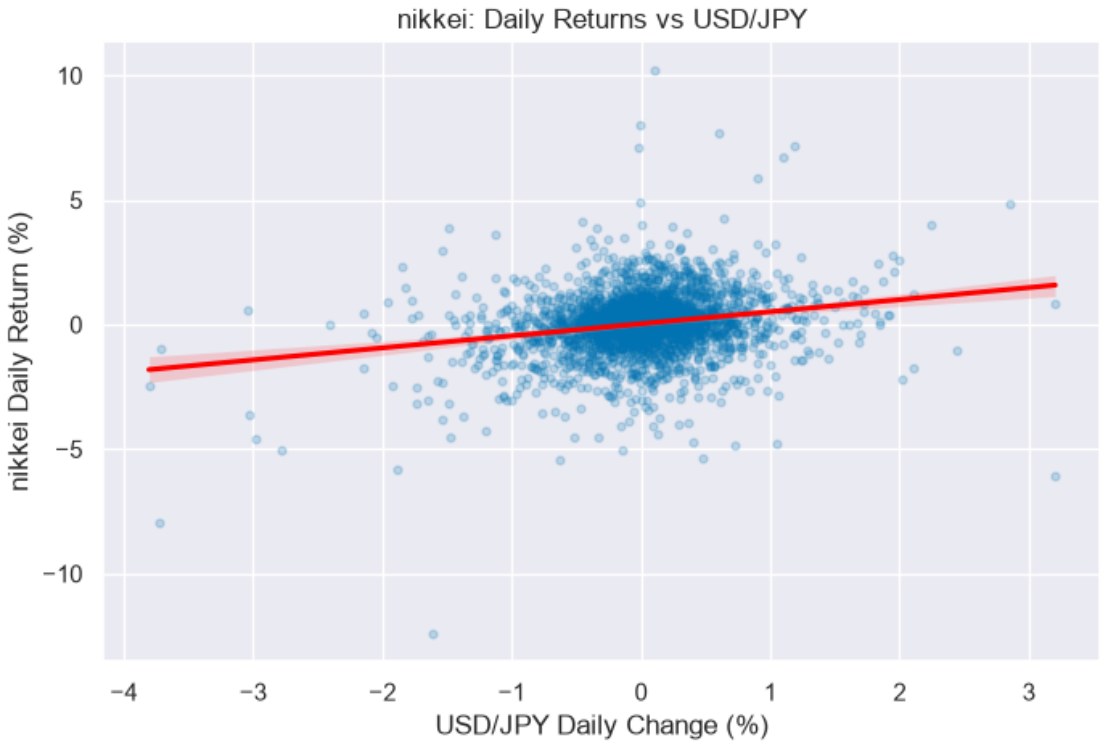

toray: Daily Returns vs USD/JPY
toray Daily Return (%)
10
5
0
−5
−10
−4
−3
−2
−1
0
1
2
3
USD/JPY Daily Change (%)

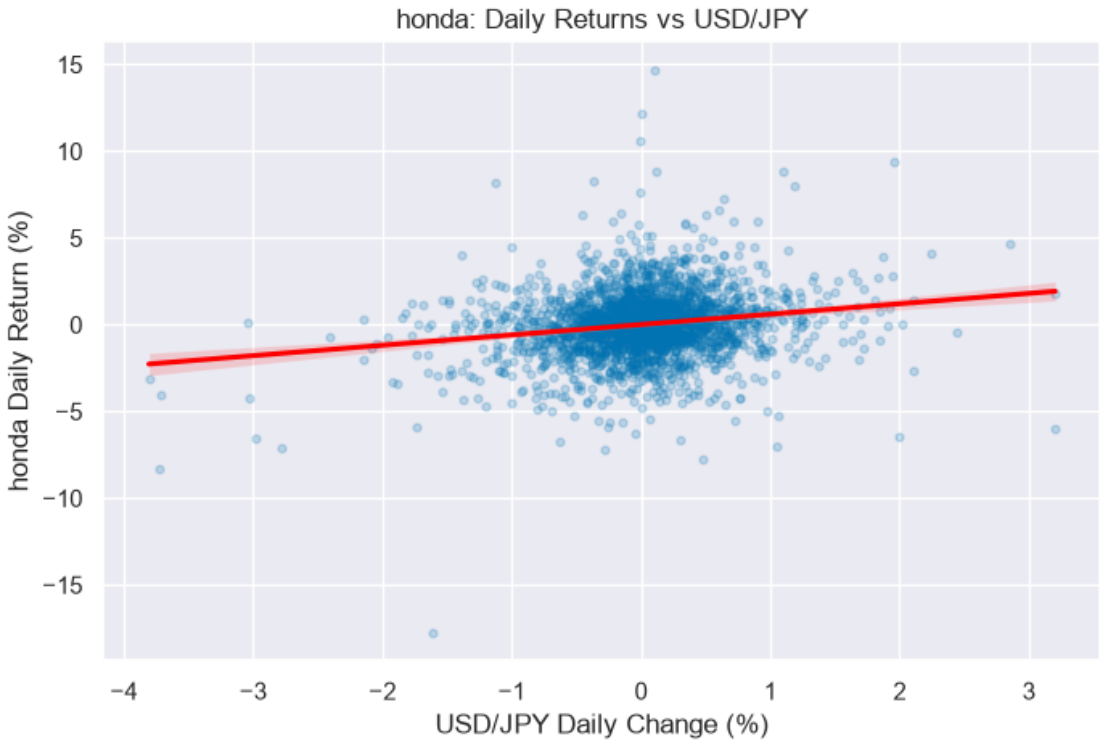

meiji: Daily Returns vs USD/JPY
meiji Daily Return (%)
15
10
5
0
−5
−10
−4
−3
−2
−1
0
1
2
3
USD/JPY Daily Change (%)

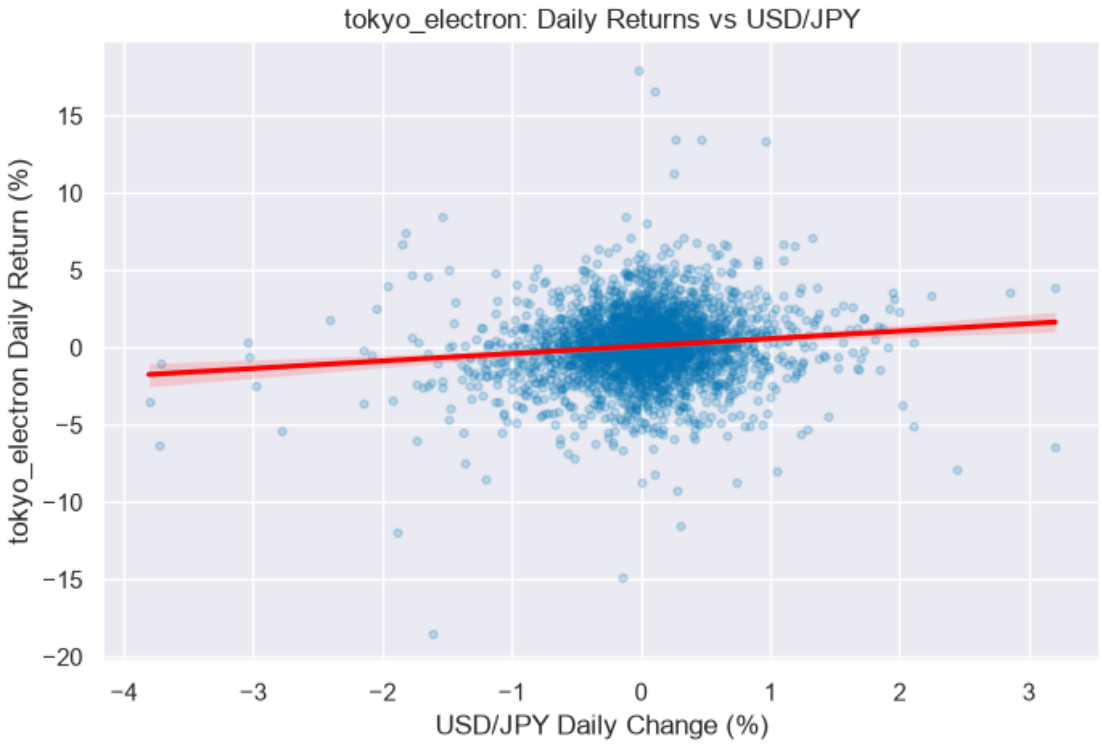

nippon_paper: Daily Returns vs USD/JPY
nippon_paper Daily Return (%)
USD/JPY Daily Change (%)
15
10
5
0
−5
−10
−15
−4
−3
−2
−1
0
1
2
3

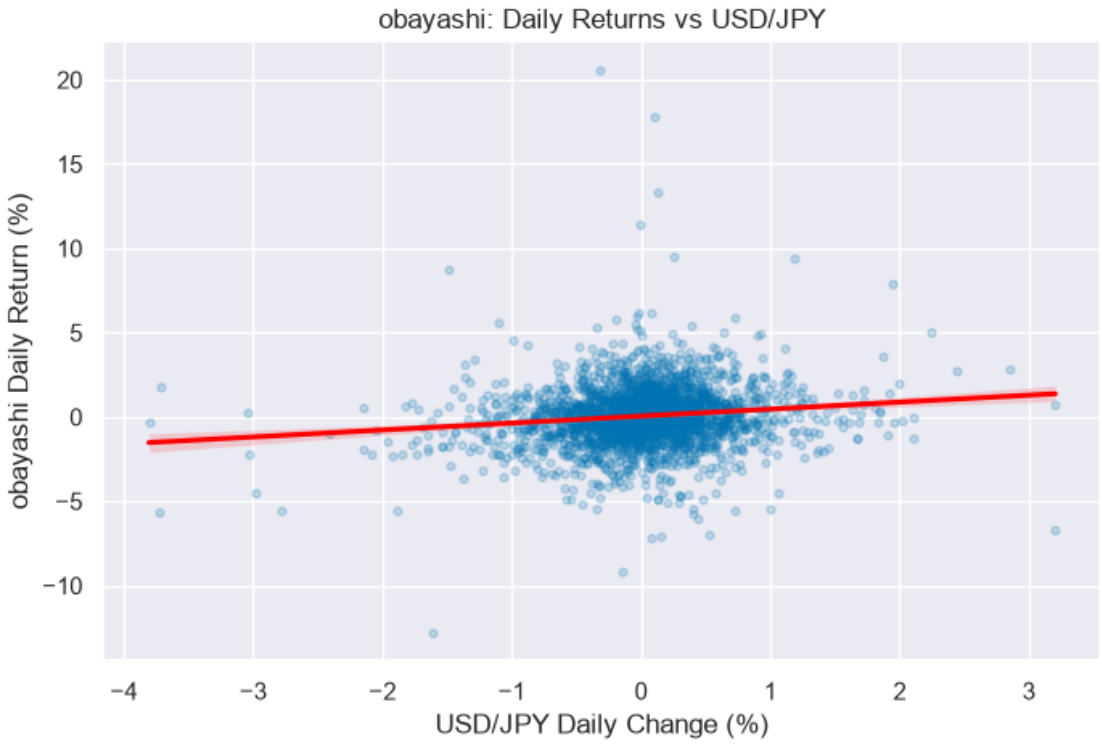

shin_etsu: Daily Returns vs USD/JPY
shin_etsu Daily Return (%)
USD/JPY Daily Change (%)
15
10
5
0
−5
−10
−4
−3
−2
−1
0
1
2
3


eneos: Daily Returns vs USD/JPY
eneos Daily Return (%)
10
5
0
−5
−10
−15
−4
−3
−2
−1
0
1
2
3
USD/JPY Daily Change (%)


nikkei: Daily Returns vs USD/JPY
nikkei Daily Return (%)
10
5
0
−5
−10
−4
−3
−2
−1
0
1
2
3
USD/JPY Daily Change (%)


honda: Daily Returns vs USD/JPY
honda Daily Return (%)
USD/JPY Daily Change (%)
15
10
5
0
−5
−10
−15
−4
−3
−2
−1
0
1
2
3


tokyo_electron: Daily Returns vs USD/JPY
tokyo_electron Daily Return (%)
USD/JPY Daily Change (%)
15
10
5
0
−5
−10
−15
−20
−4
−3
−2
−1
0
1
2
3



1.3.5 These correlations are rather compressed, but the data shows that when looking at a daily window most of the stocks have 0 to weak correlations to USD/Yen movements. Below we will be looking at the correlations with rolling windows and monthly returns

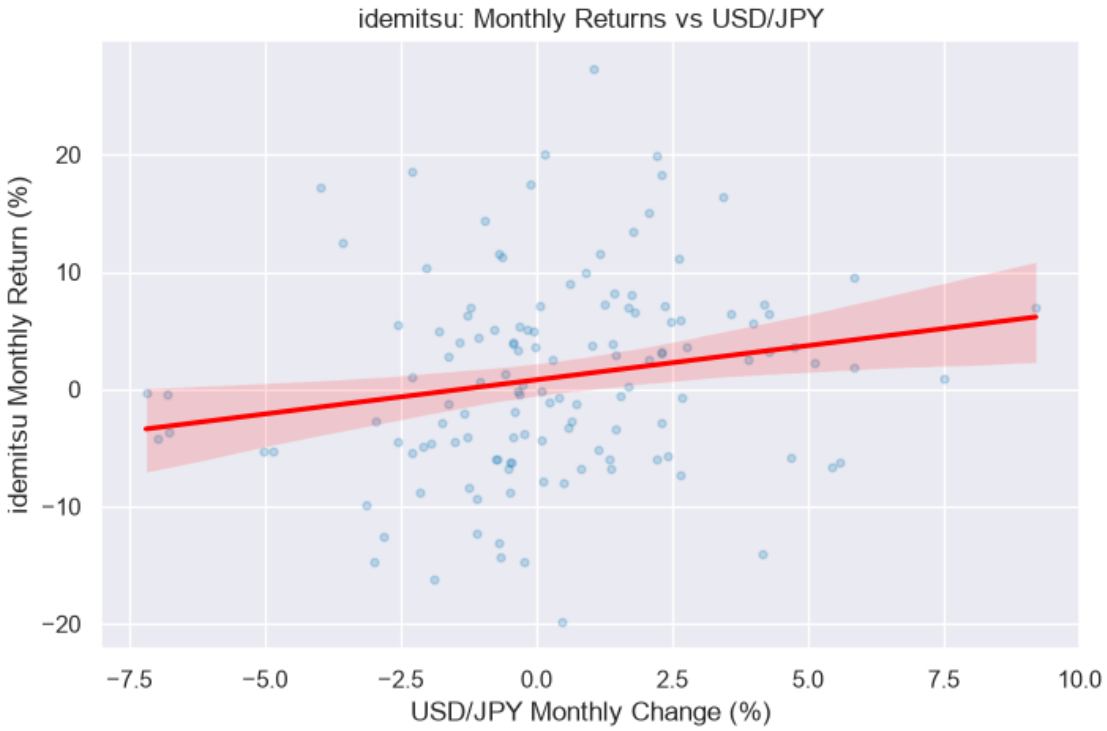
1.3.6 Correlations looking at Monthly Changes

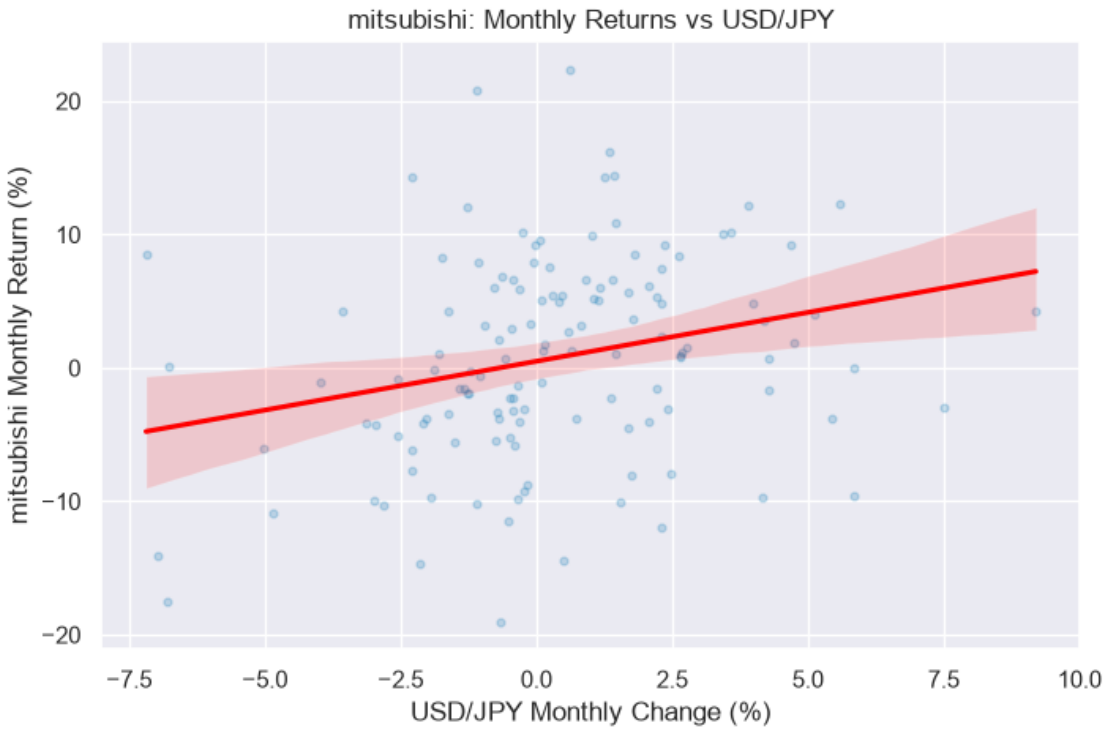
[10]:
```
for name in stocks:
#Skipping Yen_USD in graphing
    if name == "yen_usd":
        continue

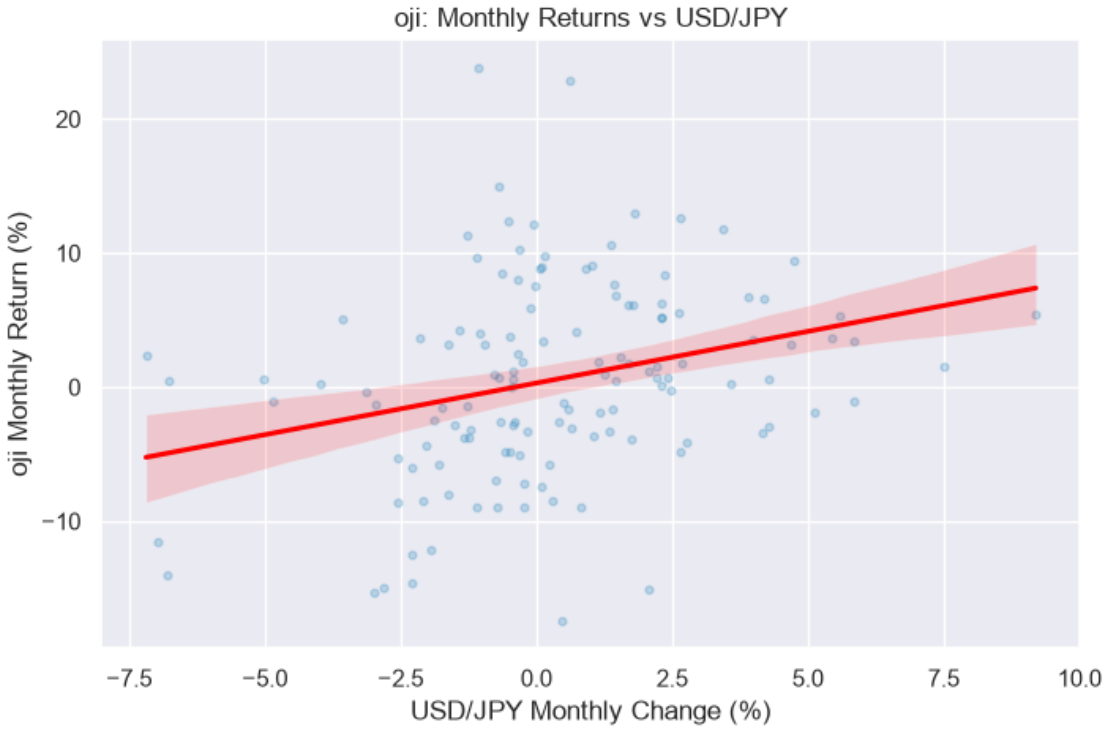
    plt.figure(figsize = (8, 5))

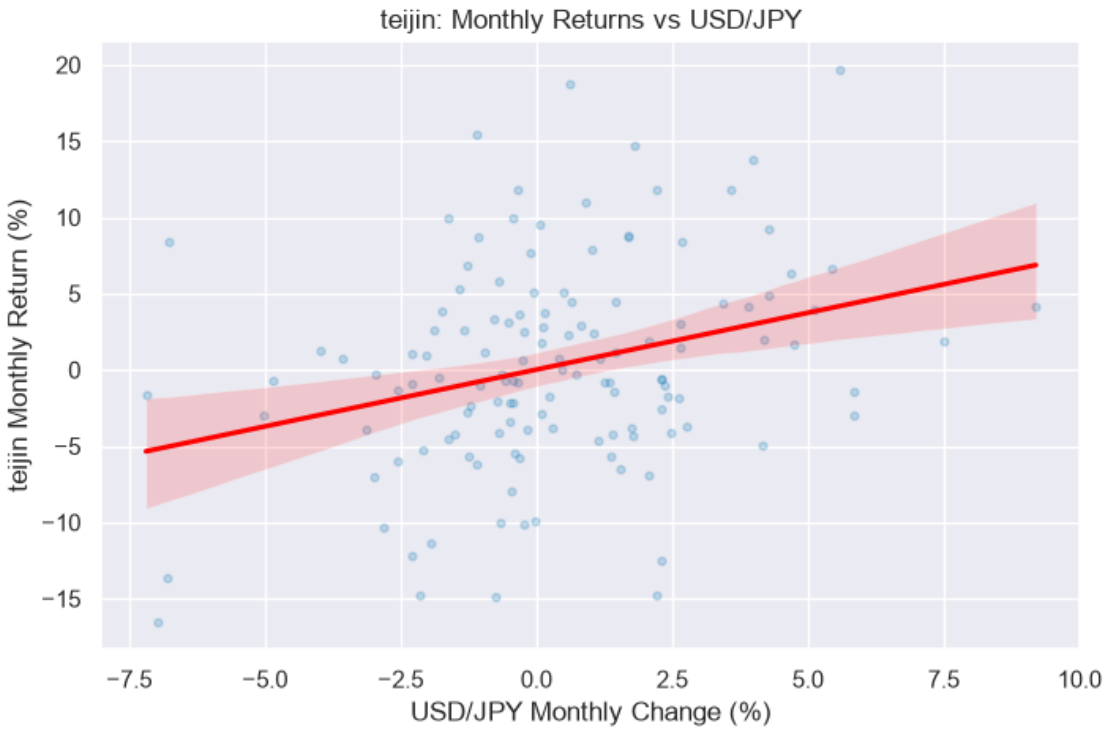
    sns.regplot(
        data = monthly_returns,
        x = "yen_usd",
        y = name,
        scatter_kws = {"alpha": 0.2, "s": 12},
        line_kws = {"color": "red"}
    )
```

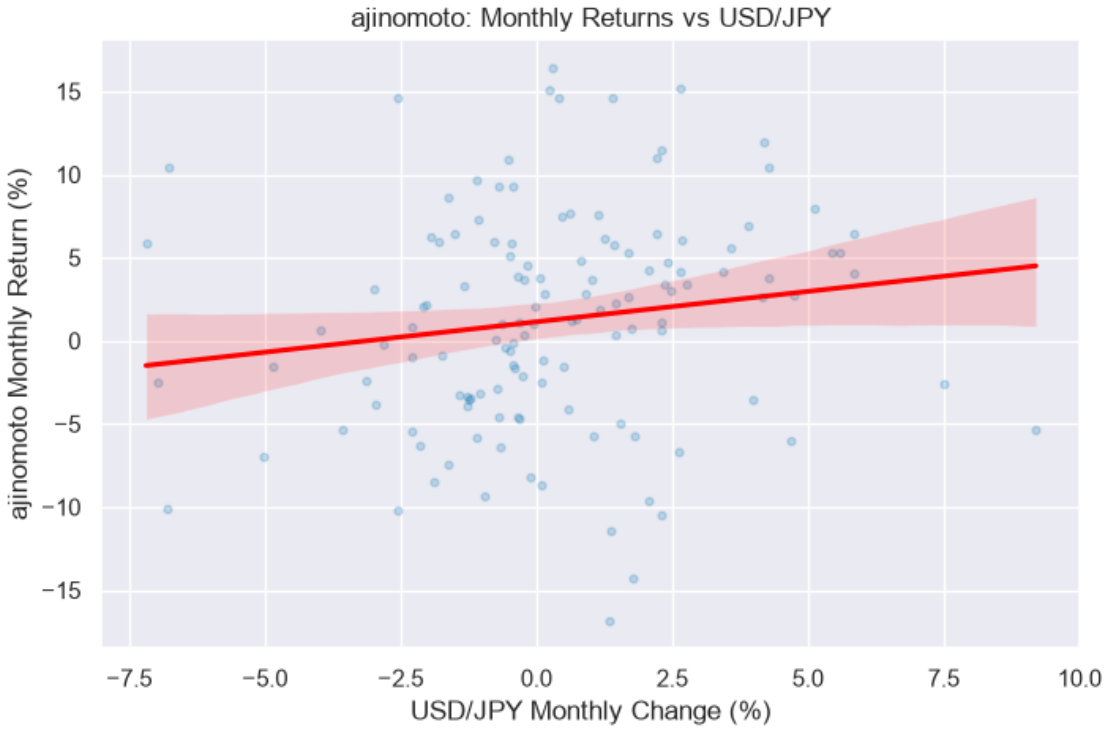
```
plt.title(f"{name}: Monthly Returns vs USD/JPY")
plt.xlabel("USD/JPY Monthly Change (%)")
plt.ylabel(f"{name} Monthly Return (%)")
plt.show()
```

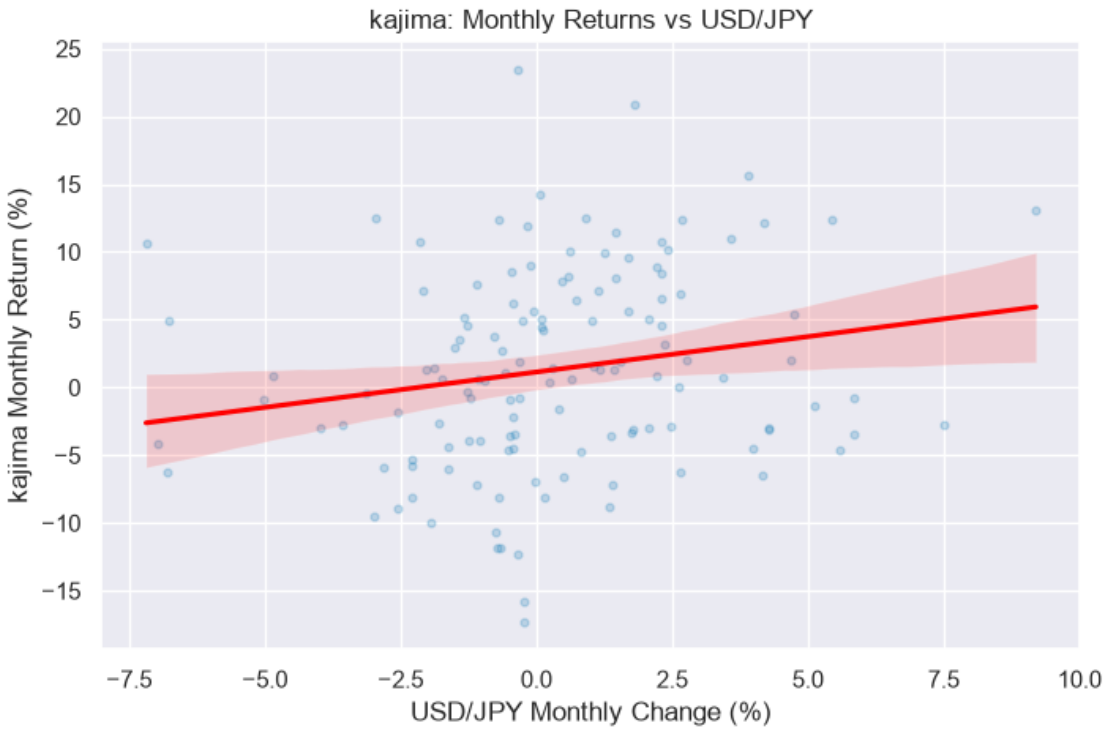


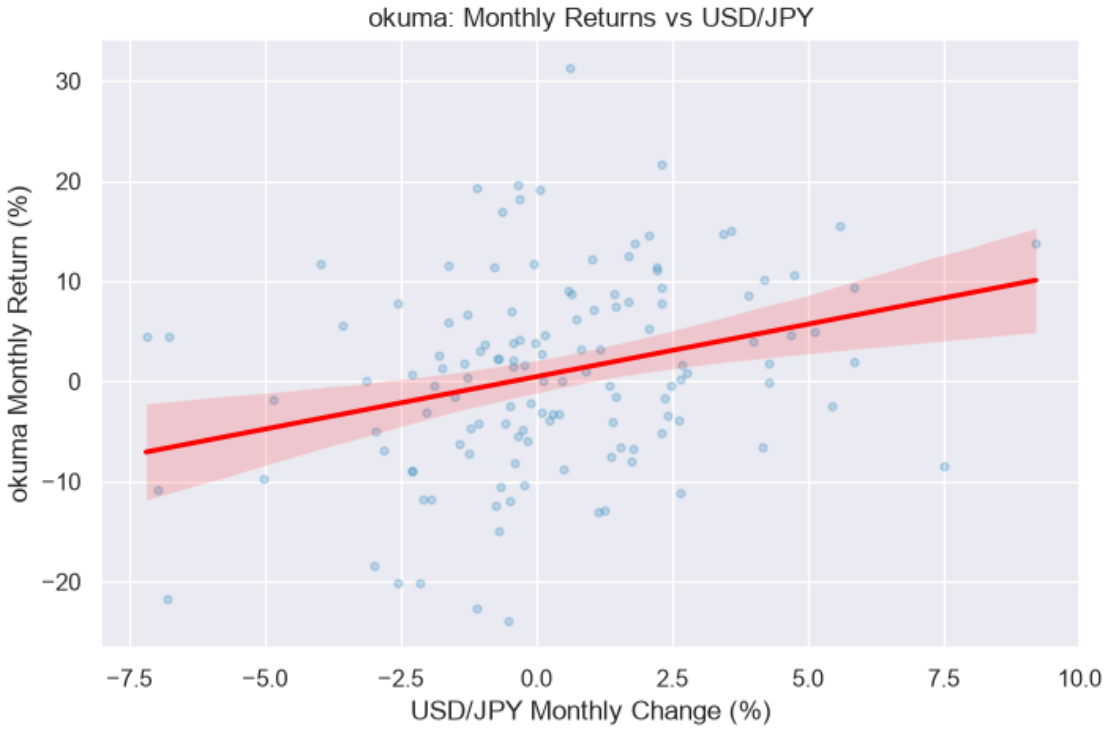

mitsubishi: Monthly Returns vs USD/JPY
mitsubishi Monthly Return (%)
20
10
0
−10
−20
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
USD/JPY Monthly Change (%)

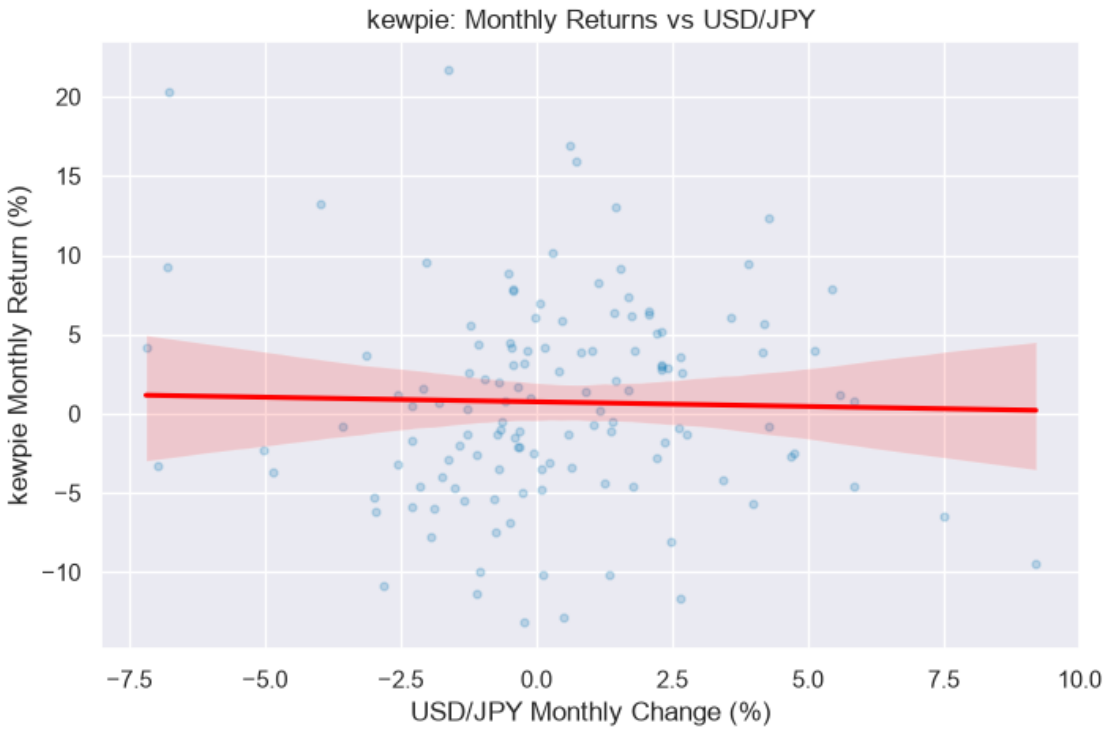

oji: Monthly Returns vs USD/JPY
oji Monthly Return (%)
20
10
0
−10
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
USD/JPY Monthly Change (%)

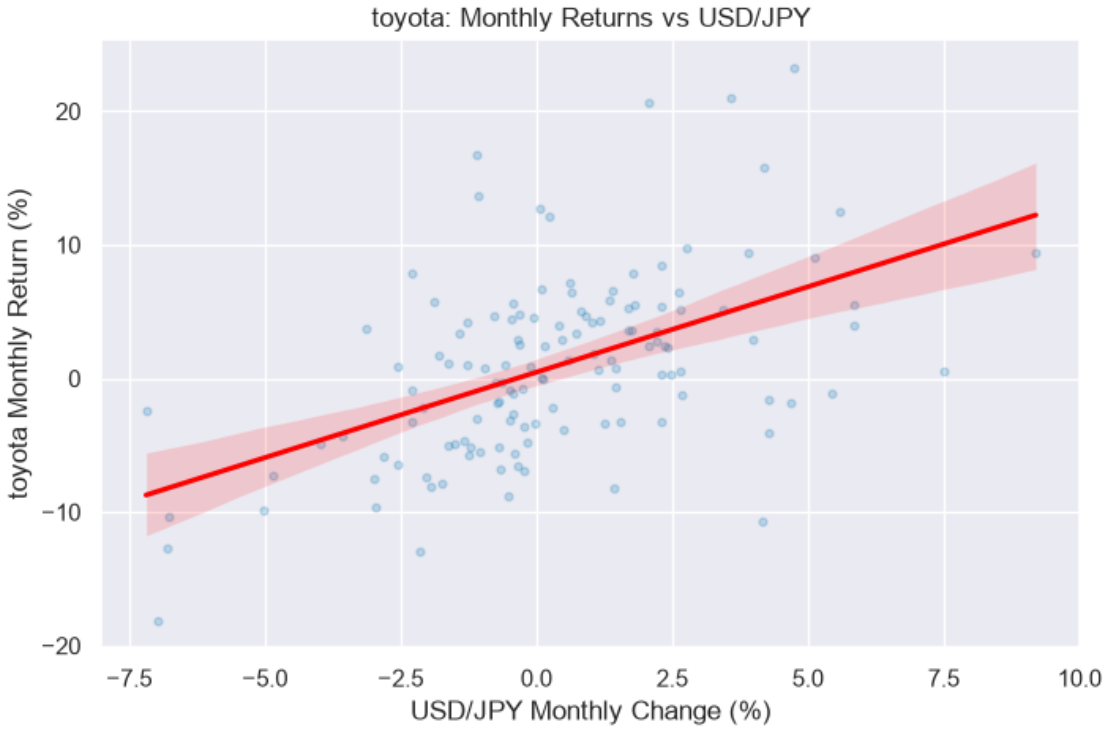

teijin: Monthly Returns vs USD/JPY
teijin Monthly Return (%)
20
15
10
5
0
−5
−10
−15
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
USD/JPY Monthly Change (%)

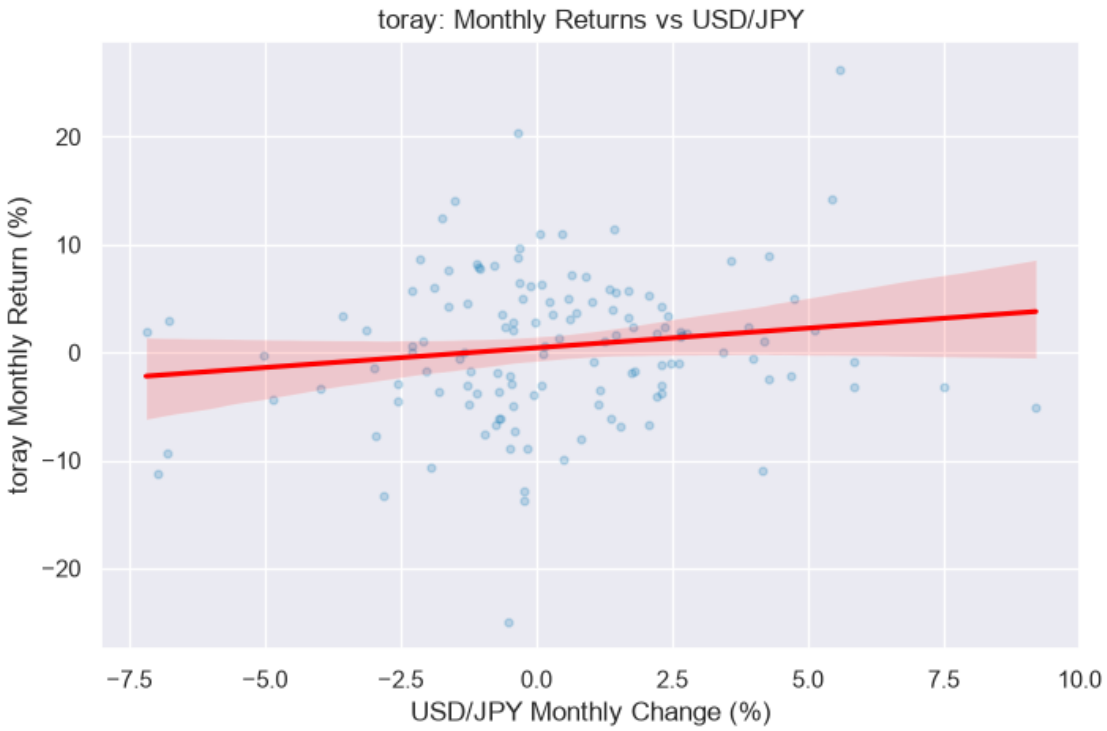

ajinomoto: Monthly Returns vs USD/JPY
ajinomoto Monthly Return (%)
15
10
5
0
−5
−10
−15
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
USD/JPY Monthly Change (%)

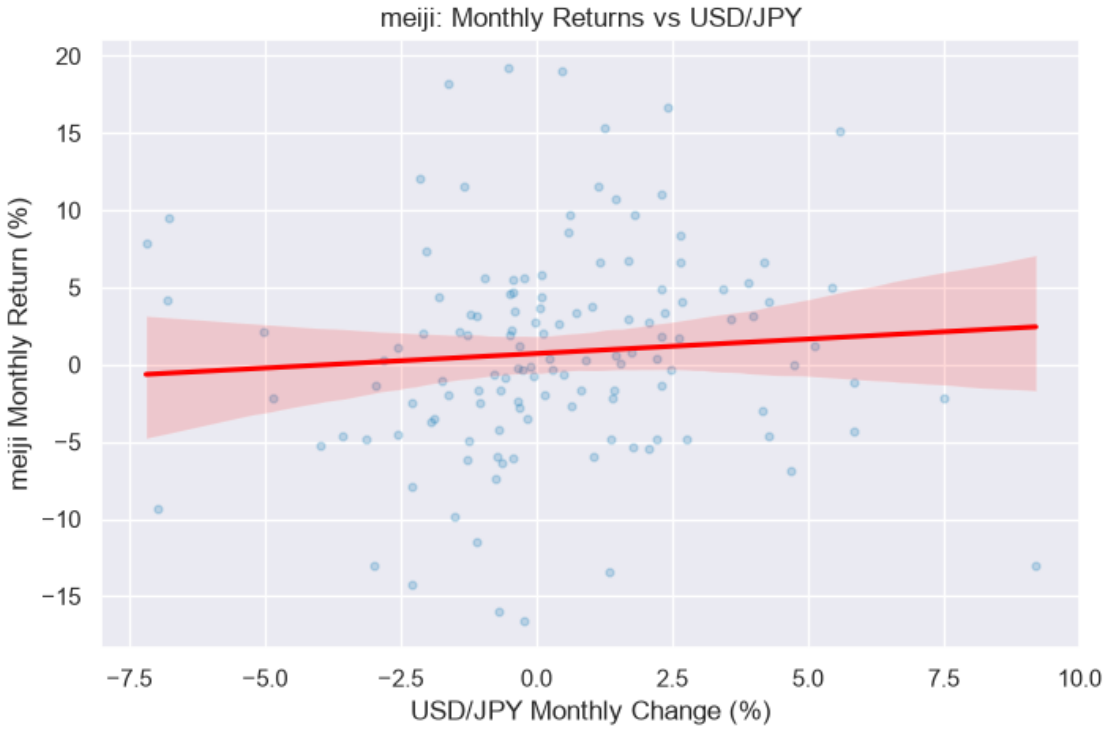

kajima: Monthly Returns vs USD/JPY
kajima Monthly Return (%)
25
20
15
10
5
0
−5
−10
−15
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
USD/JPY Monthly Change (%)

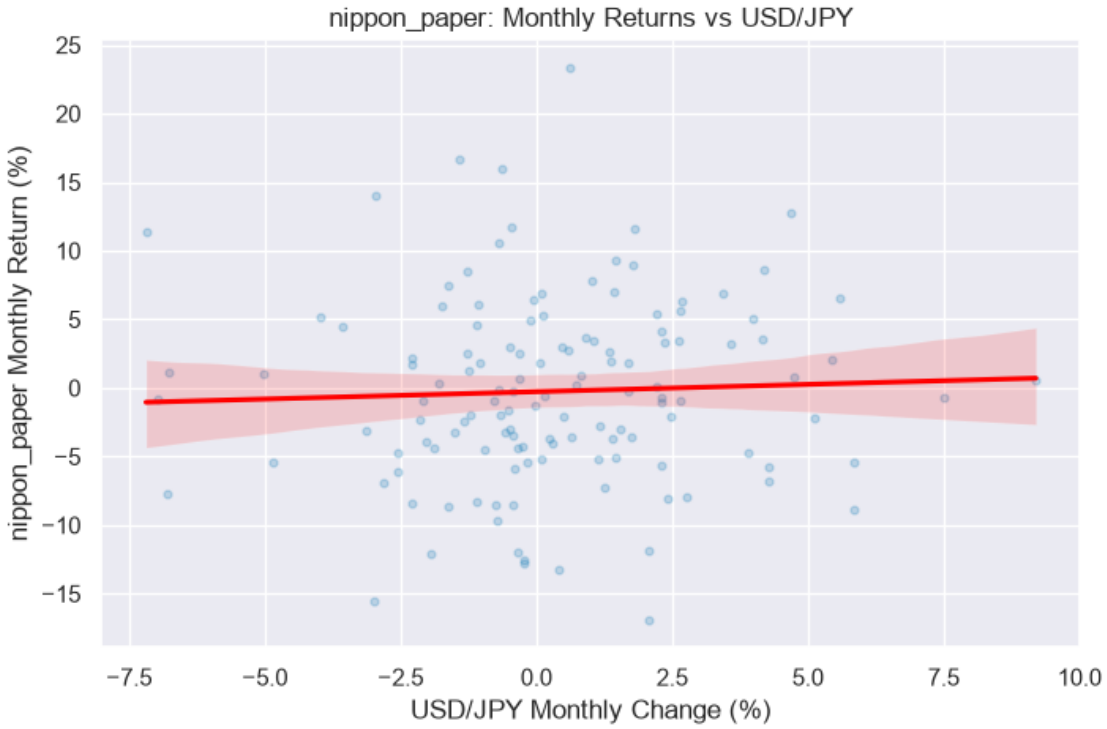

okuma: Monthly Returns vs USD/JPY
okuma Monthly Return (%)
30
20
10
0
−10
−20
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
USD/JPY Monthly Change (%)

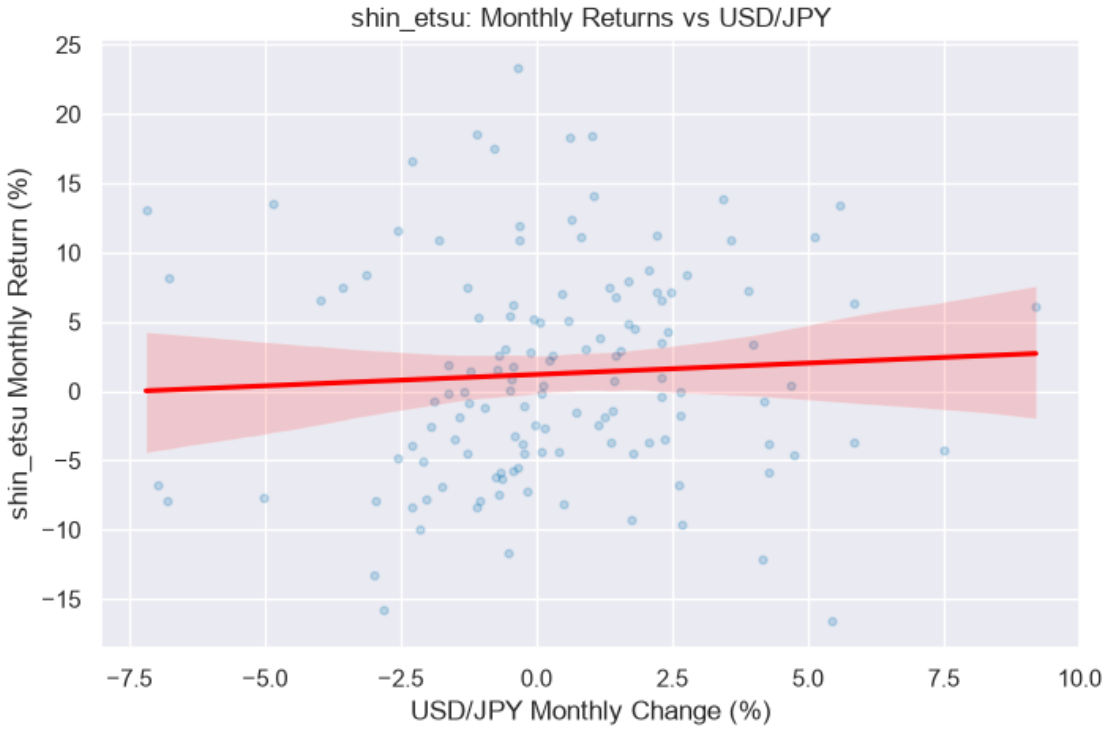

kewpie: Monthly Returns vs USD/JPY
kewpie Monthly Return (%)
20
15
10
5
0
−5
−10
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
USD/JPY Monthly Change (%)

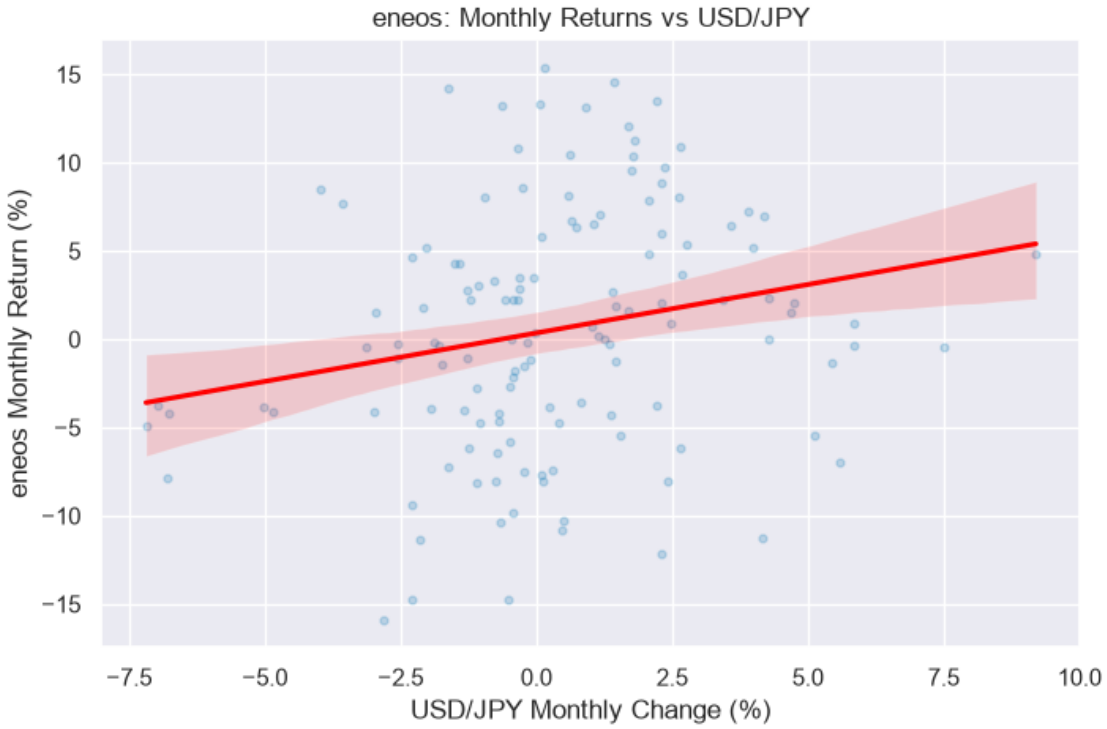

toyota: Monthly Returns vs USD/JPY
toyota Monthly Return (%)
20
10
0
−10
−20
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
USD/JPY Monthly Change (%)

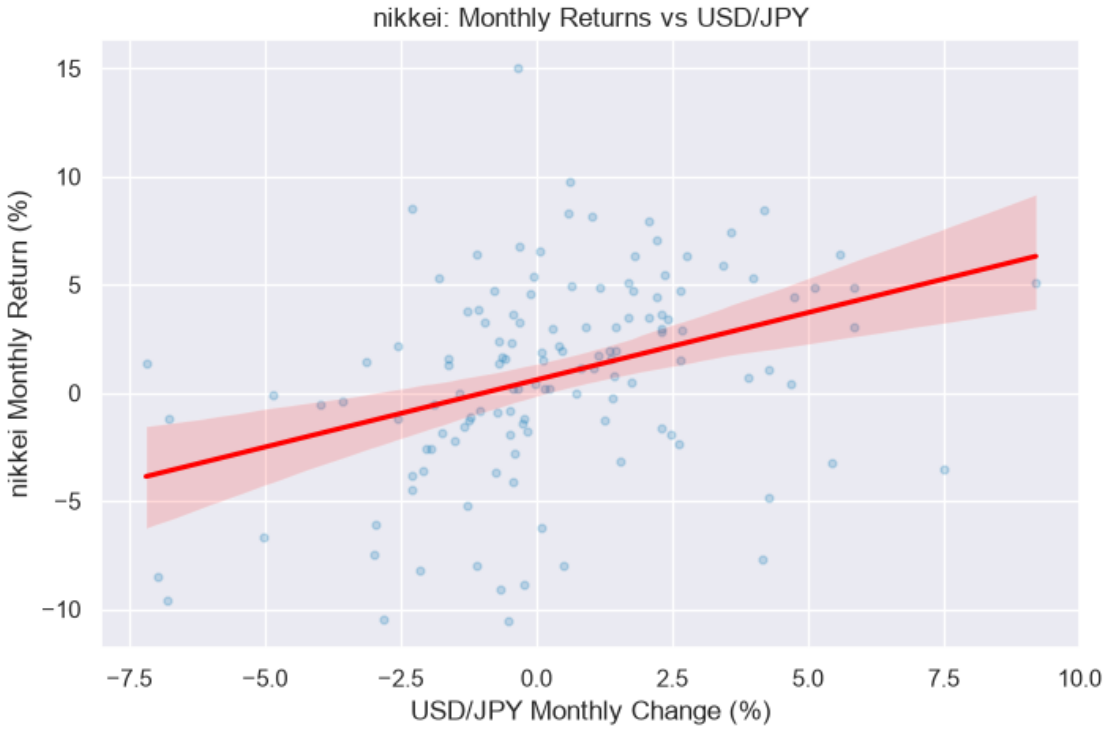

toray: Monthly Returns vs USD/JPY
toray Monthly Return (%)
20
10
0
−10
−20
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
USD/JPY Monthly Change (%)

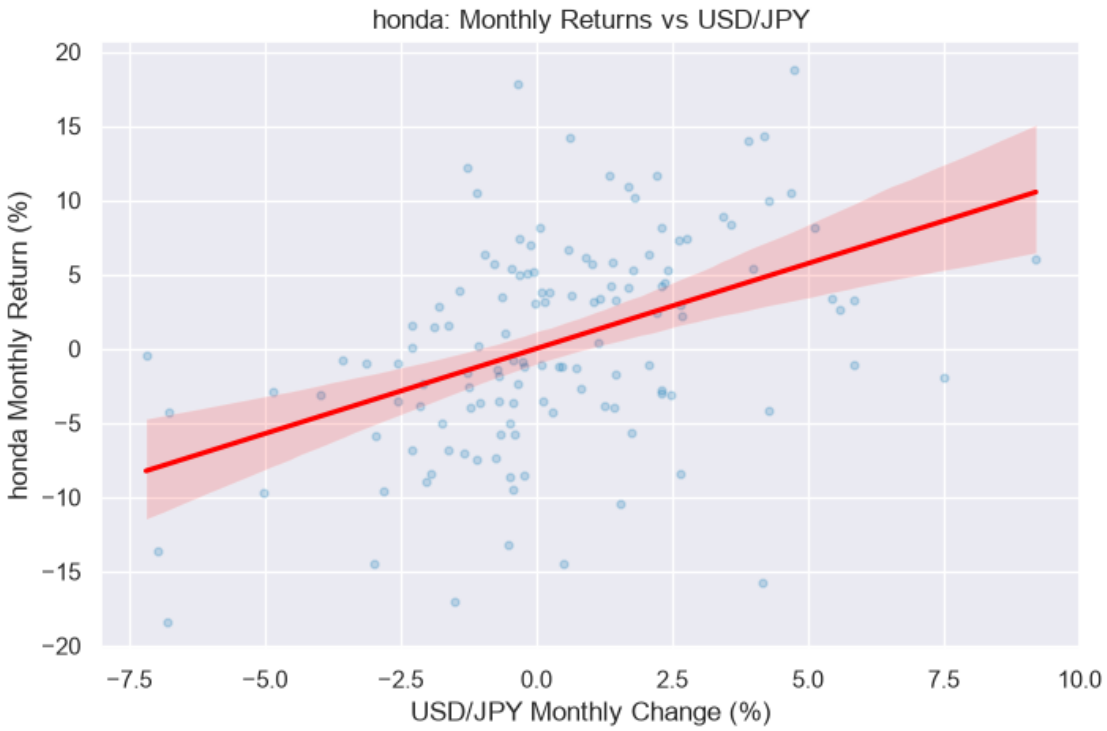

meiji: Monthly Returns vs USD/JPY
meiji Monthly Return (%)
20
15
10
5
0
−5
−10
−15
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
USD/JPY Monthly Change (%)

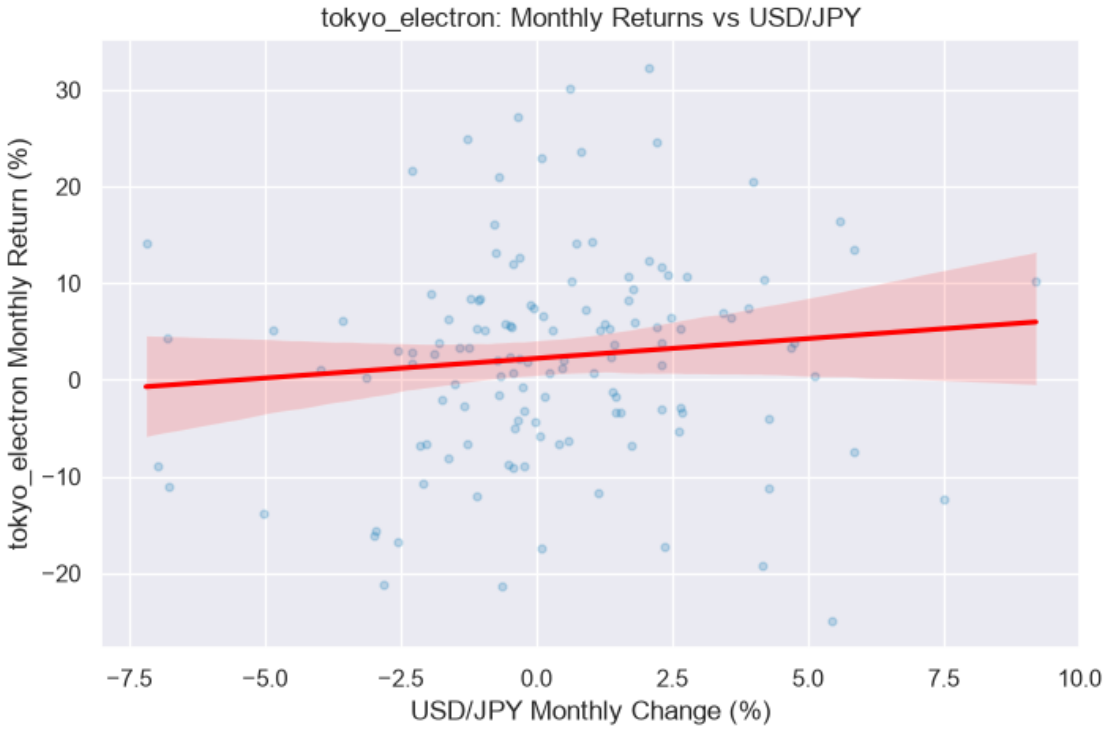

nippon_paper: Monthly Returns vs USD/JPY
nippon_paper Monthly Return (%)
25
20
15
10
5
0
−5
−10
−15
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
USD/JPY Monthly Change (%)

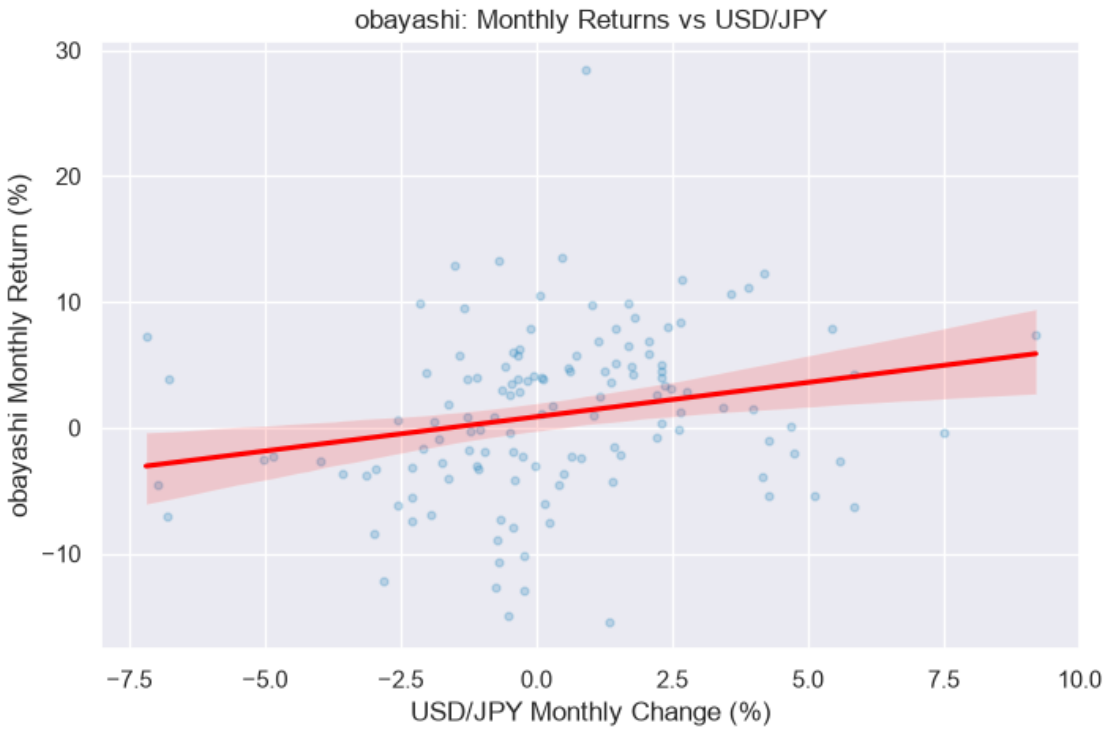

shin_etsu: Monthly Returns vs USD/JPY
shin_etsu Monthly Return (%)
25
20
15
10
5
0
−5
−10
−15
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
USD/JPY Monthly Change (%)


eneos: Monthly Returns vs USD/JPY
eneos Monthly Return (%)
15
10
5
0
−5
−10
−15
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
USD/JPY Monthly Change (%)


nikkei: Monthly Returns vs USD/JPY
nikkei Monthly Return (%)
15
10
5
0
−5
−10
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
USD/JPY Monthly Change (%)


honda: Monthly Returns vs USD/JPY
honda Monthly Return (%)
USD/JPY Monthly Change (%)
20
15
10
5
0
−5
−10
−15
−20
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0


tokyo_electron: Monthly Returns vs USD/JPY
tokyo_electron Monthly Return (%)
USD/JPY Monthly Change (%)
30
20
10
0
−10
−20
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0



1.3.7 Monthly Observations

When looking at monthly returns the error is a lot larger, however, we can see that many of the stocks are now either closer to 0 correlation, with a few firms like Toyota and Honda reflecting potentially high correlations. The nikkei index is also showing a positive correlation as well.

1.3.8 Looking at Correlation Heatmaps

Cumulative change vs USD/Yen

[11]:
```
# Selecting the total change columns
total_cols = [f"{name}_total_change" for name in stocks]
if "yen_usd_total_change" not in total_cols:
    total_cols.append("yen_usd_total_change")

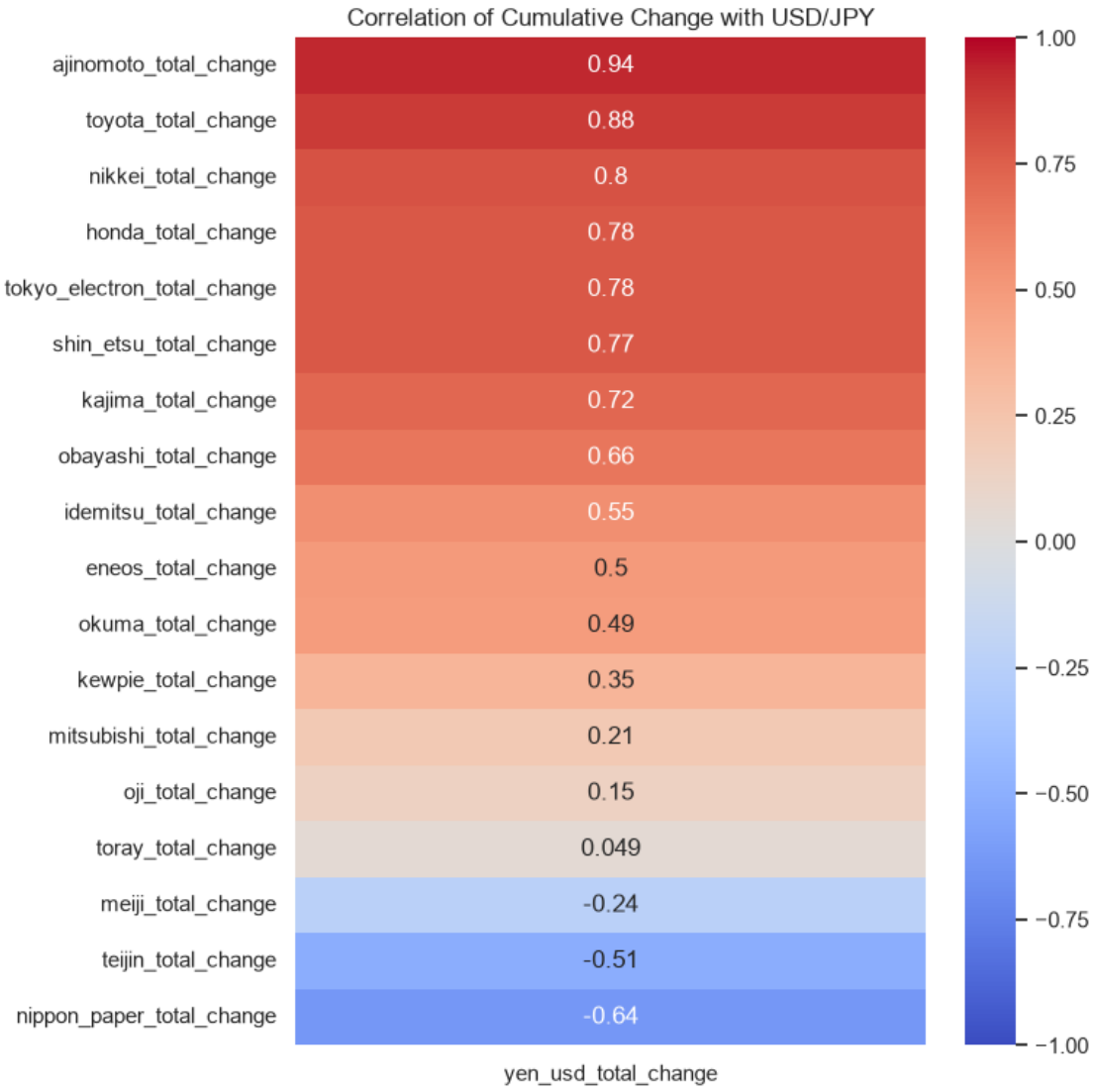
# Calculate correlations
correlations_total = stock_prices[total_cols].corr()

# Select only correlations with USD/JPY
yen_correlations_total = correlations_total[["yen_usd_total_change"]].drop(
    index = "yen_usd_total_change"
)

# Plot
plt.figure(figsize=(7, 9))
```

```
sns.heatmap(
    yen_correlations_total.sort_values(
        by = "yen_usd_total_change",
        ascending = False),
    annot = True,
    cmap = "coolwarm",
    center = 0,
    vmin = -1,
    vmax = 1
)

plt.title("Correlation of Cumulative Change with USD/JPY")
plt.show()
```



1.3.9 Cumulative Correlation Observations

This column of data is really interesting. The firm with the highest correlation is ajinomoto with a coefficient of .94. Looking at the cumulative graph we can also see that it appears to move in tandum with yen/usd movements. This might have to do with ajinomoto being a major producer of semiconductors, which are especially connected to the U.S. economy. It is also important to consider that since the time frame is so large, well performing stocks will naturally flag a high correlation with the yen/usd movements, which have also been trending upwards in the 2014 - 2024 window. Other firms that reflect large correlations are Toyota, Honda, and Tokyo electron. It appears that the highest correlation firms are within similar sectors, typically transportation or manufacturing. The lower ends like nippon paper and teijin, which are members of the paper and textile sectors. It is also important to notice that the correlations presented here are rather high compared to the findings we will make below

1.3.10 Monthly Correlation Heatmap

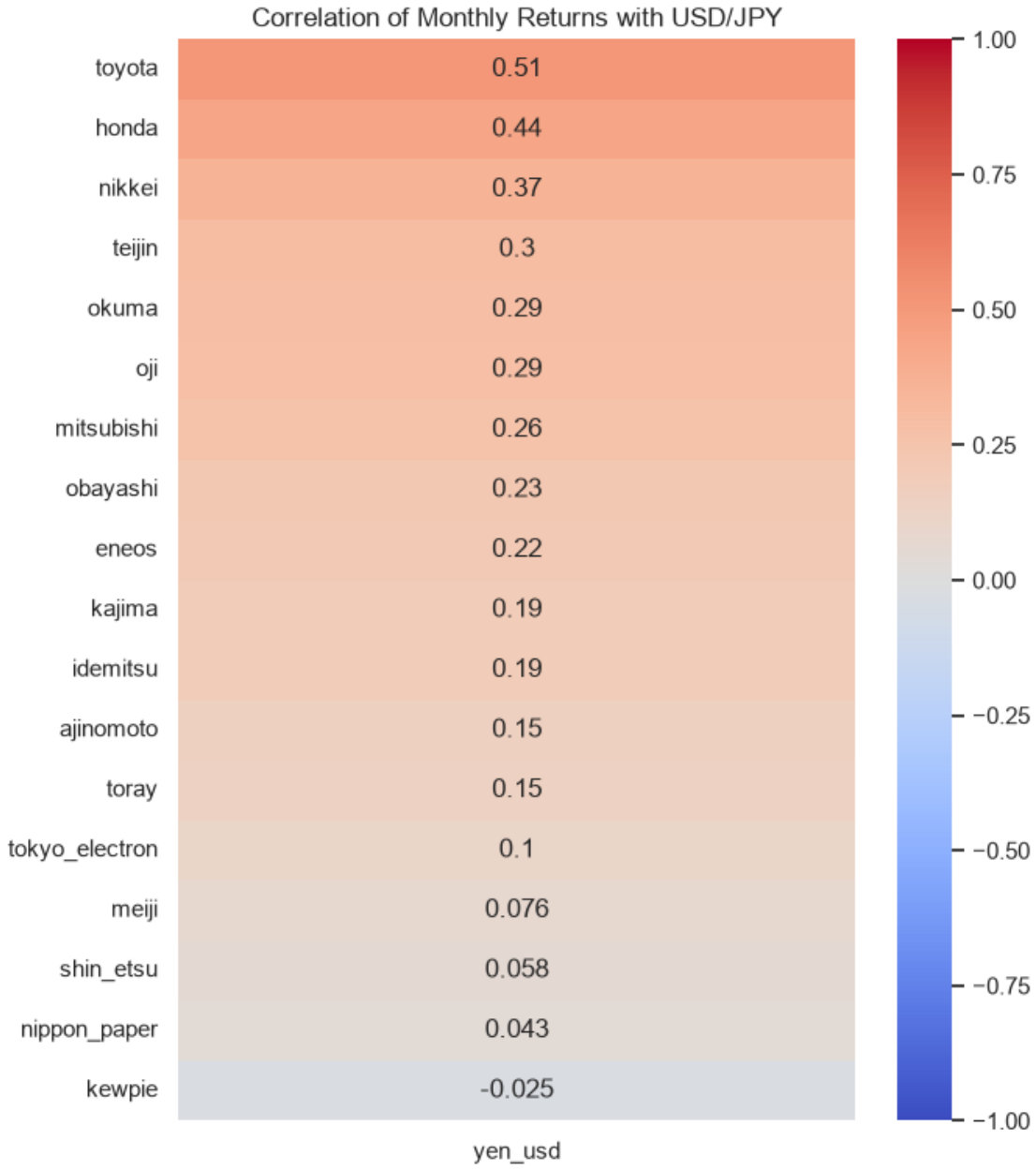
[12]:
```
# Calculate correlations
correlations_month = monthly_returns[stock_names].corr()

# Select only correlations with USD/JPY
yen_correlations_month = correlations_month[["yen_usd"]].drop(
    index = "yen_usd"
)

# Plot
plt.figure(figsize=(7, 9))

sns.heatmap(
    yen_correlations_month.sort_values(
        by="yen_usd",
        ascending=False),
    annot = True,
    cmap = "coolwarm",
    center = 0,
    vmin =- 1,
    vmax=  1
)

plt.title("Correlation of Monthly Returns with USD/JPY")
plt.show()
```



1.3.11 Monthly Return Observations

When looking at monthly changes are data changes a lot. We now have significanly more weaker correlations. Meiji, Nippon Paper Shin-etsu, and kewpie all have near 0 correlations. Signialling that these stocks might have little to no relation to Yen/USD movements.

We do also see some stocks with much higher coefficients in this window. Toyota's coefficient is now .51, indicading a moderate relationship between USD/Yen movements. Honda also experiences a large increase to .44.

Some other notable changes are mitsubishi, teijin, nikkei, oji, eneos, and okuma; which increased to the ranges of .37 to 22.

1.3.12 Daily Correlation Heatmap

[13]:
```
# Select daily change columns
daily_cols = [f"{name}_day_change" for name in stocks]

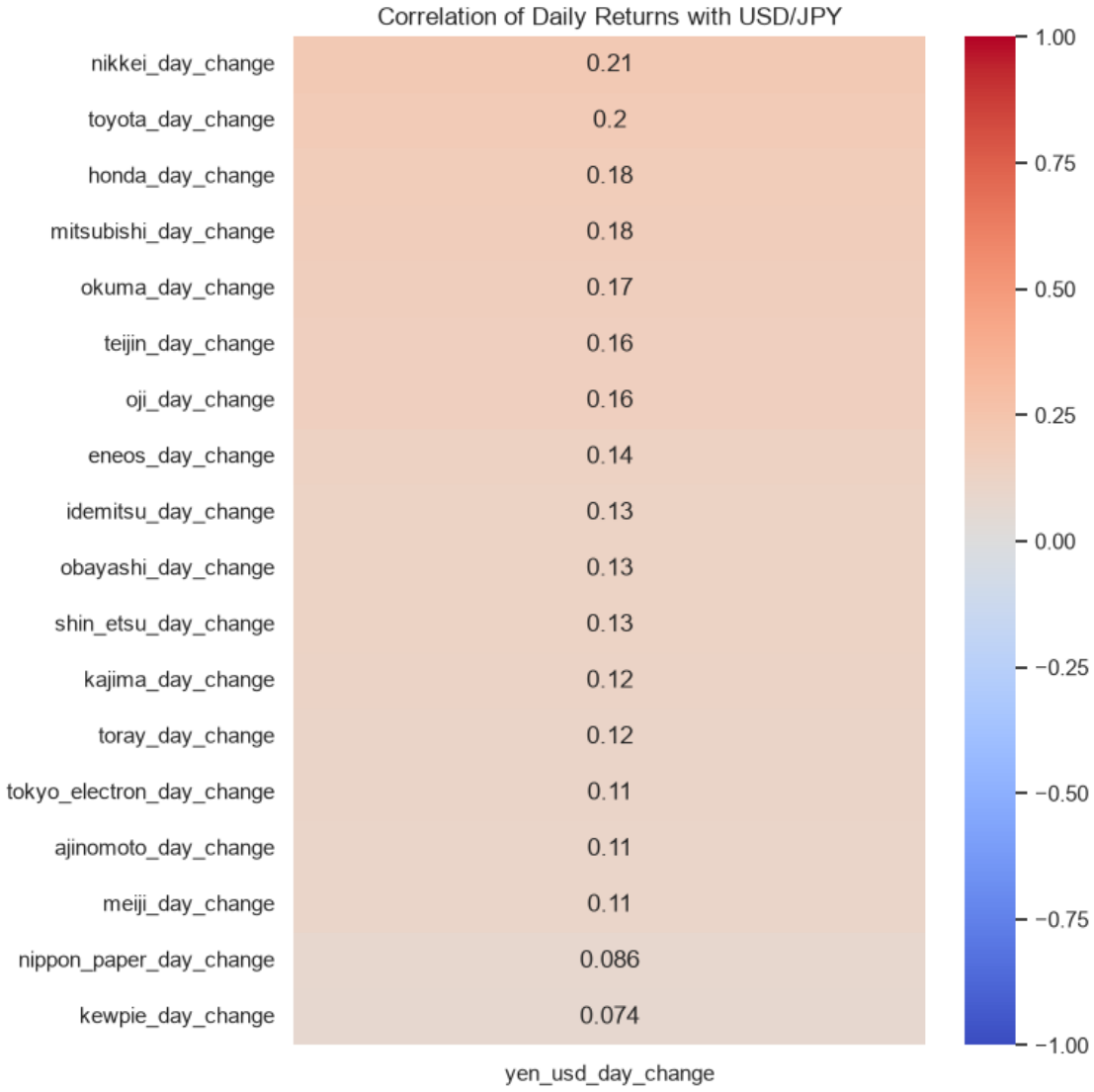
# Calculate correlations
correlations_day = stock_prices[daily_cols].corr()

# Select only correlations with USD/JPY
yen_correlations_day = correlations_day[["yen_usd_day_change"]].drop(
    index="yen_usd_day_change"
)

# Plot
plt.figure(figsize = (7, 9))

sns.heatmap(
    yen_correlations_day.sort_values(
        by = "yen_usd_day_change",
        ascending = False),
    annot = True,
    cmap = "coolwarm",
    center = 0,
    vmin = -1,
    vmax = 1
)

plt.title("Correlation of Daily Returns with USD/JPY")
plt.show()
```



1.3.13 Daily Correlation Observations

looking at daily return correlations we can see all the stocks have very low correlation coefficients to yen_usd movements. The lowest being nippon papper with .093, basically 0 correlation. The highest correlations we can see are the Nikkei with .23 and Toyota with .21. Slightly stronger correlations but still considerably weak. The rest of the stocks show coefficients .12 to .18 ranges.

1.3.14 Correlation Observations Across all time frames

Interestingly enough if we look at the cumulative correlations, the firm that had the highest ratio, ajinomoto, shifted greatly as we shrink the observation window. However, the transportation industries remained in the top 5 highest correlations throughout all 3 heatmaps. We can also see the the Nikkei remained in the top 5 as well, with it holding the top daily correlation with a coefficient of .21.

1.3.15 Creating a 90day rolling graph to see how the correlation changess over 90 day periods

[14]:
```
for name in stocks:
    if name == "yen_usd":
        continue

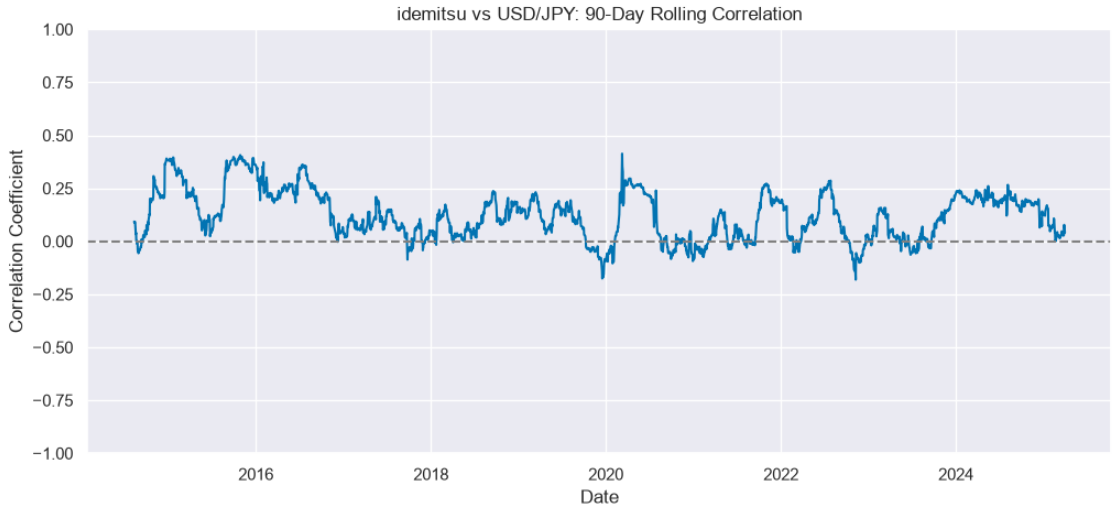
    stock_prices[f"{name}_rolling_corr"] = (
        stock_prices[f"{name}_day_change"]
        .rolling(window = 90)
        .corr(stock_prices["yen_usd_day_change"])
    )

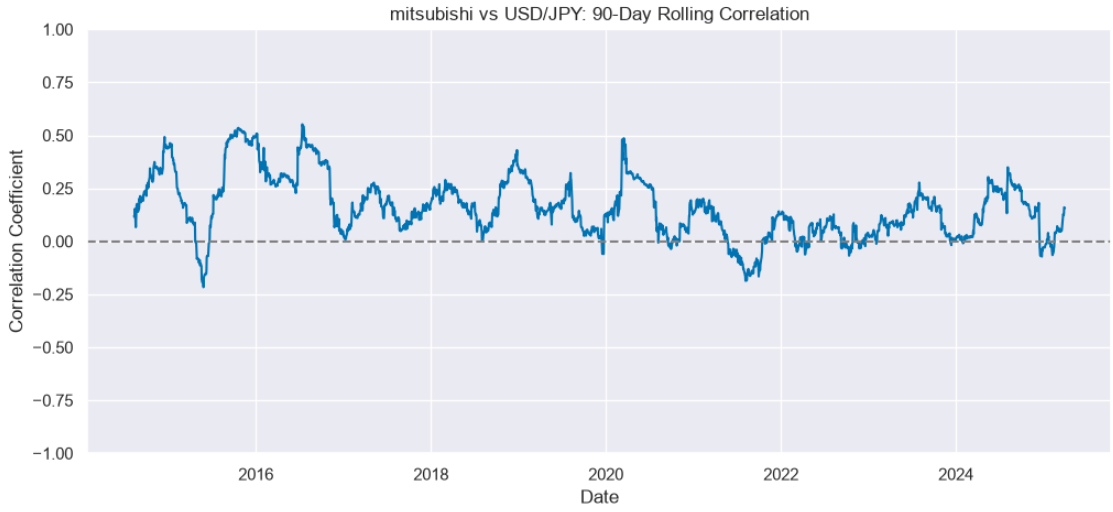

    plt.figure(figsize = (12, 5))

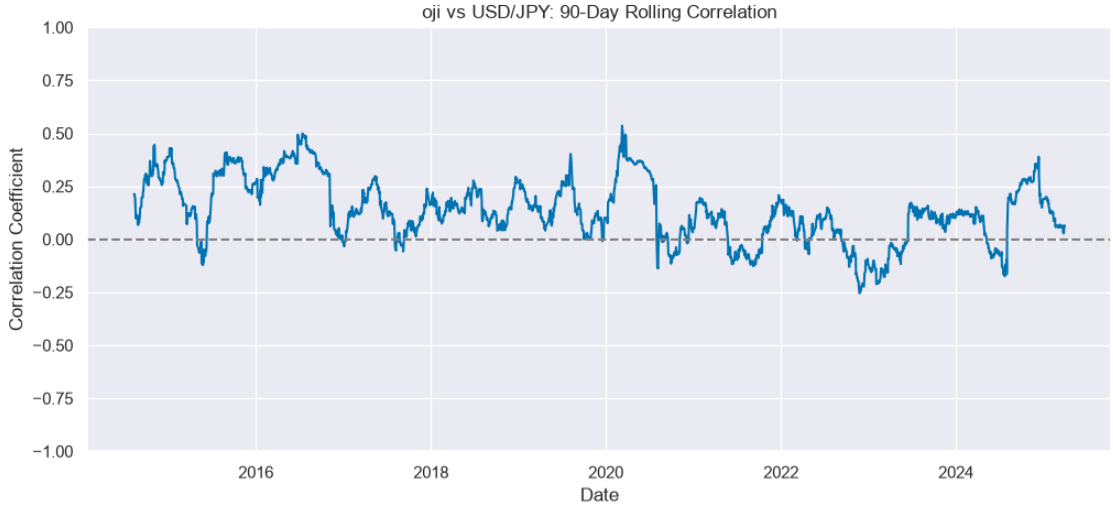
    sns.lineplot(
        data = stock_prices,
        x = "Date",
        y = f"{name}_rolling_corr"
    )

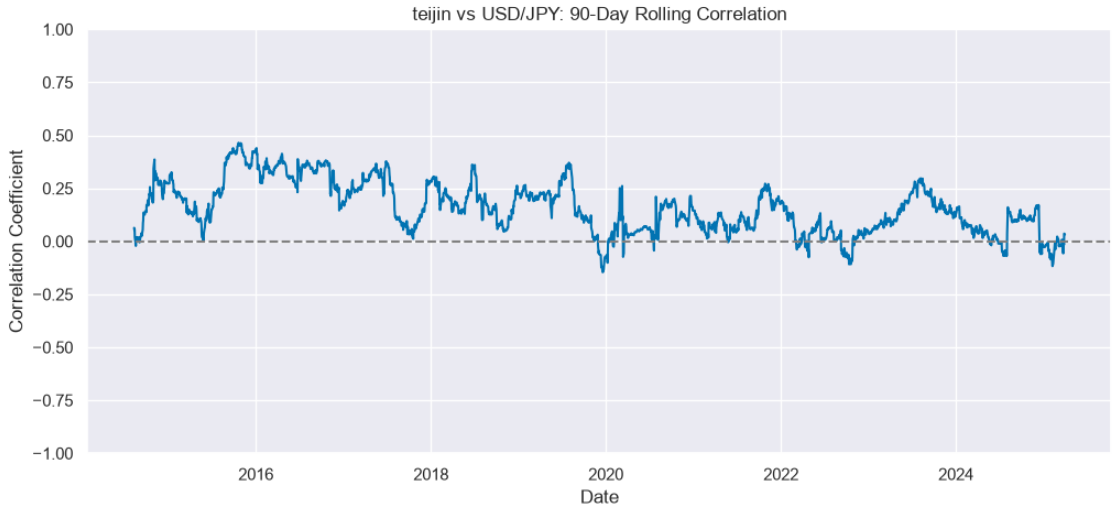
    plt.axhline(0, color = "gray", linestyle = "--")

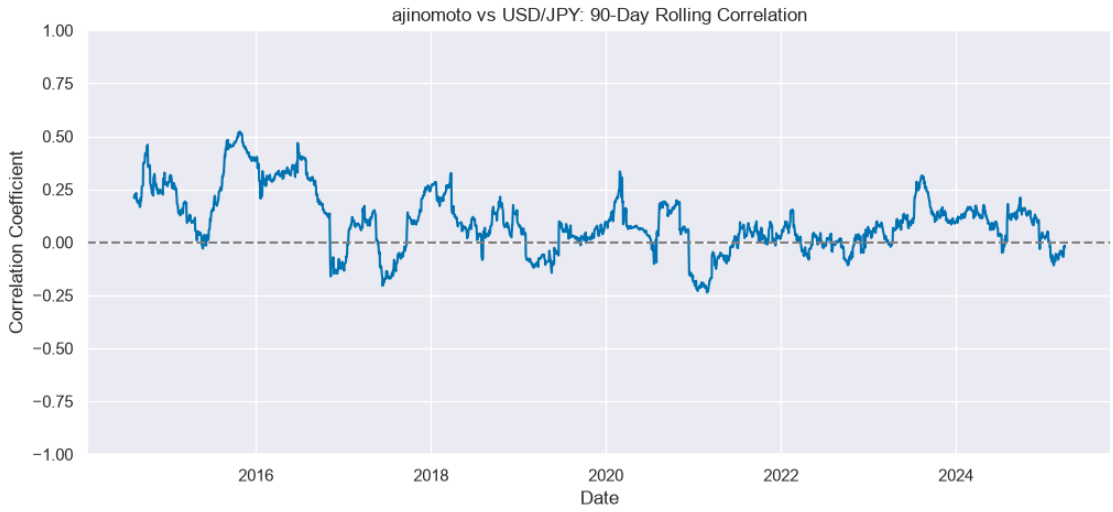
    plt.title(f"{name} vs USD/JPY: 90-Day Rolling Correlation")
    plt.ylabel("Correlation Coefficient")
    plt.ylim(-1, 1)
    plt.show()
```

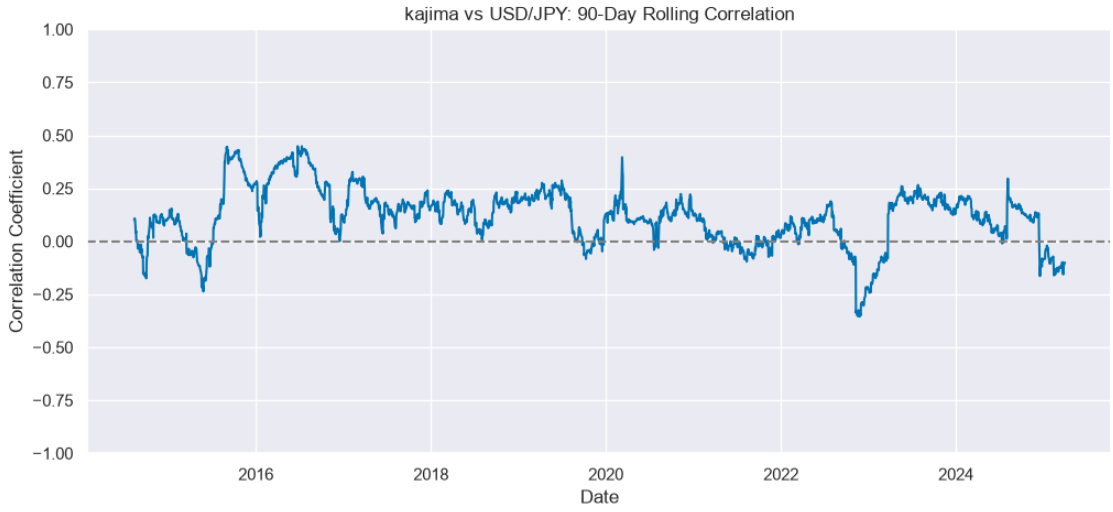


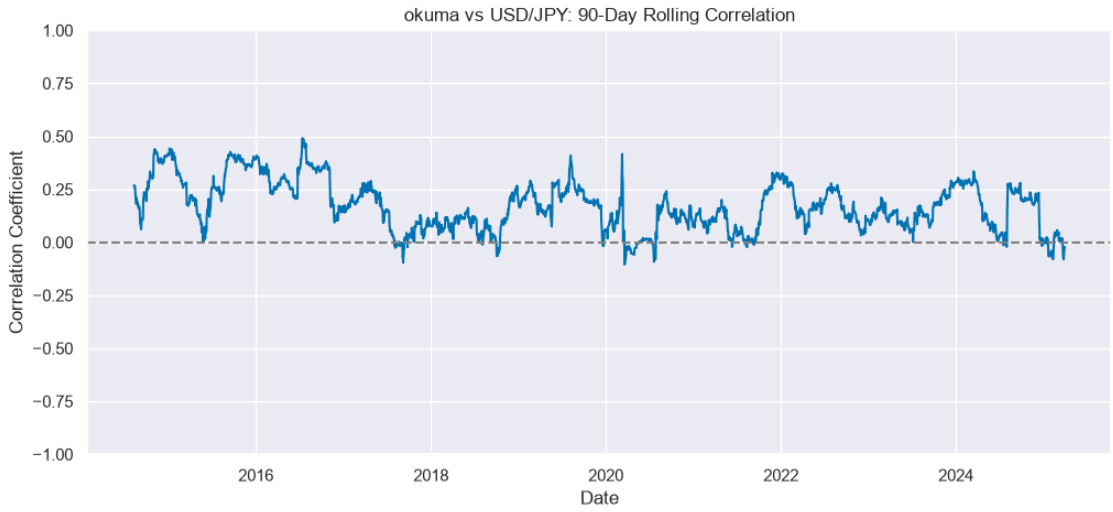

mitsubishi vs USD/JPY: 90-Day Rolling Correlation
Correlation Coefficient
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
2016
2018
2020
2022
2024
Date

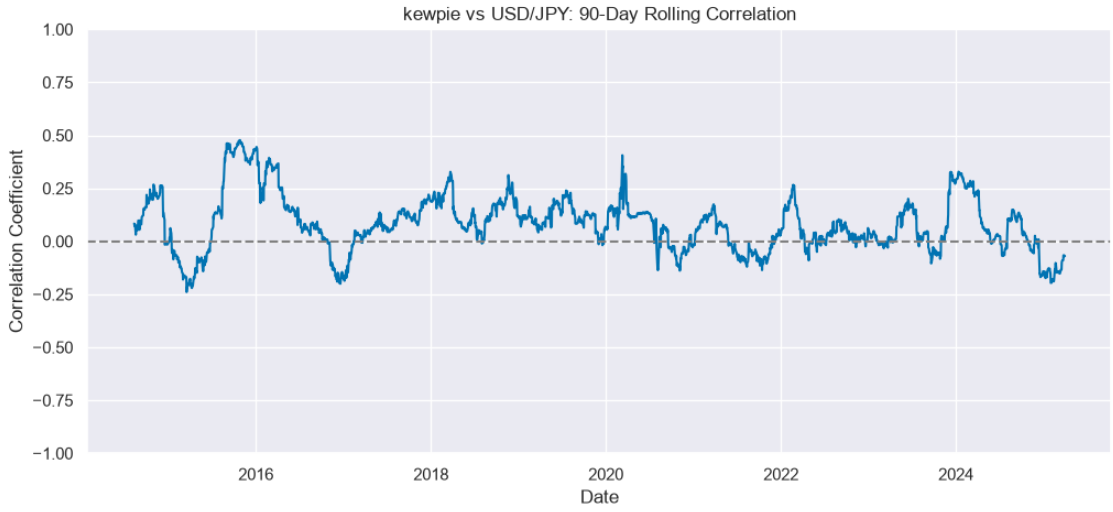

oji vs USD/JPY: 90-Day Rolling Correlation
Correlation Coefficient
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
2016
2018
2020
2022
2024
Date

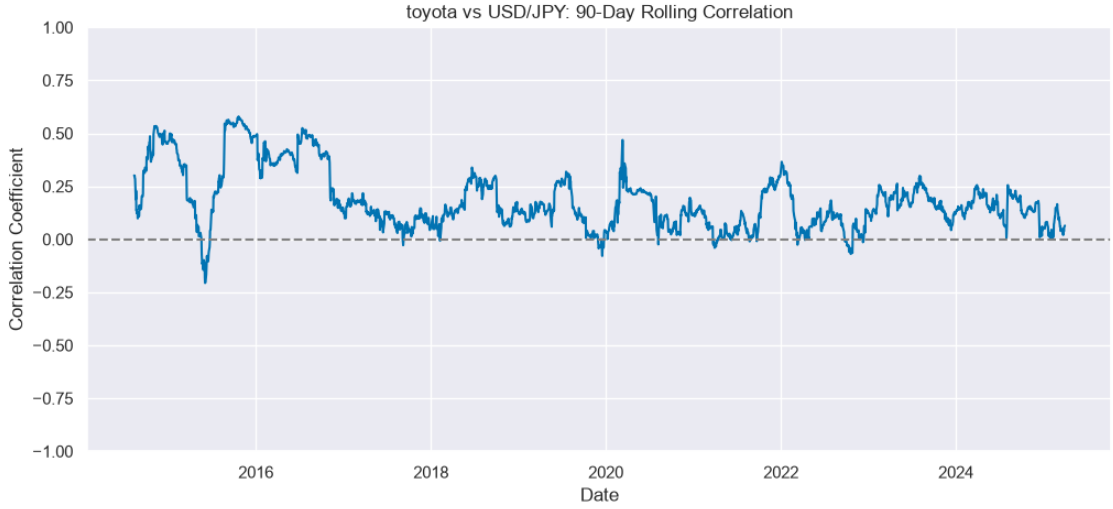

teijin vs USD/JPY: 90-Day Rolling Correlation
Correlation Coefficient
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
2016
2018
2020
2022
2024
Date

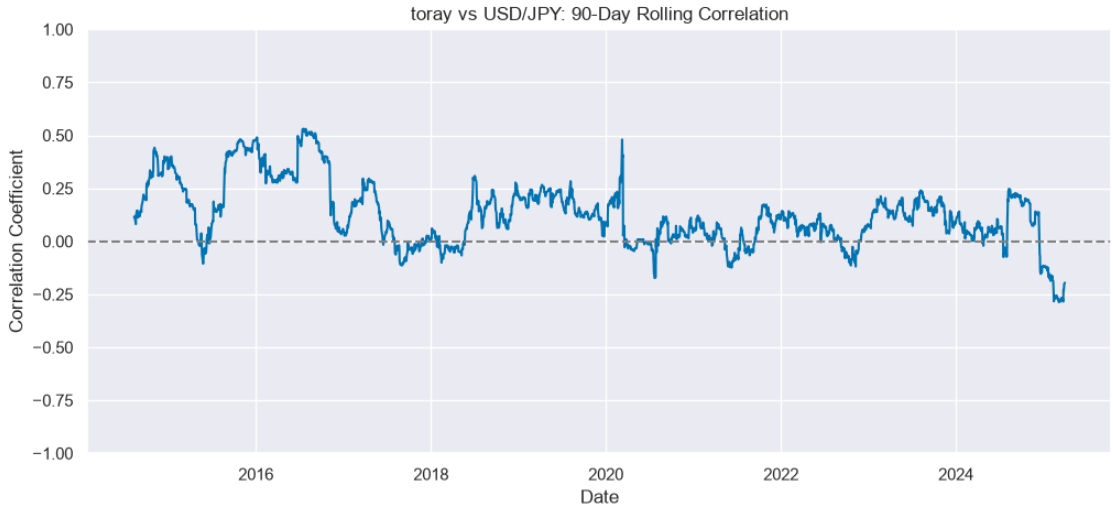

ajinomoto vs USD/JPY: 90-Day Rolling Correlation
Correlation Coefficient
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
2016
2018
2020
2022
2024
Date

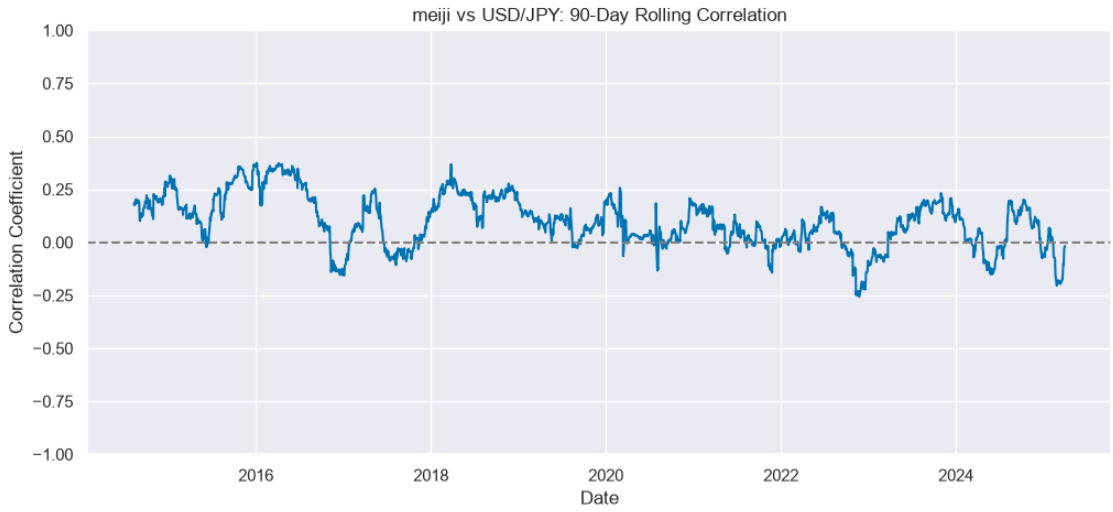

kajima vs USD/JPY: 90-Day Rolling Correlation
Correlation Coefficient
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
2016
2018
2020
2022
2024
Date

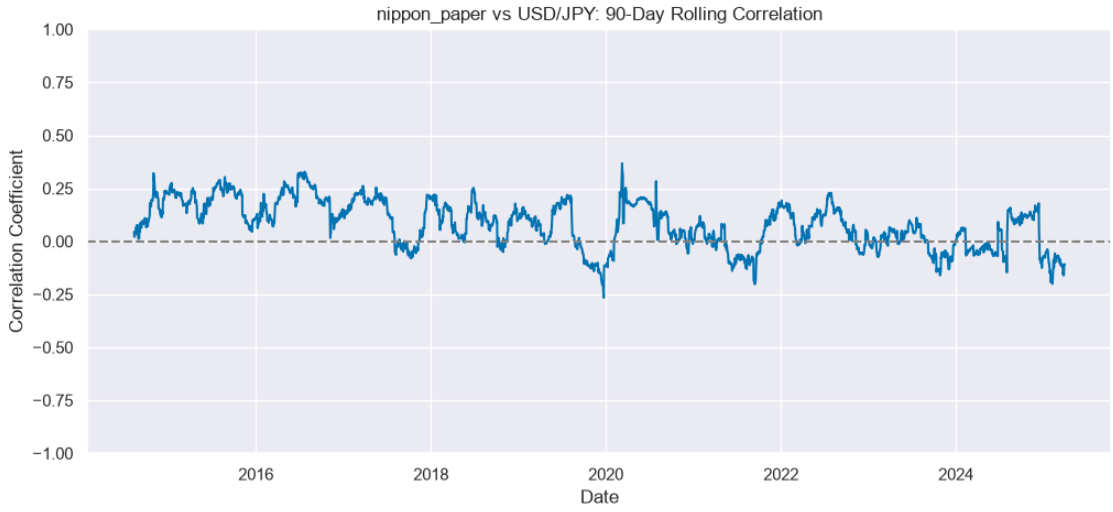

okuma vs USD/JPY: 90-Day Rolling Correlation
Correlation Coefficient
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
2016
2018
2020
2022
2024
Date

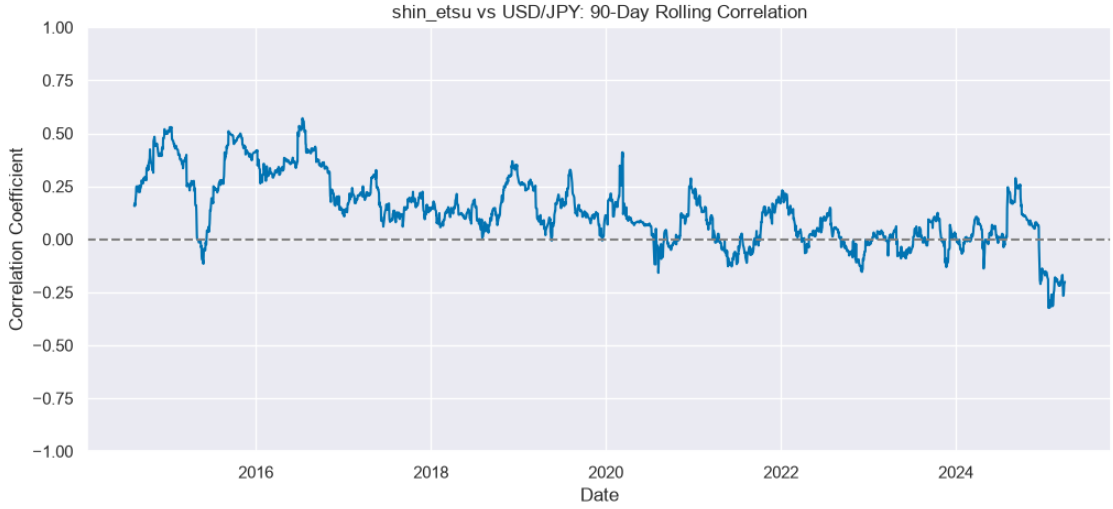

kewpie vs USD/JPY: 90-Day Rolling Correlation
Correlation Coefficient
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
2016
2018
2020
2022
2024
Date

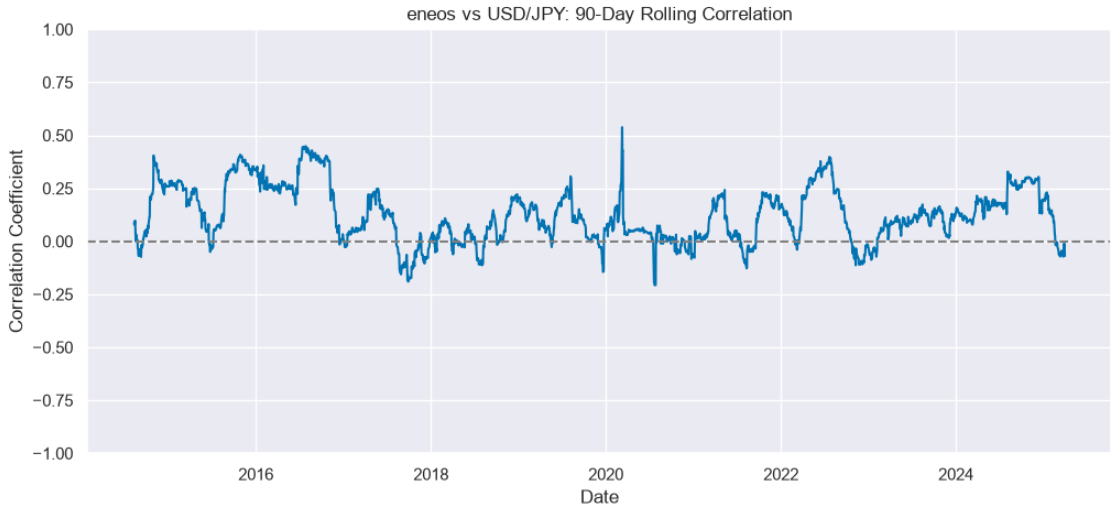

toyota vs USD/JPY: 90-Day Rolling Correlation
Correlation Coefficient
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
2016
2018
2020
2022
2024
Date

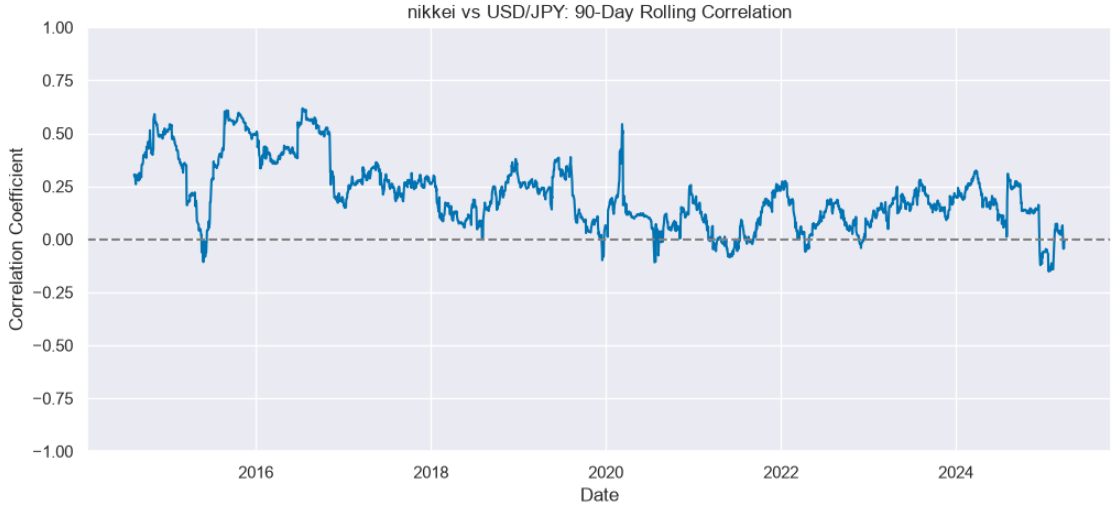

toray vs USD/JPY: 90-Day Rolling Correlation
Correlation Coefficient
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
2016
2018
2020
2022
2024
Date

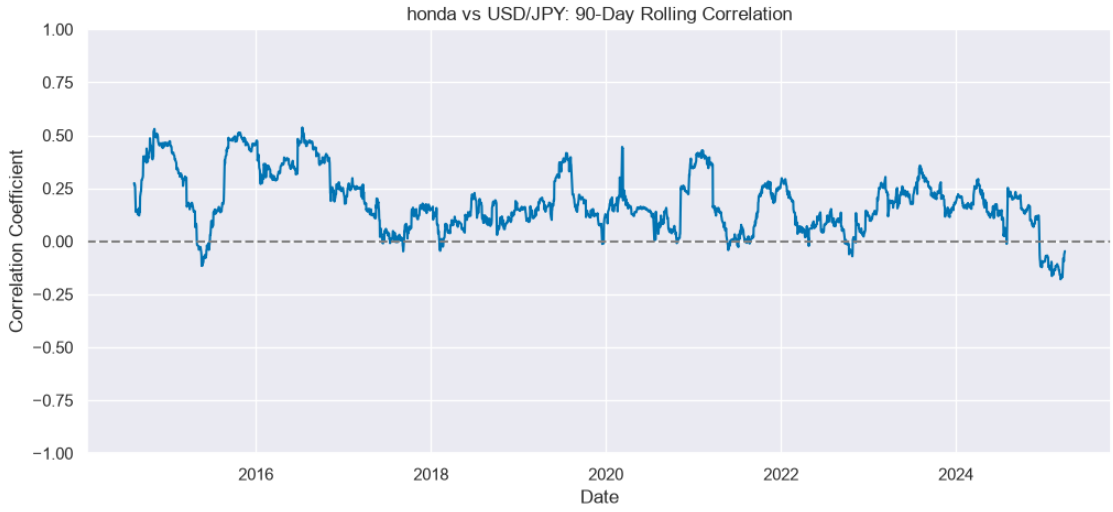

meiji vs USD/JPY: 90-Day Rolling Correlation
Correlation Coefficient
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
2016
2018
2020
2022
2024
Date

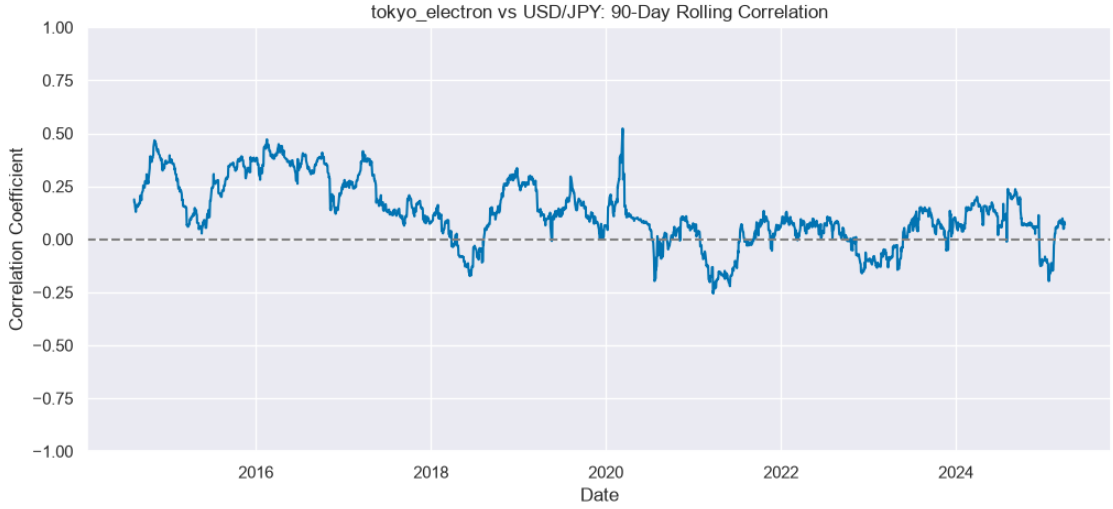

nippon_paper vs USD/JPY: 90-Day Rolling Correlation
Correlation Coefficient
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
2016
2018
2020
2022
2024
Date

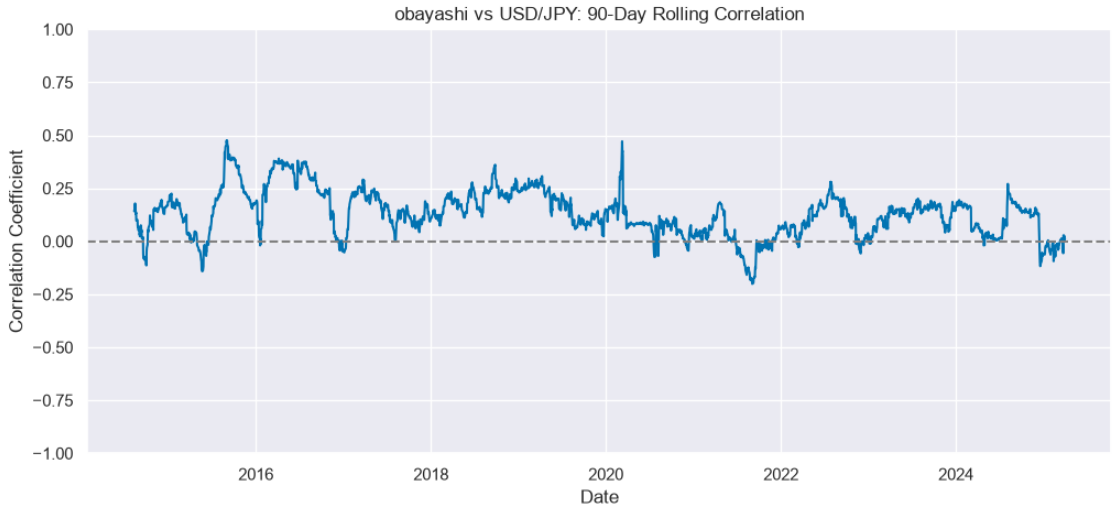

shin_etsu vs USD/JPY: 90-Day Rolling Correlation
Correlation Coefficient
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
2016
2018
2020
2022
2024
Date


eneos vs USD/JPY: 90-Day Rolling Correlation
Correlation Coefficient
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
2016
2018
2020
2022
2024
Date


nikkei vs USD/JPY: 90-Day Rolling Correlation
Correlation Coefficient
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
2016
2018
2020
2022
2024
Date


honda vs USD/JPY: 90-Day Rolling Correlation
Correlation Coefficient
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
2016
2018
2020
2022
2024
Date


tokyo_electron vs USD/JPY: 90-Day Rolling Correlation
Correlation Coefficient
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
2016
2018
2020
2022
2024
Date



1.3.16 90 Day rolling observations

Looking at the 90 day rolling correlation we see that the correlations vary widely over time. For some firms there may be a moderate positive correlation for a few years to no correlation to to moderate negative correlation. This means that the correlation between yen and usd is rather fluid, likely due to many aditional factor outside of USD/yen impacting stock returns.

2 Now lets incorporate the Government reported foreign export data into our analysis

[15]:
```
# Load the export data
industry_exports = pd.read_csv("Export Ratios.csv")

# Keep only industry-level data and necessary columns
industry_data = industry_exports.loc[
    industry_exports["level"] == "industry",
    ["fiscal_year", "industry_en", "export_ratio_pct"]
].copy()

# Standardize the industry column name
industry_data.rename(
    columns={"industry_en": "industry"},
    inplace=True
)

#Labeling stocks based on their industry"
stock_industries = {
    "idemitsu": "Petroleum & Coal",
    "mitsubishi": "Chemicals",
```

```
    "oji": "Wood, Paper & Pulp",
    "teijin": "Textiles",
    "ajinomoto": "Food Products",
    "kajima": "Construction",
    "okuma": "Production Machinery",
    "kewpie": "Food Products",
    "toyota": "Transport Equipment",
    "toray": "Textiles",
    "meiji": "Food Products",
    "nippon_paper": "Wood, Paper & Pulp",
    "shin_etsu": "Chemicals",
    "eneos": "Petroleum & Coal",
    "honda": "Transport Equipment",
    "tokyo_electron": "Production Machinery",
    "obayashi": "Construction"
}

industry_list = ["Transport Equipment",
                 "Textiles",
                 "Chemicals",
                 "Production Machinery",
                 "Food Products",
                 "Wood, Paper & Pulp",
                 "Petroleum & Coal",
                 "Construction"
                ]


avg_exports = (
    industry_data.groupby("industry")["export_ratio_pct"].mean()
)

display(avg_exports.sort_values(ascending=False))
```

```
industry
Business-Use Machinery             49.327273
Production Machinery               48.481818
Info & Communication Equipment     45.945455
Transport Equipment                45.454545
Non-Ferrous Metals                 31.300000
General-Purpose Machinery          30.809091
Electrical Machinery               29.163636
Other Manufacturing                27.454545
Iron & Steel                       26.381818
Chemicals                          25.763636
Ceramics, Stone & Clay             25.454545
Transportation (services)          21.754545
Wholesale                          16.990909
```

```
Services                            16.709091
Agriculture, Forestry & Fishery     14.527273
Textiles                            11.000000
Metal Products                      10.709091
Petroleum & Coal                    10.554545
Wood, Paper & Pulp                   8.009091
Construction                         5.718182
Mining                               4.163636
Information & Communications         3.109091
Food Products                        2.245455
Retail                               1.872727
Other Non-Manufacturing              1.209091
Name: export_ratio_pct, dtype: float64
```

2.0.1 Average Export Data

For our data have

prod. machinery | 49%

Transport | 45.45%

Chemicals | 25.79%

Textiles | 11%

Petroleum | 10.55%

Wood, Paper, Pulp | 8%

Food | 2.24%

LIMITATION
A potential limitation of this data, however, is that it's a rather crude generalization. Most of the stocks we are looking at may not neatly fall into these categories. A much more accurate strategie would be to look at each individual stocks reports on export sales, but those aren't widely available.

2.0.2 Now lets see how these numbers move over time in relation to Yen/USD

[16]:
```
# Making a yearly yen data fram
yearly_yen = stock_prices[["Date", "yen_usd"]].copy()

# Adjusting yearly yen to the Fiscal Year range of our data (April to March)
yearly_yen["fiscal_year"] = (
    yearly_yen["Date"].dt.year
    - (yearly_yen["Date"].dt.month < 4)
```

```
)

# Getting the first and last exchange rate of a fiscal year
yearly_yen = yearly_yen.groupby("fiscal_year").agg(
    yen_start=("yen_usd", "first"),
    yen_end=("yen_usd", "last")
).reset_index()

# Creating the yearly percent change
yearly_yen["yen_change_pct"] = (
    (yearly_yen["yen_end"] / yearly_yen["yen_start"] - 1) * 100
)

display(yearly_yen)
```

	fiscal_year	yen_start	yen_end	yen_change_pct
0	2014	103.66	120.13	15.888482
1	2015	119.75	112.58	-5.987474
2	2016	111.72	111.39	-0.295381
3	2017	110.90	106.24	-4.201984
4	2018	105.91	110.85	4.664338
5	2019	111.35	107.53	-3.430624
6	2020	107.17	110.72	3.312494
7	2021	110.60	121.70	10.036166
8	2022	122.53	132.76	8.348976
9	2023	132.42	151.33	14.280320
10	2024	151.65	149.97	-1.107814

2.0.3 Adjusted fiscal year yen movement observations

Looking at the fiscal year data, during out 10 year window we see yen depreciating conderably with numerous years at over 10% drops in conversion values. In 2014 the rate goes from 103 yen per dollar to 120 yen per dollar.

We can also see that the data goes from 120 down to 107 from 2014 to 2019. The yen/usd rate after those years starts to depreciate considerably hoing to a year end of 149.97 yen at the end of the 2024 fiscal year.

2.0.4 Combining yen data with industry export ratios over fiscal year

[17]:
```
# Collecting industry data of only the industries in our da
industry_data = industry_exports[
    industry_exports["level"] == "industry"
][["fiscal_year", "industry_en", "export_ratio_pct"]].copy()


industry_data = industry_data.rename(
    columns={"industry_en": "industry"}
```

```
)

# Merge industry export data with annual USD/JPY changes
export_yen = industry_data.merge(
    yearly_yen[["fiscal_year", "yen_change_pct"]],
    on="fiscal_year",
    how="inner",
    validate="many_to_one"
)

display(export_yen)
```

	fiscal_year	industry	export_ratio_pct	yen_change_pct
0	2014	Food Products	1.3	15.888482
1	2014	Textiles	10.8	15.888482
2	2014	Wood, Paper & Pulp	8.1	15.888482
3	2014	Chemicals	22.8	15.888482
4	2014	Petroleum & Coal	8.0	15.888482
..	…	…	…	…
270	2024	Transportation (services)	12.0	-1.107814
271	2024	Wholesale	13.9	-1.107814
272	2024	Retail	2.1	-1.107814
273	2024	Services	14.5	-1.107814
274	2024	Other Non-Manufacturing	1.0	-1.107814

[275 rows x 4 columns]

2.0.5 Graphing Yen Movements per year over the Export ratio changes per year

[18]:
```
for industry in export_yen["industry"].unique():

    if industry in stock_industries.values():

        # Filter for the current industry
        current_industry = export_yen[
            export_yen["industry"] == industry
        ]

        # Create a figure and first axis
        fig, ax1 = plt.subplots(figsize=(12, 5))

        # Plot export ratios
        sns.lineplot(
            data = current_industry,
            x = "fiscal_year",
            y = "export_ratio_pct",
            marker = "o",
```

```
            label = "Export Ratio",
            ax = ax1
        )

        ax1.set_ylabel("Export Ratio (%)")

        # Create a second y-axis
        ax2 = ax1.twinx()

        # Plot annual USD/JPY movement
        sns.lineplot(
            data = current_industry,
            x = "fiscal_year",
            y = "yen_change_pct",
            marker = "s",
            color = "orange",
            label = "USD/JPY Change",
            ax = ax2
        )

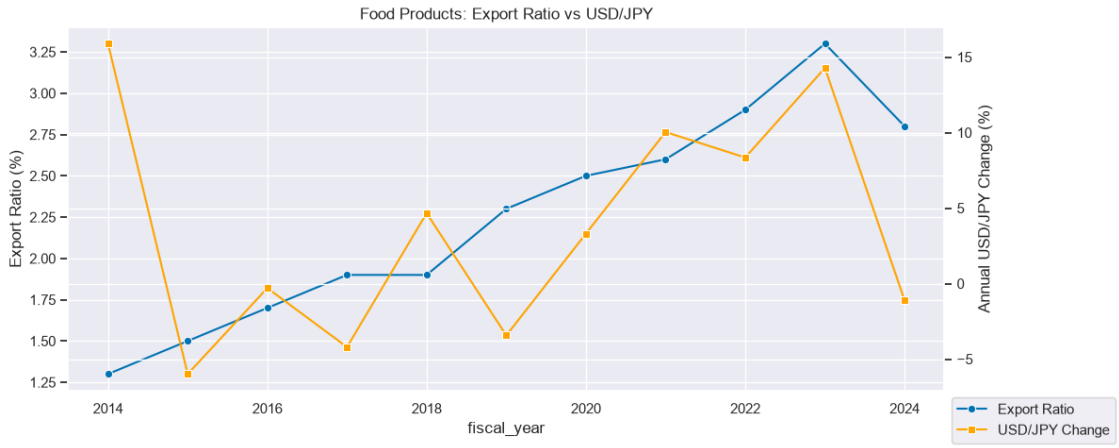
        lines1, labels1 = ax1.get_legend_handles_labels()
        lines2, labels2 = ax2.get_legend_handles_labels()

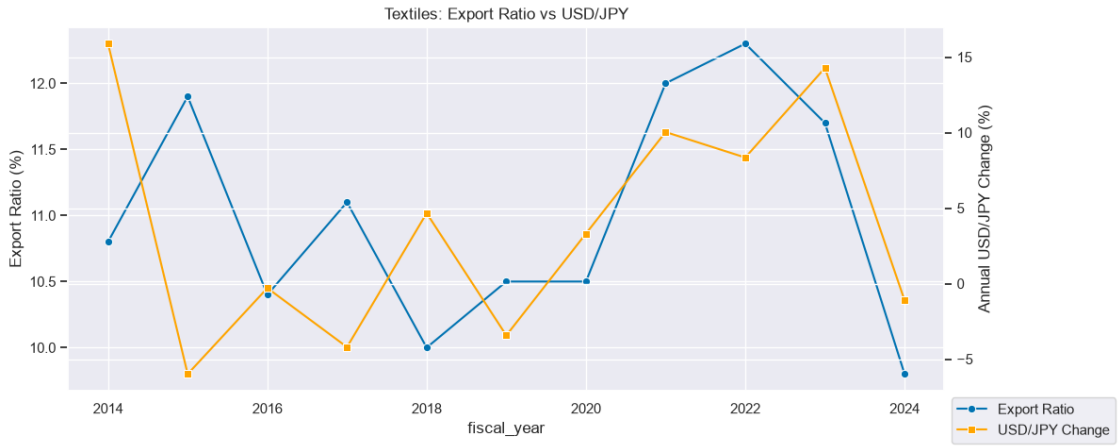
        ax1.legend(
            lines1 + lines2,
            labels1 + labels2,
        )

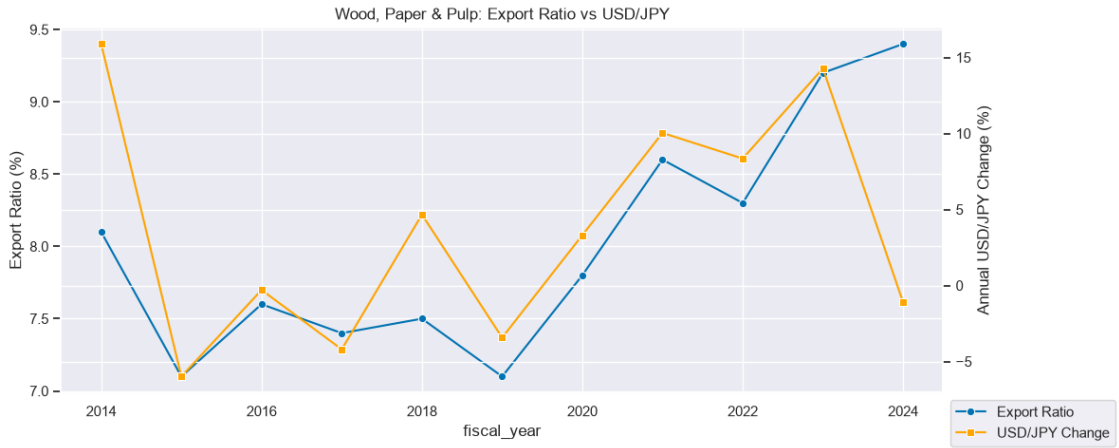
        sns.move_legend(ax1, "upper left", bbox_to_anchor=(1, 0))

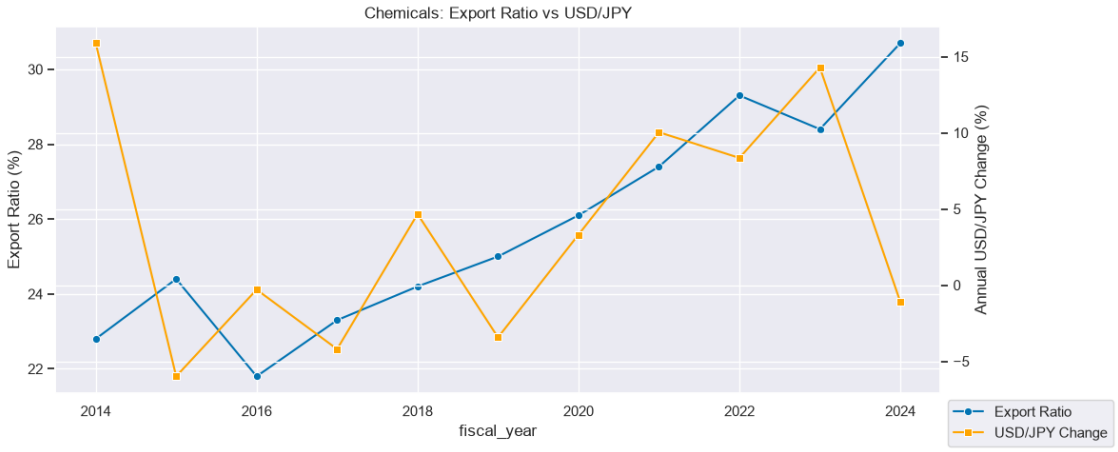
        # Remove the second legend
        ax2.get_legend().remove()

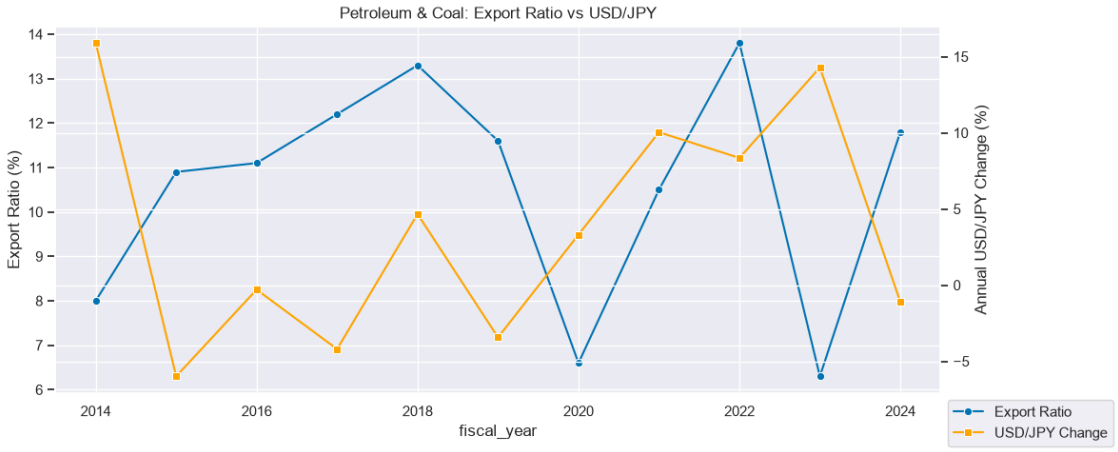
        ax2.set_ylabel("Annual USD/JPY Change (%)")
        plt.title(f"{industry}: Export Ratio vs USD/JPY")
        plt.show()
```

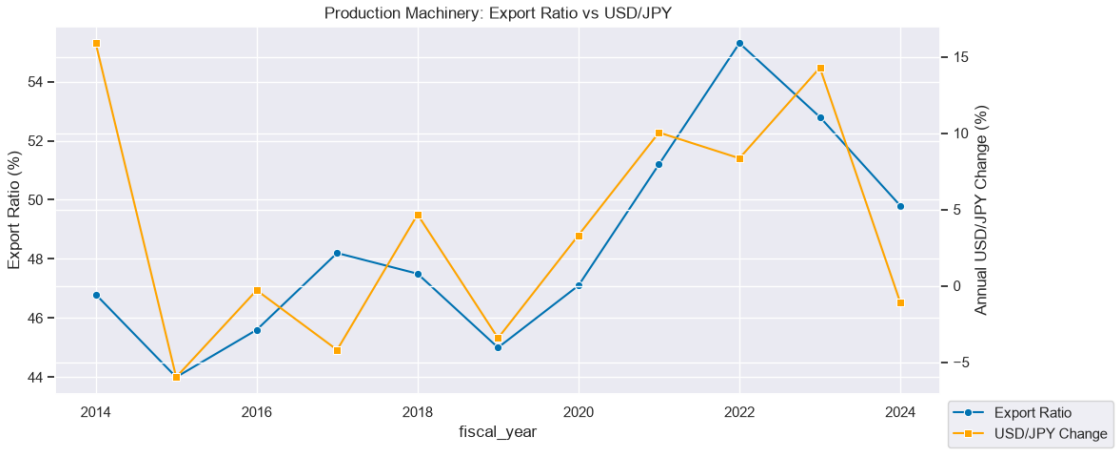

Food Products: Export Ratio vs USD/JPY
Export Ratio (%)
Annual USD/JPY Change (%)
fiscal_year
Export Ratio
USD/JPY Change

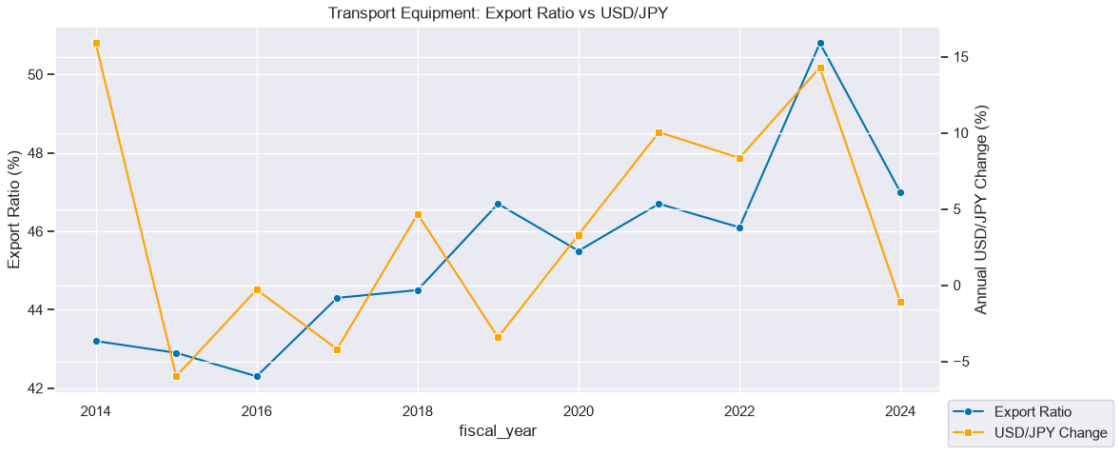

Textiles: Export Ratio vs USD/JPY
Export Ratio (%)
Annual USD/JPY Change (%)
fiscal_year
Export Ratio
USD/JPY Change

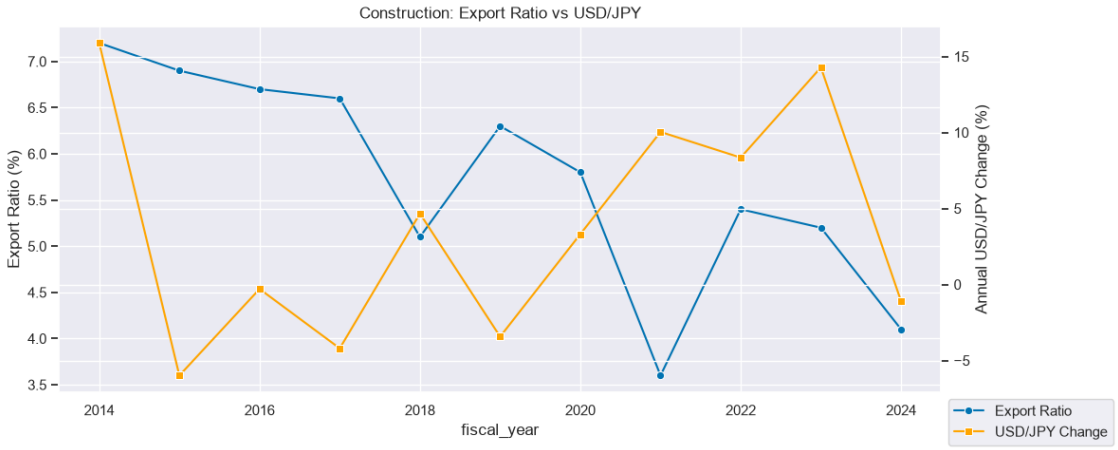

Wood, Paper & Pulp: Export Ratio vs USD/JPY
Export Ratio (%)
Annual USD/JPY Change (%)
fiscal_year
Export Ratio
USD/JPY Change


Chemicals: Export Ratio vs USD/JPY
Export Ratio (%)
Annual USD/JPY Change (%)
fiscal_year
Export Ratio
USD/JPY Change


Petroleum & Coal: Export Ratio vs USD/JPY
Export Ratio (%)
Annual USD/JPY Change (%)
fiscal_year
Export Ratio
USD/JPY Change


Production Machinery: Export Ratio vs USD/JPY
Export Ratio (%)
Annual USD/JPY Change (%)
fiscal_year
Export Ratio
USD/JPY Change





2.1 Export vs Yen Yearly Observations

When looking at the changes in exports we can see some interesting trends

Many of the graphs again tend to either follow the or mirrior the trends of USD/Yen. For example looking at transportation equipment we run into various years, like 2016- 2020 where export ratios would drop when yen/usd exchange rates would increase (Positive Change in USD/JPY means Yen is getting weaker). This was a bit surprising because my gut reaction was that weaker yen leads to more exports, but for this industry a weakening yen years would have cooling export ratios.

Looking at Food products we see it is steadily increasing over the decade from 1.25% to a max of 3.25%. The export ratios dont appear to be shifting too much in response to USD/JPY movements, the biggest reaction appears to be in 2018 and 2019 where a weakening yen in 2018 saw a flat export ratio and the following year of strenghtening yen saw a increase in export ration of around .6%.

Interestingly enough, it looks like paper and pulp shifter alongside yen/usd movements the most

2.1.1 Correlation heatmap of industry export ratio changes

[19]:
```
# Sort by industry and fiscal year
export_yen = export_yen.sort_values(
    ["industry", "fiscal_year"]
)

# Calculate annual changes in export ratios
export_yen["export_ratio_change"] = (
    export_yen.groupby("industry")["export_ratio_pct"]
    .diff()
)

#Removing Industries that are not in our data
export_yen = export_yen[
    export_yen["industry"].isin(stock_industries.values())
].copy()

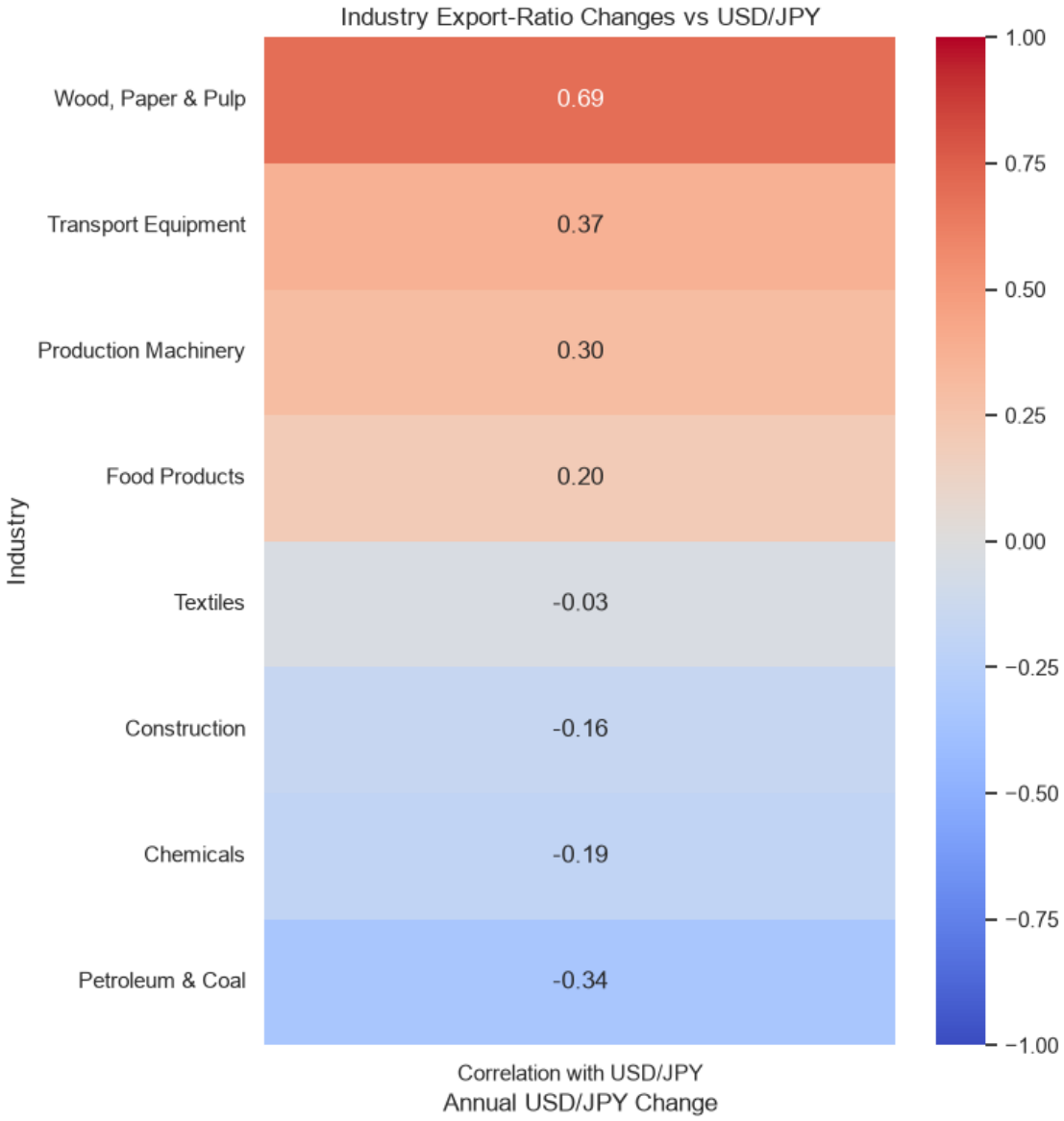

industry_corr = (
    export_yen.groupby("industry")
    .apply(
        lambda df: df["yen_change_pct"].corr(
            df["export_ratio_change"]
        )
    )
)

industry_corr = industry_corr.to_frame(
    name = "Correlation with USD/JPY"
)

plt.figure(figsize=(7, 9))

sns.heatmap(
    industry_corr.sort_values(
        by = "Correlation with USD/JPY",
        ascending = False
    ),
    annot = True,
    cmap = "coolwarm",
    center = 0,
    vmin = -1,
    vmax = 1,
    fmt = ".2f"
)
```

```
plt.title("Industry Export-Ratio Changes vs USD/JPY")
plt.xlabel("Annual USD/JPY Change")
plt.ylabel("Industry")
plt.show()
```



2.1.2 Correlation Observations of industries

Looking our heat map, a positive number indicated that yen weakening compared to USD correlates with an increase in exports. On the flip side, a negative number indicates that a strengthening yen correlates with an increase in exports.

The industry with the highest positive correlation is Wood, Paper & Pulp with a .69 correlation.

This was very interesting considering it has a rather low export percentage. The wood and pulp firms were also some of the worst performing stocks in relation to Yen/USD, so it is a bit surprising to see the correlation coefficient being so high.

2.1.3 Organizing stock by fiscal year and labeling the type of industry

[20]:
```
stock_prices["fiscal_year"] = (
    stock_prices["Date"].dt.year
    - (stock_prices["Date"].dt.month < 4)
)


yearly_correlations = []

for year, df in stock_prices.groupby("fiscal_year"):

    for name in stock_industries:

        stock_col = f"{name}_day_change"

        # Calculate correlation within this fiscal year
        corr = df[stock_col].corr(
            df["yen_usd_day_change"]
        )

        yearly_correlations.append({
            "fiscal_year": year,
            "company": name,
            "industry": stock_industries[name],
            "stock_corr": corr
        })

yearly_corr_df = pd.DataFrame(yearly_correlations)

display(yearly_corr_df.head())
```

	fiscal_year	company	industry	stock_corr
0	2014	idemitsu	Petroleum & Coal	0.210401
1	2014	mitsubishi	Chemicals	0.197418
2	2014	oji	Wood, Paper & Pulp	0.192892
3	2014	teijin	Textiles	0.179247
4	2014	ajinomoto	Food Products	0.185059

2.1.4 Adding export ratio percentages to the Dataframe

[21]:
```
# Merging yearly comparison with industry data
yearly_comparison = yearly_corr_df.merge(
    industry_data,
    on = ["fiscal_year", "industry"],
    how = "left",
    validate = "many_to_one"
)

display(yearly_comparison)
```

```
     fiscal_year         company              industry  stock_corr  \
0           2014        idemitsu      Petroleum & Coal    0.210401
1           2014      mitsubishi             Chemicals    0.197418
2           2014            oji    Wood, Paper & Pulp    0.192892
3           2014          teijin              Textiles    0.179247
4           2014       ajinomoto         Food Products    0.185059
..           …             …                   …          …
182         2024       shin_etsu             Chemicals   -0.005241
183         2024           eneos      Petroleum & Coal    0.185073
184         2024           honda   Transport Equipment    0.041815
185         2024  tokyo_electron  Production Machinery    0.069170
186         2024        obayashi          Construction    0.097209

     export_ratio_pct
0                 8.0
1                22.8
2                 8.1
3                10.8
4                 1.3
..                …
182              30.7
183              11.8
184              47.0
185              49.8
186               4.1

[187 rows x 5 columns]
```

2.1.5 Correlations grouped by industry

[23]:
```
#Creating a new df to look at correlations grouped by industry
industry_year_corr = (
    yearly_comparison.groupby(["fiscal_year", "industry"])
    .agg(
        mean_stock_corr=("stock_corr", "mean"),
        export_ratio_pct=("export_ratio_pct", "first")
```

```
    )
    .reset_index()
)

industry_year_corr
```

[23]:

	fiscal_year	industry	mean_stock_corr	export_ratio_pct
0	2014	Chemicals	0.251147	22.8
1	2014	Construction	0.054296	7.2
2	2014	Food Products	0.109959	1.3
3	2014	Petroleum & Coal	0.206589	8.0
4	2014	Production Machinery	0.245041	46.8
..	…	…	…	…
83	2024	Petroleum & Coal	0.163948	11.8
84	2024	Production Machinery	0.090405	49.8
85	2024	Textiles	0.023082	9.8
86	2024	Transport Equipment	0.072912	47.0
87	2024	Wood, Paper & Pulp	0.078460	9.4

[88 rows x 4 columns]

2.1.6 Graphing Industry Export ratio compared to Average stock correlation

[26]:
```
# iterating over unique industries
for industry in industry_year_corr["industry"].unique():

# Selecting the correlation coefficient
    corr = industry_year_corr[
        industry_year_corr["industry"] == industry
    ]

# Plotting the correlation for the stock
    fig, ax1 = plt.subplots(figsize=(12, 5))

    sns.lineplot(
        data= corr,
        x = "fiscal_year",
        y = "mean_stock_corr",
        marker = "o",
        ax = ax1,
        label = "Average Stock Correlation"
    )

# Labeling axis and joining the graphs
    ax1.set_ylabel("Stock Correlation")

    ax2 = ax1.twinx()
```

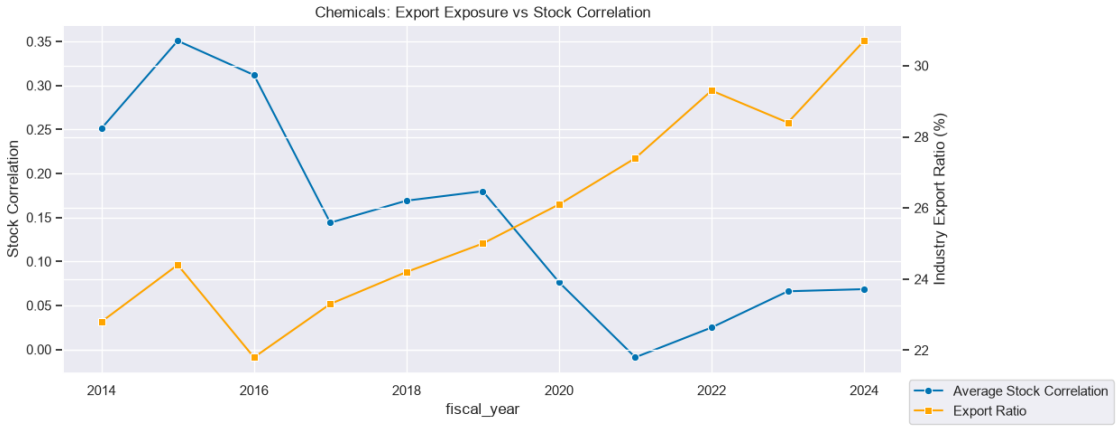
```
    sns.lineplot(
        data = corr,
        x = "fiscal_year",
        y = "export_ratio_pct",
        marker = "s",
        color = "orange",
        ax = ax2,
        label = "Export Ratio"
    )

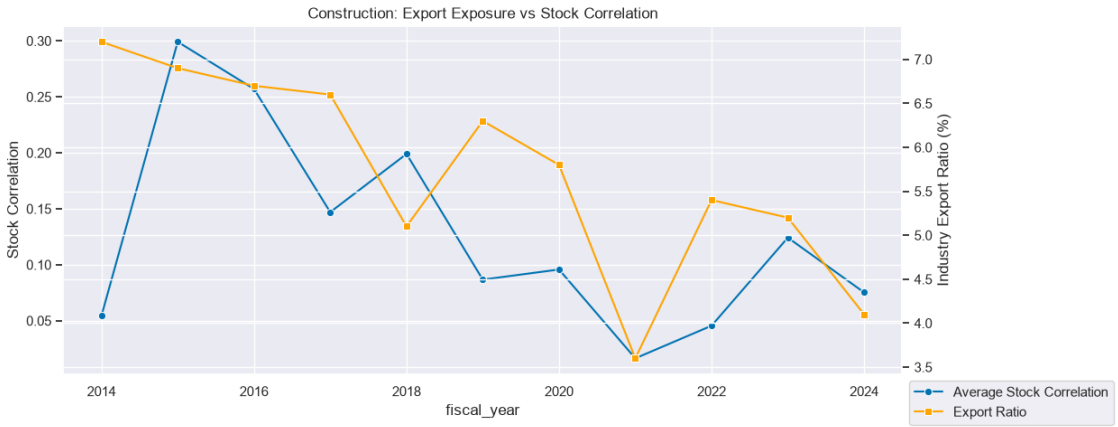
    lines1, labels1 = ax1.get_legend_handles_labels()
    lines2, labels2 = ax2.get_legend_handles_labels()

    ax1.legend(
        lines1 + lines2,
        labels1 + labels2,
    )

# Moving the legend to a less obstructive position
    ax2.get_legend().remove()
    sns.move_legend(ax1, "upper left", bbox_to_anchor=(1, 0))

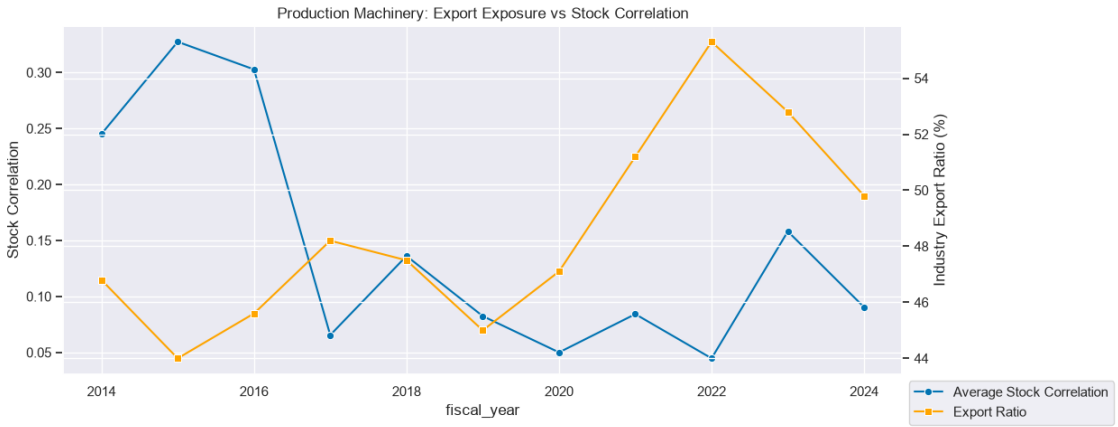

    ax2.set_ylabel("Industry Export Ratio (%)")
    plt.title(f"{industry}: Export Exposure vs Stock Correlation")
    plt.show()
```

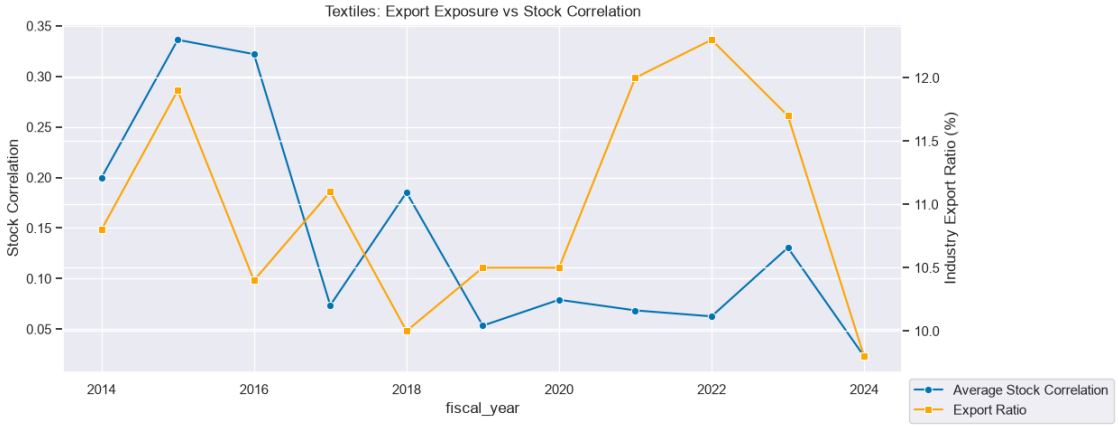


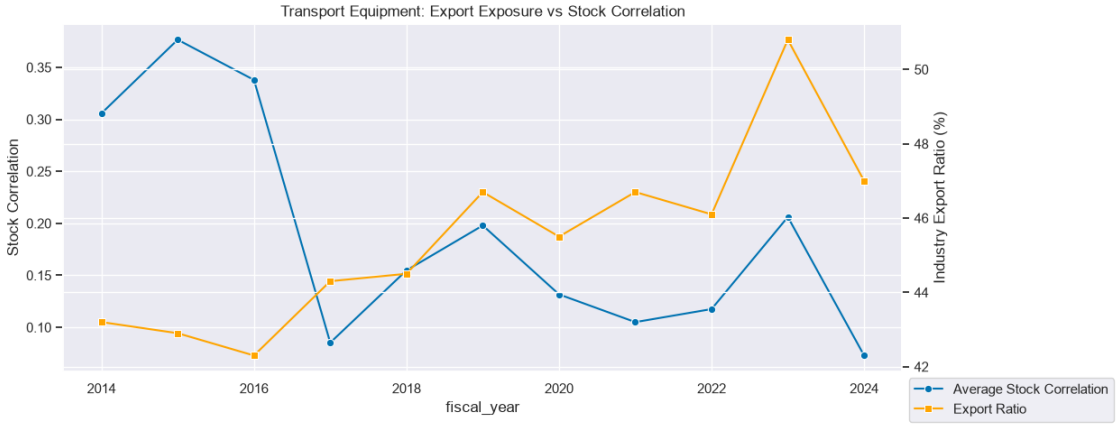

Construction: Export Exposure vs Stock Correlation
Stock Correlation
Industry Export Ratio (%)
fiscal_year
Average Stock Correlation
Export Ratio

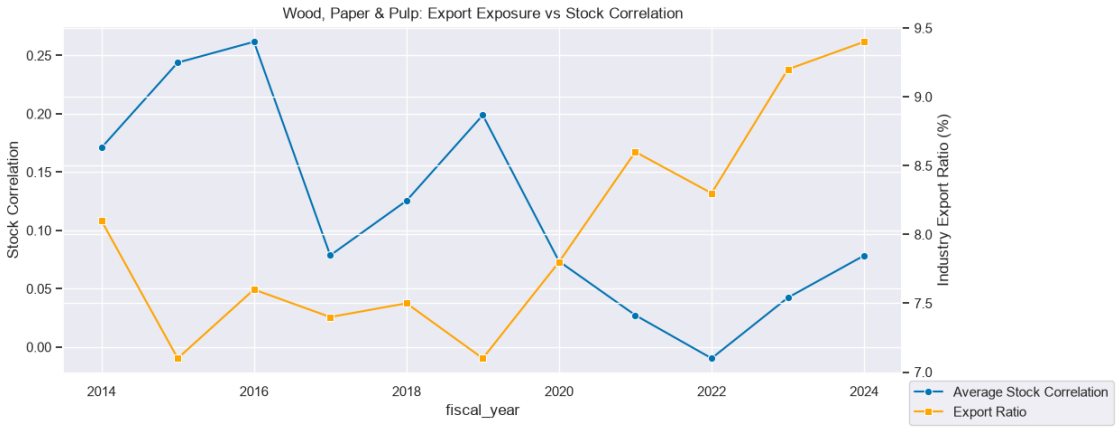

Food Products: Export Exposure vs Stock Correlation
Stock Correlation
Industry Export Ratio (%)
fiscal_year
Average Stock Correlation
Export Ratio


Petroleum & Coal: Export Exposure vs Stock Correlation
Stock Correlation
Industry Export Ratio (%)
fiscal_year
Average Stock Correlation
Export Ratio


Production Machinery: Export Exposure vs Stock Correlation
Stock Correlation
Industry Export Ratio (%)
fiscal_year
Average Stock Correlation
Export Ratio


Textiles: Export Exposure vs Stock Correlation
Stock Correlation
Industry Export Ratio (%)
fiscal_year
Average Stock Correlation
Export Ratio


Transport Equipment: Export Exposure vs Stock Correlation
Stock Correlation
Industry Export Ratio (%)
fiscal_year
Average Stock Correlation
Export Ratio



2.1.7 Industry Average Correlation vs Export Ratio Obsevations

Among these graphs we can notice some interesting trends.

- **Chemicals**
  - Stock correlation drops as export ratio rises, reaching its minimum at 2021 (Probably due to covid?). We saw a max correlation of .35 and a minimum of slighly below 0. This industry experienced a wide variance of correlation movement
- **Construction**
  - Stock correlation seems to decrease with export ratio, going from 7% - 4% and experiencing a correlation decline of .3 to .1
- **Food Producs**
  - Export ratios seemed to not budge much at all in relation to export ratio. The biggest similarity was a peak and drop from 2023 to 2024
- **Pretol & Coal**
  - Export ratios seem to inverse stock correlations. When stock correlation decreased it often followed with an inrease in export ratio.
- **Production Machinery**
  - Export ratios varied widely between 14 and 6% but their relation to mean correlation does not appear to be too strong, we can see some mirrioring and some inverse reactions.
- **Textiles**
  - Export ratios in this graph seemed to move hand in hand with correlation but slightly lagged. Where we can see correlation mimicking the movements of export ratios of prior movements
- **Transport**
  - Export ratios are seen to be steadily increasing over time while average correlation drops from .36 to below .1
- **Wood, Pulp, Paper**
  - Export ratios are increasing over time and we see the correlation following usually in the inverse of this trend

2.1.8 Scatter plot of Export exposure vs USD/JPY correlation grouped by industry type

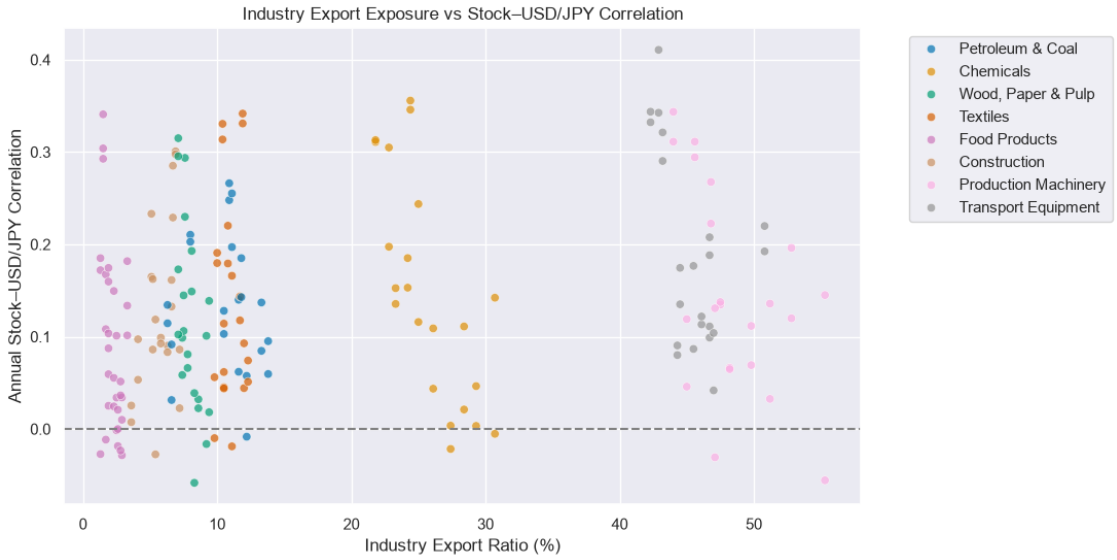
[25]:
```
plt.figure(figsize=(10, 6))

sns.set_theme(palette = "colorblind")

sns.scatterplot(
    data = yearly_comparison,
    x = "export_ratio_pct",
    y = "stock_corr",
    hue = "industry",
    alpha = 0.7
)


plt.axhline(0, color="gray", linestyle="--")

plt.title("Industry Export Exposure vs Stock-USD/JPY Correlation")
plt.xlabel("Industry Export Ratio (%)")
plt.ylabel("Annual Stock-USD/JPY Correlation")
plt.legend(bbox_to_anchor=(1.05, 1), loc="upper left")
plt.show()
```



2.1.9 Indstry Export vs Stock-USD/JPY Correlation observations

Looking at the graph we can clearly see that industries tend to hover around the same export ratios.

Reflecting our ealier data Food, Wood, Petroleum, Textiles, populate the lower export rations while transportation and production machinery populate the higher export ratio percentage.

It appears that there is a slight increase in responsiveness over export ratio. We can also see that there are less data points near zero correlation the higher up in industry export ratio we go up.

3 Findings Summary

In this project we analyzed the relationship between stock movements, USD/Yen movements, and industry export ratios.

We found that many stocks do appear to have some correlation with USD/Yen movements. When observing month to month movements, we observed that transportation related stocks like Toyota and Honda had the highest correlation coefficients of .51 and .44. The lowest in this time interval was Kewpie and Nippon Paper with .045 and -.025.

When we look at export ratios based on industry, we see that transportation has a high ratio of 45.45%, with the lower end being food and paper products with 2.24% and 8%.

From these two points we can make the observation that businesses with a higher export ratio may correlate more with movements for USD/Yen movements.

When we look at the correlation between industry export ratio changes and USD/yen movements we actually see Wood, Paper, & Pulp with a higher coefficient of .69 than transport equipment with a coefficient of .37. This observation was surprising because the ratio of exports done by that industry is only 8%.

We then conclude our analysis with an Industry Export Exposure vs Stock-USD/JPY Correlation graph. From this graph we can observe that industry does appear to have a small positive association on annual stock correlation. Industries with a higher export ratio tended to chart higher in correlation coefficient compared to lower ratios.

3.1 Business Suggestion

Since we observed higher export ratio industries haveing a higher association with yen movements, I would recommend that businesses look at thier own export ratios to decide how to react to exchange rate changes

Firms with high export ratios could benefit to monitoring yen/USD movements regulary and adjusting business strategies based on the strength or weakness of the Yen

Firms with low export ratios may be less directly associated with yen/USD movements, our findings showed even low export firms had some association with yen/USD movement. Therefore, I would suggest that firms in this bracket monitor yen/USD movements if their inputs are sourced primarily from abroad.

3.1.1 Limitations

- Some of the data sets were missing dates, forcing other data sets to drop those missing dates as well.
- Daily correlation may be subject to time lags, in a future analysis it may be interesting to see how introducing a one day or two day lag impacts the data

- Covid 19 pandemic is within our date ranges. We can see ratios, averages, and returns dropping significantly in the 2020 - 2022 period that may influence our analysis
- Many of the stocks in this data set dont perfectly fit into one specific industry, Ajinomoto and Meiji for example, are very diverse companies that produce food and a wide variety of other goods. It would have been more insightful to use company reported export ratios but that data isn't as widely availble as the industry wide ratios provided by the Japanese government
- Large Scope | This analysis attempts to go over numerous variables at once on a surface level. It is likely worthwhile to narrow the scope and focus purely on yen/usd movements or a smaller set of stocks, then widen the scope later in a seprate analysis

3.1.2 Data sources

- Stock CSV data collected from : https://www.wsj.com/market-data
- Industry export ratios collected from : https://www.e-stat.go.jp/stat-search/files?page=1&toukei=00550120&tstat=000001011012

3.1.3 References

- General Firm and product info (Profile tabs for each stock): https://www.wsj.com/market-data
- Ajinomoto proucts (semiconductors): https://www.ajinomoto.com/innovation/our_innovation/buildupfilm
- Meiji product (Food / Pharmasuticals) : https://www.meiji.com/global/food/